Climate change increases bilateral trade costs

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Abstract

It is well established that climate change affects productivity, but its effects on trade costs have not been studied. I combine international trade and weather data covering 190 years and show how an augmented gravity framework can be used to estimate the impact of climate change on bilateral trade cost. I find that climate change, as measured by decadal temperature shifts at the origin or destination country, increases them. Adaptation to these changes seems slow. They appear to be driven by impacts on maritime trade in particular, perhaps due to the vulnerability of seaports to climate change. Combining these results with a standard international trade model, I find that 2010s welfare would increase by 1.7 percent if we could undo the impact of climate change on trade cost over the preceding 100 years. Welfare gains depend not only on countries' own climate trends, but also on their neighbors' trajectories. Smaller economies, which are more reliant on international trade, are especially affected. Ignoring this trade cost channel and focusing only on productivity impacts leads to a nine percent underestimate of the welfare effect of climate change, with larger differences for smaller open economies. Because it is based on a gravity framework, my methodology can easily be embedded in studies of the impact of climate change.

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Analyses of the economic effect of climate change typically take trade costs as given and focus on the impact on productivity. For example, they might estimate the impact of weather shocks or climate change on productivity in different countries or sectors, use a climate scenario to project future changes in productivity and feed these through a trade model to assess damages across countries. The trade network remains unchanged in this projection. Trade costs, however, are shaped by the same economic forces through which climate change affects productivity, namely labor productivity and the availability of labor and capital. Does climate change, then, directly affect trade costs, just as it does other forms of economic activity? If so, does ignoring this mean we underestimate the impact of climate change?

I use trade and weather data spanning 190 years to show that climate change increases bilateral trade cost. I show this using an augmented gravity framework with one simple addition: an interaction between distance and decade-to-decade changes in average temperature at the origin and destination countries. I find that climate change significantly raises trade cost. The core identification concern is that countries which see more rapid climate change are different along other dimensions as well, and would have seen trade cost increases even absent climate change. I show that my results are robust to allowing for rich heterogeneity in trade cost levels and trends based on countries' climatic environment, allaying these identification concerns. I find that impacts are especially severe for countries with lower benchmark temperatures and countries which see especially rapid climate change. This suggests that countries can adapt to these effects, but the adaptation process is slow.

I present additional reduced form results suggesting the impact is driven by maritime trade in particular. My findings here align with recent research on weather impacts on ports and anecdotal evidence from policymakers and port operators. I discuss the importance of including climate variables in the gravity framework in a way that captures how physical climate processes affect the economy. This builds on, but leads to markedly different modeling choices than the literature on estimating the impacts of non-discriminatory trade policies (such as across-the-board, rather than bilateral, tariffs). There, all international trade is equally affected while domestic trade cost (presumably) remains unaffected. Climate change, however, affects both domestic and international trade cost, and likely affects some kinds of international trade more than others, especially where climate-driven disruptions to infrastructure are especially likely (e.g., in maritime trade).

I embed my estimates in a standard model of international trade (Eaton & Kortum, 2002) to quantify welfare impacts. I find that welfare in the 2010s would have been 1.7 percent higher if

climate change had not increased trade cost over the preceding 100 years. This is due only to the resulting reduction in trade cost, shutting down any productivity impacts. Welfare gains depend not just on countries' own climate trends, but also on their neighbors' trajectories: When country i's neighbors face more drastic climate change than i itself, it experiences a relative trade cost reduction. Reverting climate change thus benefits i less, since its relative position declines. Smaller economies see especially large welfare gains, because they are more reliant on international trade. Finally, I show that ignoring this trade cost channel and focusing solely on productivity leads to a nine percent underestimate of the welfare impact of climate change. Even using a more conservative calibration for the model's core parameter, I still find a five percent increase. The underestimate is again larger for some countries, especially small open economies. (Note that my welfare calculations focus solely on the impact of climate change on trade cost and productivity. I set aside feedback loops with emissions or health benefits of reduced warming.)

Since my reduced form results rely on a simple augmented gravity specification, the effect I find can easily be included in other studies of the impact of climate change. A broad class of trade models yield gravity equations that can be estimated using simple regression techniques, outside of computationally costlier estimation loops. Any study using one of these models could incorporate my methodology and capture the impact of climate change on trade cost without incurring much computational overhead.

I contribute to the literature on the impacts of climate change in equilibrium. Existing studies generally estimate how trade affects productivity (e.g., Costinot, Donaldson, & Smith, 2016; Cruz & Rossi-Hansberg, 2024; Dallmann, 2019; Dellink, Hwang, Lanzi, & Chateau, 2017; Desmet, Kopp, Kulp, Nagy, Oppenheimer, Rossi-Hansberg, & Strauss, 2021; Huppertz, 2025; Jones & Olken, 2010; Nath, 2020; Porteous, 2024; Schlenker & Osberghaus, 2025). They model climate change counterfactuals with reduced productivity but an *unchanged* trade network. That is, while different countries (or firms) become less productive in these counterfactuals, it is no more difficult for those countries (or firms) to ship goods across the globe. I show that this is too optimistic a baseline. We should expect that under climate change, trade networks will deteriorate. Using current trade networks to assess the impact of climate change thus underestimates that impact. My results also show that the impacts of weather on yearly trade flows (Gassebner, Keck, & Teh, 2010; Martínez-Martínez, Esteve-Pérez, Gil-Pareja, & Llorca-Vivero, 2023) and ports (Astier, 2025; Izauirre, Losada, Camus, Vigh, & Stenek, 2021; Massoni, 2025; Sytsma, 2020), which affect global supply chains (Bai, Fernández-Villaverde, Li, & Zanetti, 2024), translate into long-term climate

change damages to trade costs.

In this literature, I also highlight some important issues in isolating trade cost from productivity. These specifically include the necessity to include origin- and destination-period fixed effects at the frequency at which trade and temperature are observed. This sets my approach apart from papers using trade data or considering exporting firms, but which still focus on productivity rather than trade cost (e.g., Dallmann, 2019; Jones & Olken, 2010; Schlenker & Osberghaus, 2025).

Relatedly, I contribute to the literature on estimating the effect of country-specific variables in a gravity context (for a discussion, see, e.g., Yotov, Piermartini, Monteiro, & Larch, 2016). Specifically, I show that it is still possible to recover an impact of climate change on trade cost with these fixed effects in place, even though climate change by itself is a country-specific phenomenon which gets absorbed by those fixed effects. Adapting approaches from the literature on non-discriminatory trade policy effects, I argue that interacting climate variables with an appropriate bilateral measure of trade cost allows estimation of trade cost impacts. Given the physical process behind climate change and how it affects the economy, I argue that the standard division of domestic and international trade does not work well in this context. For example, climate change may well affect domestic trade, and it does not necessarily affect all international trade equally, e.g., because certain modes of transport face more bottlenecks than others (seaports are few and far between compared to roads, for example). Rather, I propose distance as a parsimonious first step that fits the climate context, and show how those results can then inform further analyses honing in on one specific mechanism. (In my case, focusing on maritime trade.) I also show that the exact functional form of the bilateral variable is not a concern, as long as climate-related variables are plausibly exogenous.

I further contribute to the debate about the magnitude of the welfare impacts of climate change. Using a long time horizon and considerable variation in decade-level average temperatures, I can directly estimate the impact of climate change on trade cost, without having to extrapolate from weather fluctuations to climate change impacts. The welfare impacts I find for these trade cost changes are comparable in magnitude to the welfare impacts from productivity impacts found in Costinot et al. (2016) and Nath (2020), for example. They are only about ten percent of direct estimates of climate productivity damages (e.g., Burke, Hsiang, & Miguel, 2015, 2026; Ramey, Nath, & Klenow, 2025), however. They are also only about five percent of the welfare impact of one degree of warming estimated in Bilal and Känzig (2026), who use global GDP and temperature and thus combine trade cost and productivity impacts. In workhorse models of international trade, the welfare impacts of trade cost changes (the gains from trade) tend to be small relative to the impacts

of productivity changes. The impacts I find are thus consistent with overall larger estimates of the impact of climate change.

Finally, my results also relate to the literature on the carbon cost of trade. Trade itself generates considerable carbon emissions (Cristea, Hummels, Puzzello, & Avetisyan, 2013; Shapiro, 2016) and interacts with environmental regulations (Shapiro & Walker, 2018). As a consequence, as Farrokhi and Lashkaripour (2021) point out, trade policy is one tool that could be used to curb global emissions. My results suggest that, because climate change increases trade cost, it can also reduce carbon emissions from trade. This novel channel might be important to take into account when modeling the impacts of carbon taxes, for example. I do not focus on feedback loops like this directly, however.

The paper most closely related to mine is Martínez-Martínez et al. (2023). They use year-to-year weather variation for 67 countries over the 1986–2016 period to estimate an impact of an array of different weather and weather-related shocks on international compared to domestic trade. They find both positive and negative impacts depending on the specific shock, with overall significance depending on the inclusion of China. My findings suggest that some of the impacts they find accumulate over time into detectable impacts of climate change on trade. My results are less rich in heterogeneity than theirs, though I find a clearer pattern of impacts which is robust to dropping China from the estimation sample. This may be because, using decadal temperature shifts, I get closer to a climate signal rather than weather variation. Climate change might have larger and more consistently negative impacts on the economy compared to one-time weather shocks. For example, a one-time warmer year in a colder country might be beneficial for trade. A sustained change in weather patterns, as indicated by a decadal shift in temperature, however, could still have negative effects, since it could lead to more frequent storms, droughts and other weather-related disruptions to trade. I also note the importance of choosing a suitable measure of trade cost that accounts for the physical process behind climate change. It is not obvious that climate change would affect only international trade and not domestic activity, so I focus on impacts on longer-distance and eventually maritime trade instead. (Indeed, my results suggest the impact is less clear when focusing only on trade among close neighbors.) I further note the importance of checking for robustness to differential trade cost trends across climatic environments, rather than just levels. Finally, the structural model I use allows me to calculate welfare impacts and compare these to the productivity channel. Overall, the two papers are highly complementary. Martínez-Martínez et al. (2023) provide a detailed overview of the impact of weather shocks. I show that longer-term variation and a focus

on a specific mechanism find significant climate impacts arising from those shocks.

Another closely related paper is Gassebner et al. (2010), who estimate the impact of general natural disasters on yearly trade flows, though without focusing specifically on trade cost. They find a consistently negative effect. Since trade cost is not the focus of their analysis, their estimation strategy does not isolate trade cost from productivity impacts. This could lead to confounding trade and productivity effects. I avoid this in my approach, since my focus is trade cost specifically. They also do not use a pseudo-Poisson maximum likelihood estimator, which has since become the standard in gravity estimation. Their log-log approach can lead to problems with dropped zero flows (Santos Silva & Tenreyro, 2006). Again, however, the two papers are highly complementary: Gassebner et al. (2010) show that year-to-year natural disasters, some of which are linked with climate change, affect trade flows. I show that this aggregates up to longer-term climate impacts on trade cost.

Note that throughout the paper, my measure of climate change are decade-to-decade changes in average temperature. These of course go along with changes in extreme weather events (J. Liu, Leung, Zhao, Xu, Huang, Zang, Tian, Ma, Li, & Zhang, 2026; Q. Liu, Fu, Xu, & Ding, 2025; Qi, Liu, Jiang, & Liu, 2026; Rahmstorf & Coumou, 2011). As I discuss further below, I capture the aggregate impact in changes across all weather dimensions, loaded onto temperature changes as a primary indicator of climate change. When I say climate change, I thus mean changes in the joint distribution of all possible weather variables, but I capture these through changes in average temperature across decades.

The remainder of the paper proceeds as follows: Section 1 discusses the data I use and presents descriptive statistics, Section 2 describes the gravity equation framework I use for my reduced form estimation, Section 3 presents results of the reduced form estimation, Section 4 explores mechanisms behind those results, Section 5 estimates the welfare impacts of trade cost increases due to climate change and compares these to productivity impacts, and Section 6 concludes.

1 Data and descriptive statistics

1.1 Data sets

This section presents the data sources I use and some descriptive statistics. I use data on trade flows from the CEPII TRADHIST (TRADHIST) database of historical international trade data (Fouquin & Hugot, 2016). These data cover yearly international bilateral trade flows from 1827 until 2014 and

contain additional information necessary for estimating gravity equations. (For example, measures of distance and other bilateral factors that could affect trade cost, such as shared languages, colonial past, or memberships in international trade agreements.). All trade flows are in nominal British pounds (GBP). I convert these to real values using data on UK GDP deflators from the Bank of England (Thomas & Dimsdale, 2017).[1] For a few exercises that require domestic distance measures, I further use CEPII's GeoDist database (Mayer & Zignago, 2011). TRADHIST includes these for its country pairs, but it only covers international trade.

I combine these trade flows with Berkeley Earth (BKE) data on monthly average temperature (Rohde, Muller, Jacobsen, Muller, Perlmutter, Rosenfeld, Wurtele, Groom, & Wickham, 2013). The temperature data go as far back as 1753 for some areas, achieve significant global coverage starting in 1850 and full global coverage beginning in 1960. I have weather data for virtually all countries in the trade data beginning in the 1850s. I use mainly BKE's combined land and ocean temperature data set, but augment this with their land only data set, since the latter goes further back in time. In order to link trade and temperature data, I use country boundaries from the Global Administrative Areas database (GADM) (Global Administrative Areas, 2022). I read in BKE temperature rasters for each month and calculate averages for each country based on its GADM area. GADM covers only currently existing countries, while TRADHIST also contains information on countries which no longer exist, such as West and East Germany, for example. For these countries, I create boundaries based on the GADM data and historical maps (shown in Appendix D).

For counterfactual exercises, I require current domestic trade flows to be able to calculate current domestic trade shares. (I do not require past domestic trade shares.) I therefore also use the International Trade and Production Database for Estimation (ITPD) (Borchert, Larch, Shikher, & Yotov, 2021, 2022). This database covers both international and domestic trade flows for a wide range of countries. For a robustness check looking at sectoral composition, I use World Bank data on sectoral composition (World Bank, 2026b). These include sectoral shares and value added for agriculture, industry and services, plus a final miscellaneous/other category. To investigate mechanisms, I additionally use data on container port throughput (United Nations Trade and Development [UNCTAD], 2026a). This contains the number of containers processed by country and year expressed in 20-foot equivalents (a measure of volume), going back to 2010. I also use data on

[1] Especially for earlier years, TRADHIST contains trade flows from some origins and destinations which are not countries. For example, it contains information on trade flows out of colonial administrative areas or individual cities. When I use the word 'country' in this paper, I always also mean these kinds of non-country reporters unless otherwise specified.

current global port locations and total outgoing trade flows from each port (World Bank, 2026a). For some supplementary descriptives, I use UNCTAD data on modes of transport (UNCTAD, 2026b).

1.2 Descriptive statistics

For the TRADHIST data, Figure 1 shows the number of countries observed by year. For each year, I count countries which appear at least once with a non-missing trade flow, since those are the only observations I can use in estimations. I separately show the number of origin and destination countries in the data, though the numbers barely diverge. The number of countries appearing in the data increases until around 1900 and stays roughly stable afterward. Figure 2 shows the number of observed trade flows by year. The number of flows observed per year is a lot higher after 1950. This suggests that post-1950 data give a more complete picture of each year's trade network. My analyses rely on analyzing individual trade flows, however, so this is not a limitation.

For the BKE data, Figure 3 shows the number of countries with non-missing weather observations by year. To focus purely on changes in weather coverage over time, rather than countries in the trade data, I count only currently existing countries that ever appear in the TRADHIST data. For the 1750s, I have weather coverage for a little over 60 countries. This increases over time, rising sharply in the 1850s. Starting in the 1880s I have truly global weather coverage.

To look at global climate trends, Figure 4 shows area-weighted average temperature in degrees Celsius for this same set of countries across years, plus a ten-year moving average. (I start the figure in 1880 because I have global weather coverage starting at that time.) Over time, average temperature rises from around 12.3°C in the 1880s to around 14.25°C in the 2010s. As the moving average shows, global mean temperature increases for most years after 1900. I see an especially fast increase and generally above-trend temperatures beginning in the 1980s. Note the lull in global temperature from roughly the 1950s to the 1980s, likely driven by aerosol emissions and associated 'global dimming' (Wild, Ohmura, & Makowski, 2007, 2021). I will sometimes use this as a benchmark period to look at heterogeneous impacts on colder and hotter countries.

To show the match between trade and weather data, Figure 5 shows the percentage of countries which appear in the trade data but have missing weather information across years. Prior to 1850, I am able to match between 60 and 80 percent of all trade flows. Starting in 1850, I have non-missing weather information for virtually all countries in the trade data. This is because BKE provides much better coverage starting in 1850.

2 Gravity estimation framework

This section presents the estimation framework I use for my core reduced form results. Gravity equations describe trade flows X_{nit} between an origin i and destination n during period t as (Head & Mayer, 2015)

$$X_{nit} = G_t S_{it} M_{nt} \phi_{nit}$$

where G_t captures global factors during the period, S_{it} and M_{nt} are exporter and importer specific terms, also called multilateral resistance terms (Anderson & van Wincoop, 2003), and ϕ_{nit} is a measure of trade cost between the two countries, called a bilateral resistance term. Because trade costs include lots of hard-to-measure components, e.g., personal relationships, the cost of time in port, or the cost of obtaining a permit, these are generally unobserved and have to be modeled (Anderson & van Wincoop, 2004).

Many international trade models yield a gravity equation of this form, with different interpretations of what the multi- and bilateral resistance terms comprise. For estimation purposes, the bilateral resistance term is often modeled as

$$\phi_{nit} = d_{ni}^{\alpha_t} e^{\mathbf{C}'_{nit}\boldsymbol{\beta}_t}$$

with d_{ni} a measure of physical distance between the two countries and $\mathbf{C}_{nit}$ a collection of bilateral variables that affect trade between the two countries, such as sharing a border or colonial history. The elasticity of trade flows with respect to distance captures a major determinant of trade cost. It varies over time to capture global changes in trade cost, e.g., innovations in shipping technology or the policy environment.

I augment this basic specification, allowing the effect of distance to vary with changes in origin and destination temperature,

$$\phi_{nit} = d_{ni}^{\alpha_t + \delta_1 \Delta T_{it} + \delta_2 \Delta T_{nt}} e^{\mathbf{C}'_{nit}\boldsymbol{\beta}_t} \tag{1}$$

Here, T_{ct} is average temperature in country c during period t, and $\Delta T_{ct} \equiv T_{ct} - T_{ct-1}$ is the change from period $t-1$ to period t. I include origin-period and destination-period fixed effects to model the multilateral resistance terms (Anderson & van Wincoop, 2003). Accordingly, I drop the level

effects of ΔT_{ct} which are captured by those fixed effects. This yields an estimating equation

$$\begin{aligned}\mathbb{E}\left(X_{nit}|\mathbf{D}_{nit}\right) &= e^{\gamma_{it}+\xi_{nt}+\log(\phi_{nit})} \\ &= \exp\left\{\gamma_{it}+\xi_{nt}+\alpha_t\tilde{d}_{ni}+\delta_1\tilde{d}_{ni}\Delta T_{it}+\delta_2\tilde{d}_{ni}\Delta T_{nt}+\mathbf{C}'_{nit}\boldsymbol{\beta}_t\right\} \end{aligned} \tag{2}$$

with origin-period and destination-period fixed effects γ_{it} and ξ_{nt}, using $\tilde{d}_{ni} \equiv \log\left(d_{ni}\right)$ to denote log distance and letting $\mathbf{D}_{nit}$ denote the set of n, i, t covariates. (Note that the global factors G_t are not separately identified from the origin-period and destination-period fixed effects, so I drop them.) To deal with instances where trade flows are equal to zero, rather than taking logs of both sides and using the resulting linear model, this is commonly estimated in its exponentiated form using pseudo-Poisson maximum likelihood estimation (PPML) (Santos Silva & Tenreyro, 2006), which I follow here. Note that this specification could be applied to any trade model that yields a gravity equation, so my estimation results apply to any model in this large class (see., e.g., Arkolakis, Costinot, & Rodríguez-Clare, 2012, for an overview).

2.1 Why interact temperature changes with distance?

The commonly used origin- and destination-period fixed effects are especially important for identification in my case, because they capture any impacts of climate change on country productivity. This allows me to isolate trade cost impacts. Using only origin and destination fixed effects, rather than origin- and destination-period fixed effects, would instead risk confusing the productivity and trade cost impacts of climate change.

Including these fixed effects of course means that I cannot include level temperatures in the estimation. This is a well-known problem in the gravity literature: It is difficult to estimate the impacts of country-specific variables, as opposed to bilateral variables. As (Yotov et al., 2016) note, there are two well established solutions to this problem. The first would be to use a two-stage estimation, where the first stage estimates multilateral resistance terms as fixed effects and the second stage regresses these on country-specific variables. In my case that is not feasible, because the multilateral resistance terms depend on productivity as well as trade cost. I would therefore again risk confusing productivity and trade cost effects.

The second approach is to interact the country-specific variable of interest with a bilateral variable. This is the approach I take here. This problem is common when studying the effects of non-discriminatory trade policy (i.e., trade policy instruments that affect all trade partners equally).

In that literature, if domestic trade data are available, the most common approach is to interact country-specific policy measures with an indicator variable for international trade (Yotov et al., 2016). Martínez-Martínez et al. (2023) use this approach to study the impact of different weather shocks and disasters on international compared to domestic trade. As Yotov et al. (2016) note, this interaction makes sense for non-discriminatory trade policy, since that affects all international trade costs equally and does not affect domestic trade cost at all.

In the case of climate change, however, it is not obvious that it would affect only international trade. It is also not clear that it would affect all international trade costs equally. For example, it might disrupt road traffic if roads wash out, or it might disrupt seaports, and hence maritime transport, due to storm surges or river droughts. These effects might be worse for some international trade routes than domestic trade, but the physical processes driving them do not start and stop at national borders the way trade policy does. Assuming no effect on domestic trade and a one-time change at the border, as international trade indicator interaction does, is thus not at all an obvious choice in the climate context. (Indeed, as I show in Section 4, the impacts I find are driven by country pairs where maritime trade is more attractive, rather than by trade over shorter land distances. The international indicator might miss this.) Finally, from a practical standpoint, I do not have domestic trade data going back far enough to study longer-term climate impacts.

I therefore interact temperature changes with distance. This is attractive for two reasons. First, it is probably the most well-known workhorse specification in gravity estimation. Understanding how climate change affects trade through the lens of this well-studied model is the obvious first step towards understanding all the nuances of its impact on different countries and different modes of trade. Second, it is intuitive to think that climate change could have differential impacts across routes depending on their distance. For example, longer routes often use different modes of transport, especially maritime compared to road freight, that might be especially susceptible to climate-related disruptions. Indeed, my results are consistent with this. Accordingly, when investigating mechanisms in Section 4, I switch from using distance to functional forms which isolate differential impacts on maritime trade compared to land-based transportation. As a starting point, however, distance is a parsimonious way to capture many different factors that make it harder to ship goods from the US to Australia than from the US to Canada, for example. I discuss in Section 2.3 below why the exact functional form in which distance enters the model is less of a concern in my context.

The interaction with a bilateral variable means my estimates are necessarily a lower bound on the impact of climate change on trade cost. If climate change shifted trade cost across all routes

equally, this would be absorbed by the fixed effects and I would miss it here — a shift in trade cost across the board is indistinguishable from a shift in productivity. I can only pick up whether climate change affects some kinds of trade more than others. This is not specific to the climate context. It is also an issue when estimating country-specific effects of non-discriminatory trade policy. If we estimated the impact of an across-the-board export tariff using an interaction with international trade, for example, it would not be clear from the results whether it only lowers international trade, or whether it also raises domestic trade. (It could even reduce both forms of trade due to input-output-linkages, but reduce international trade more.) This is an unfortunate but unavoidable problem when estimating trade cost impacts. It makes this problem markedly different from using trade data to estimate productivity effects (e.g., Dallmann, 2019; Jones & Olken, 2010; Schlenker & Osberghaus, 2025). Because productivity impacts do not require country-period fixed effects for identification, these papers can include level temperatures. (Though this in turn means they cannot necessarily distinguish between productivity impacts and across-the-board trade cost shocks.)

Note also that, while temperatures are interacted with distance, this specification captures changes in both the fixed and variable costs of trade. In a model such as Melitz (2003), for example, ϕ_{nit} depends both on the product of both the variable and fixed costs of trade. The specification I use simply uses distance to approximate that bilateral resistance term, regardless of what fraction of it is due to variable or fixed costs of trade.[2] The thought experiment is this: Every country is separated from every other country by a set of bilateral distances. Shipping goods requires bridging those distances and that is costly. As a country — Germany, for example — experiences climate change, the specification I use can tell whether it becomes more costly for Germany to bridge those distances and send goods abroad. Likewise, it can tell whether it becomes more costly for other countries to bridge that distance and send goods to Germany. The model thus allows temperature to change the cost of bridging a given distance, whether that is due to increased variable or fixed costs of trade.

Finally, it would be interesting to think about climate change along the route between origin and destination. That route, however, is (i) unobserved and (ii) even if I observed it, it would be endogenous to climate change. Suppose, for example that extreme weather made the Suez Canal

[2] What is a fixed or variable cost additionally depends on the time horizon under consideration. For each good in a single shipment, time spent loading, unloading or in port storage might be a fixed cost. Across all shipments in a year, however, total time spent waiting in port clearly scales with the number of shipments, and is thus variable. Across an entire decade, lots of trade costs related to physical processes, such as loading and unloading or customs inspections, might eventually become variable. Understanding regulations to enter a foreign market, on the other hand, probably remains fixed.

impassable. Shipping operators could then divert their route, for example, by going around the southern coast of Africa. Measuring climate impacts along that new route, however, misses the first order impact of the Suez Canal having become impassable. There are only two points along the route which are not endogenous — the origin and destination. There is no way to go from Italy to India, for example, that does not start in Italy and end in India. My specification therefore uses the only two non-endogenous points along the route to measure climate change. These are also where all loading and unloading cost is accrued, which form a considerable part of the overall cost of international trade.

2.2 Capturing climate change via decadal temperature changes

To assess the impact of climate change, I estimate this model across decades rather than using year-to-year weather variation. That is, I calculate decadal averages of all variables for each origin-destination pair and use those in the estimation. This allows me to capture longer-term, more permanent changes in countries' climates, rather than one-off shocks. Using decadal averages is also attractive if trade data are interpreted as a (noisy) measure of the true underlying trade network, since decadal averages are closer to the true underlying value than yearly data. The exact choice of aggregation period is of course somewhat arbitrary. Very short periods measure weather rather than climate, while very long periods lose a lot of potentially interesting variation in the data. As a robustness check, I also run all estimations using shorter and longer aggregation periods (five years and 30 years). I refer to these when presenting results in Section 3.

Using average temperatures does not mean that I try to capture the impact of temperature rather than precipitation, storms, droughts, heatwaves, floods or other extreme weather phenomena. Rather, I use decadal temperatures as an indicator of changes in countries' climate. These changes go along with increases in all kinds of extreme weather events, including heat waves and storms (J. Liu et al., 2026; Q. Liu et al., 2025; Qi et al., 2026; Rahmstorf & Coumou, 2011). A regression using average temperature loads all of these effects onto the coefficient on temperature, weighted by their covariance with temperature.[3]

Using average temperatures also has the advantage of being agnostic about which events are most important in driving climate change impacts. Further, they are potentially less noisy than extreme events, and data are available across more countries and further back in time. On the

[3] Specifically, using standard econometric arguments, the coefficient on temperature will converge in probability to $\delta_c \to_p \gamma_1 + \boldsymbol{\gamma}_2 {}^{\mathbb{C}\text{ov}(\Delta T_{ct}, \mathbf{x}_{ct})}/_{\mathbb{V}(\Delta T_{ct})}$, $c \in \{i, n\}$, where γ_1 is the direct effect of changes in average temperature and $\boldsymbol{\gamma}_2$ are the direct effects of all other weather phenomena $\mathbf{x}_{ct}$, e.g., storms, heat waves, droughts and floods.

other hand, focusing on specific extremes could lead to more accurately estimating their potentially larger impacts. This is why the small but growing literature on year-to-year weather impacts on ports, for example, has focused on extreme events (Astier, 2025; Izauirre et al., 2021; Massoni, 2025; Sytsma, 2020). Since mine is the first paper looking at climate impacts on seaports, I opt for the more parsimonious specification using average temperature, but with the understanding that this captures the effect of overall climate change, not just of temperature changes.

Averaging temperatures across a given country by area of course means potentially underweighting highly populated or active areas. On the other hand, it provides an exogenous weighting scheme that is less likely to be skewed towards finding an effect. (For example, climate-drive outmigration could mean the most affected areas get lower weights in population or GDP weights, thus missing some of the worst impacts of climate change.) On balance, I treat my main results as likely somewhat of a lower bound in the sense that I might be under-weighting populated areas and thus have a noisier signal of climate change at hubs of economic activity as well. When investigating mechanisms, I show that focusing on temperatures around ports, for example, leads to larger impact estimates when considering maritime trade specifically, but does not affect distance-based specifications much. (I also show in robustness checks that using population weighted distance does not make much of a difference for my main results.)

2.3 Identification

One potential identification concern could be whether I am capturing the correct functional form of trade costs, an important issue when trying to assess level differences in costs across countries (Anderson & van Wincoop, 2004; Bernhofen, El-Sahli, & Kneller, 2016; Hummels & Skiba, 2004; Tolva, 2026). I am interested, however, in identifying shocks to trade costs. Importantly, the counterfactuals I present in Section 5 also only depend on trade cost changes, not their levels, due to an insight by Dekle, Eaton, and Kortum (2008). Identifying trade cost shocks is possible even if the level is misspecified.

Specifically, I can obtain a consistent estimate of the coefficients on the interactions of interest as long as temperature changes are independent of the error term and distance. To see this, consider the correlation between the interacted variable $\tilde{d}_{ni}\Delta T_{ct}$, $c \in \{i, n\}$, and the error term ε_{nit} in the log version of (2). If this is zero, I can consistently estimate the interaction coefficient. With

independent temperature changes, it is

$$\mathbb{C}\text{ov}\left(\tilde{d}_{ni}\Delta T_{ct}, \varepsilon_{nit}\right) = \mathbb{E}\left(\tilde{d}_{ni}\Delta T_{ct}\varepsilon_{nit}\right) - \mathbb{E}\left(\tilde{d}_{ni}\Delta T_{ct}\right)\underbrace{\mathbb{E}\left(\varepsilon_{nit}\right)}_{=0} = \mathbb{E}\left(\Delta T_{ct}\mathbb{E}\left(\tilde{d}_{ni}\varepsilon_{nit} \mid \Delta T_{ct}\right)\right)$$
$$= \mathbb{E}\left(\Delta T_{ct}\mathbb{E}\left(\tilde{d}_{ni}\varepsilon_{nit}\right)\right) = \underbrace{\mathbb{E}\left(\Delta T_{ct}\right)}_{=0}\mathbb{E}\left(\tilde{d}_{ni}\varepsilon_{nit}\right) = 0$$

Here, I use that the error term is mean zero. (This is without loss of generality since the model contains an intercept — in fact, fixed effects for every origin- and destination-period. I simply save notation by not explicitly noting that this is all conditional on the model's fixed effects.) I also rely on the fact that the expectation of $\tilde{d}_{ni}\varepsilon_{nit}$, while not zero if the functional form is misspecified, is independent of the temperature changes. Finally, the mean temperature change is zero because I de-meaned the variable. (Again, this is without loss of generality as long as the model includes an intercept and the level effect of distance.)

This is true for misspecification due to any omitted variable related to distance. That includes nonlinearities in the distance effect, where the omitted variable might be $\tilde{d}_{ni}{}^2$, but it also captures other forms of heterogeneity in the distance effect, since any of these can be expressed as an omitted variable v_{nit} capturing heterogeneous effects of distance by country pair and period. Appendix C shows simulation results highlighting that while the estimated coefficient on distance will be biased if distance is endogenous, the estimated coefficients on the interactions of interest remain consistent in the face of heterogeneous effects of distance and endogeneity of distance.

Whether temperature changes are exogenous is obviously untestable, since I do not observe the errors. One testable implication I can explore, however, is that distance should not predict the interaction between distance and temperature changes, because with independent temperature changes, the correlation between the two is

$$\mathbb{C}\text{ov}\left(\tilde{d}_{ni}\Delta T_{ct}, \tilde{d}_{ni}\right) = \mathbb{E}\left(\tilde{d}_{ni}\Delta T_{ct}\tilde{d}_{ni}\right) - \mathbb{E}\left(\tilde{d}_{ni}\Delta T_{ct}\right)\mathbb{E}\left(\tilde{d}_{ni}\right)$$
$$= \mathbb{E}\left(\Delta T_{ct}\mathbb{E}\left(\tilde{d}_{ni}^2 \mid \Delta T_{ct}\right)\right) - \mathbb{E}\left(\Delta T_{ct}\mathbb{E}\left(\tilde{d}_{ni} \mid \Delta T_{ct}\right)\right)\mathbb{E}\left(\tilde{d}_{ni}\right) = \mathbb{E}\left(\Delta T_{ct}\right)\mathbb{V}\left(\tilde{d}_{ni}\right) = 0$$

A regression of the distance-temperature interaction on distance should therefore asymptotically yield a coefficient of zero.[4] This is an estimation I can run. Table 1 shows results for regressing

[4] I implicitly used this when showing consistency above: Without de-meaning the temperature changes, this correlation would not be zero even for independent shocks. One would then have to use the Frisch-Waugh-Lovell approach to partial out level distance first.

the interaction terms on log distance. The estimations also include the model's origin- and destination-period fixed effects. The coefficients are both small and not statistically significant. This is encouraging, but neither necessary nor sufficient for consistency. (The interaction could be uncorrelated with distance but still correlate with the error term, and it could be correlated with distance but not the error term.)

The core remaining identification concern is therefore that temperature changes might not be exogenous. This could be the case if countries which have different climatic environments, and hence see more rapid climate change, also have different trade cost trends for other reasons. These might be due correlations between income and geographic location, for example. (Hotter countries are also poorer, on average.) This would create a spurious correlation between decadal temperature changes and trade flows.

To address this concern, I capture countries' climatic environments in three ways — using their average temperature between 1950 and 1980, temperature change between the first decade of the 1900s and the 2000s decade, and latitude. I calculate deciles of each of these variables and interact them with distance, flexibly allowing for different levels of trade cost for countries with different climates. I also estimate specifications which add trade cost time trends for each decile. (Trends are potentially crucial to account for in the climate context since temperature changes tend to be positive over time, due to global warming.) The full specification with different trade cost levels and trends is

$$\begin{aligned}&\mathbb{E}\left(X_{nit}|\mathbf{D}_{nit}\right)\\&= \exp\left\{\gamma_{it} + \xi_{nt} + \alpha_t \tilde{d}_{ni} + \left(\sum_{D=1}^{10} \alpha_D \tilde{d}_{ni} + \tau_D t \tilde{d}_{ni}\right) + \delta_1 \tilde{d}_{ni}\Delta T_{it} + \delta_2 \tilde{d}_{ni}\Delta T_{nt} + \mathbf{C}'_{nit}\boldsymbol{\beta}_t\right\} \qquad (3)\end{aligned}$$

Here, α_D allows for separate coefficients on distance — separate trade costs — for each decile D of a given measure of climatic environment, e.g., 1950–1980 average temperature. τ_D allows for separate time trends in the coefficients on distance — separate time trends of trade costs — across deciles.

Note that the specification using 1900s to 2000s climate change as a measure of climatic environment is very conservative, perhaps even overly conservative. It assumes trade cost differences between countries with different long-run climate trends are not driven by climate change (even though they might be). It then uses only the remaining decadal variation among countries with similar long-term climate trends to estimate the impact of climate change. This inherently discards a lot of the identifying variation and limits the potential impact of climate change. Estimates from

this specification are therefore a lower bound on the impact of climate change on trade cost.

An alternative approach to controlling for the specific heterogeneity I know to be a potential confounder — climatic environment — would be to include fixed effects for every origin-destination pair and pair-specific time trends.[5] This controls for unstructured heterogeneity in countries' trade policy, for example. To estimate the causal effect of trade policy instruments, which are not plausibly exogenous, this can strengthen claims of causality (Yotov et al., 2016). The trade-off is that it has much less power to detect an effect in my setting, because I have relatively few decades to use per country pair. This makes results from the pairwise fixed effect specification potentially noisier and driven by smaller subsamples of the data.

In the climate context, variation is more plausibly exogenous. I am also able to control for the potential source of confounding, climatic environment, directly. This makes me prefer my main specification and robustness checks over the pairwise fixed effects setup. Nevertheless, as a very conservative robustness check, I also report results from a specification using pairwise fixed effects and pair-specific time trends. I defer this to the discussion of the mechanisms behind my main results in Section 4.2, however, because the patterns I find provide additional evidence for the mechanism I suggest is driving my main results.

Finally, one might worry about reverse causality: More trade leads to higher emissions, which lead to further global climate change. This is true but not an identification concern because my identification comes from local temperature changes at the origin and destination countries. These are not caused by local emissions directly. Rather, local emissions mix with emissions from other locations in the atmosphere and lead to global warming. Local warming, in turn, depends on climate patterns and geography, not on local emissions. This results in essentially every location across the globe seeing a disparity between the emissions it causes and the warming it experiences (Van Houtan, Tanaka, Gagné, & Becker, 2021).

2.4 Heterogeneity

I also look at heterogeneity of the impact of climate change on trade cost. As I explain below, this can help shed some light on patterns of adaptation to climate change. To do this, I modify equation (2), letting the coefficients on distance by decade and the core interaction terms vary by

[5] I thank a referee for suggesting this.

climatic environment,

$$\mathbb{E}\left(X_{nit}|\mathbf{D}_{nit}\right) = \exp\left\{\gamma_{it} + \xi_{nt} + \alpha_t \tilde{d}_{ni} + \left(\sum_{D=1}^{10} \alpha_D \tilde{d}_{ni} + \tau_D t \tilde{d}_{ni}\right) + \delta_1 \tilde{d}_{ni}\Delta T_{it} + \delta_2 \tilde{d}_{ni}\Delta T_{nt} + \delta_3 \tilde{d}_{ni}\Delta T_{it}\mathcal{T}_i + \delta_4 \tilde{d}_{ni}\Delta T_{nt}\mathcal{T}_n + \mathbf{C}'_{nit}\boldsymbol{\beta}_t\right\} \tag{4}$$

Here, I use similar measures of heterogeneity as I did to control for climatic environment above — 1950–1980 average temperature and 1900s–2000s climate change. I again allow for differential levels and trends in trade cost for each decile. To preserve some power for heterogeneity estimation, and for comparability with existing work on the heterogeneous impacts of weather shocks on productivity (e.g., Ramey et al., 2025), I then interact temperature changes with the level variable $\mathcal{T}_c$ used to create the bins, e.g., 1950–1980 average temperature. (I use de-meaned temperature changes for estimation throughout the paper, which centers interactions at the mean decadal temperature shift.)

3 Gravity estimation results

3.1 Impact of climate change on trade cost

This section presents my core reduced form results, based on the gravity estimation framework laid out in the previous section. I estimate all regressions via PPML, using the `R` package `fixest` to deal with the high dimensional fixed effects involved (Bergé, 2018). I capture d_{ni} using the great circle distance between the origin and destination countries in kilometers. TRADHIST also contains a population-weighted distance measure, but this is available only for a subset of countries and usually missing for historical countries. I therefore opt for unweighted distance, which is available for all countries, and show robustness to using the population-weighted distance measure.

As temperature measures, I use the decadal mean of the yearly average of daily average temperatures in °C. Decadal temperature changes are winsorized at the 1st and 99th percentiles to guard against outliers, though this does not impact my results. Figure 6 shows a histogram of the winsorized decadal temperature changes, highlighting the source of my identifying variation. The additional bilateral controls $\mathbf{C}_{nit}$ contain a common language indicator, sharing a border indicator and indicators for current and past colonial relationships. I take decadal means for all continuous variables and decadal medians for indicators.

Figure 7 shows a coefficient plot for the coefficients of interest — those on the interaction

between distance and temperature change — across a number of specifications. These include estimating (2), the climatic environment robustness checks following (3), some additional robustness checks. Table 2 shows the results in table form, including additional coefficient estimates (for example, on log distance) and a benchmark specification excluding temperature variables, with p-values in brackets. Standard errors are clustered by country pair, since that is the unit at which treatment $d_{ni}T_{ct}$, $c \in \{i, n\}$ is assigned (Abadie, Athey, Imbens, & Wooldridge, 2023). Note that I de-mean temperature changes and log distance, so interactions are centered at sample means. (For log distance, I de-mean by subtracting log mean distance.)[6]

The first pair of coefficients (*Basic*) is for the basic model (2). The following pairs are for the climatic environment decile regressions specified in (3). These allow the effect of distance — and hence trade cost levels and trends — to differ by countries' 1950–1980 average temperature decile (T_{50-80}, with *t.t.* indicating the specification with time trends), their 1900s to 2000s climate change (CC) and their latitude (Lat). I present some additional robustness checks as well. These use population-weighted great circle distance instead of the unweighted measure (d_{popw}), subset to the 1950s onwards ($t \geq$ '50s), drop China from the data set (*No CHN*), interact the additional bilateral controls with temperature changes ($C_{nit} \times \Delta T_{ct}$), or include additional bilateral cost measures, namely indicators for both countries being in the EU, OECD, or GATT (C^{+}_{nit}). I also calculate a rough proxy for domestic trade by subtracting total exports from GDP, as in Yotov (2012). I then allow for differential impacts of distance by period for both international and domestic trade.[7] This ensures I do not simply capture patterns of globalization, which could spuriously vary across climatic environments (*Dom.*). A final robustness check shown here uses residualized temperatures from an autoregression (i.e., the filter proposed by Hamilton, 2018), akin to Ramey et al. (2025).[8] This is a different approach to addressing potential spurious correlation between climate and trade

[6] Appendix Figures 33 and 34 present versions of this estimation that aggregate to half-decades (five-year period) and three-decades (30-year periods) instead of decades. This shows that results are robust to the exact choice of temporal aggregation. Impacts grow with period length, though not one-for-one. For the basic specification (2), the half-decade aggregation estimates roughly two-thirds smaller, the three-decade aggregation roughly 50 percent larger impacts. Estimates are noisier using the shorter aggregation period.

[7] That is, I add the term $\mathbb{1}[i = n]\alpha_t \tilde{d}_{ni}$ to (2). This estimation uses population-weighted distance to retain a well-defined notion of internal distance. I thank a referee for the suggestion to consider domestic trade and patterns of globalization.

[8] The specification for the Hamilton filter is

$$T_{it} = \left(\sum_{l=1}^{3} \beta_{1l} T_{it-l} + \beta_{2l} \bar{T}_i^{50\text{-}80} T_{it-l} \right) + \gamma_i + \nu_t + \varepsilon_{it}$$

using three lags of decadal average temperature interacted with countries' 1950–1980 average temperature, plus country and decade fixed effects. I then use residuals from this regression instead of ΔT_{it}.

cost trends by de-trending temperature.

A final robustness check tests whether, rather than capturing a trade cost shock, I might capture that certain sectors are more affected than others and also trade more over shorter or longer distances.[9] Since trade data by sector do not go back very far, I instead check whether effects are driven by countries that are currently heavily specialized in certain sectors. A worrying pattern would be if sectors which tend to be trade across long distance see larger effect. If, for example, countries specializing in manufacturing were driving the impacts, this could suggest I am simply capturing a larger vulnerability of manufacturing to climate change. To operationalize this, I use data on countries' current (2010s) sectoral composition, both the share of different sectors (the World Bank provides data on agriculture, industry, services and miscellaneous/other) as well as their log value added. I interact these with the temperature interactions of interest, with the miscellaneous/other sector left out. (Otherwise shares would sum to one, and due to country-period fixed effects, increasing value added in all sectors at once is not possible either.) I also allow the effect of distance to vary by sector shares and value added. I de-mean sector shares and value added. Interaction terms are thus centered at sample means, as with prior estimations.

Figure 8 shows the results. I see similar main effects to before, suggesting no specific sector is driving the results. Using sectoral shares, which directly capture specialization, I find that all sectors have a slightly lower impact (a small but positive coefficient) than the left-out miscellaneous category. However, the impact is small (noting that I use shares, not percentages). Using log value added, agriculture does not have a significantly different impact, and industry and services again have a slightly smaller impact. This suggests that there is no great heterogeneity between countries that specialize in sectors trading over longer distances (especially industry) and those trading over shorter distances (especially agriculture). That bolsters the case that I am picking up trade cost effects rather than sectoral heterogeneity. It is also consistent with research showing that, if anything, agriculture is more affected by weather shocks than industry (Nath, 2020).

To provide context for the magnitude of the coefficients in Figure 7, I can compare them to the impact of distance on trade. I consistently find a negative and significant effect of distance on trade flows, as expected from the gravity literature. My baseline specification yields that, at zero change in origin and destination temperatures, a one percent increase in distance in the 2010s decreases trade flows by 0.552 percent. This distance effect is roughly comparable to the estimates from Santos Silva and Tenreyro (2006), who find that a one percent increase in distance decreases

[9] I thank a referee for suggesting this.

trade flows by 0.784 percent. Figure 9 shows the coefficients on distance across decades, highlighting that there is a slight but imprecisely estimated decrease in trade cost over time. Figure 10 shows a similar figure for the benchmark model excluding temperature variables.

The novel empirical result in this paper is that temperature changes at both the origin and destination increase this negative effect of distance. That is, rising temperatures make it harder to cross a given distance — climate change hence increases trade cost. Across specifications, I consistently find negative and significant effects. Even using the most conservative specification, which allows for different trade cost levels and trends based on countries' long-term climate trends (and thus discards part of the variation of interest), I find that a one degree increase in temperature at the origin decreases trade flows by a further 0.071 percent. Similarly, a one degree increase in temperature at the destination decreases trade flows by an additional 0.048 percent. Overall, I thus find that climate change increases trade cost.[10]

To put these numbers into perspective, between the 1910s and the 2010s, for example, the average country saw a temperature increase of about 1.31°C. Comparing the estimated impact of temperature from the most conservative specification with the average impact of distance on trade flows, over the last 100 years, the average destination country saw the effect of distance on trade flows increase by about 11.5 percent ($\approx 1.31 \times 0.048/0.552$). The average destination country saw an increase of about 16.9 percent (calculated analogously).

Note that, because of the long time horizon of the data I use, these results incorporate adaptation to climate change. Since I actually observe climate change directly, rather than having to make inferences about the impact of climate change from a short period's worth of weather data, past adaptation will be incorporated into my coefficient estimates. This is similar to the long differences used in Burke and Emerick (2016).

3.2 Patterns of adaptation to climate change

I briefly discuss countries' ability to adapt to the impact of climate change on trade cost. To this end, I analyze how the effect of climate change on trade cost varies with climatic environment. As I

[10] Theoretically, it is possible that climate change could decrease trade cost across all distances but less so for longer distance trade. That would also generate a negative coefficient on longer-distance trade. I would then misinterpret this as an increase in trade cost, instead of a lower decrease. This is highly unlikely, however, due to the evidence on mechanisms I discuss below. I present evidence for several channels through which climate change could increase trade cost. There is no evidence I am aware of for channels that would systematically decrease trade cost. The one channel through which it might decrease trade cost for certain country pairs in the future is an ice-free Northwest Passage. Brancaccio, Kalouptsidi, and Papageorgiou (2020) explore this in a counterfactual. This would, if anything, tend to benefit longer-distance trade more, though. Furthermore, it has not happened yet and would only apply to certain routes. On balance, therefore, a negative impact is overwhelmingly more likely.

explained above, I do this by estimating equation (4), a version of my main estimating equation (2) which lets the impact of climate change on trade cost vary across climatic environments. Analogous to my core robustness checks above, I consider two dimensions of climate heterogeneity. First, I assess how the impact of climate change varies across countries' 1950–1980 average temperature. Second, I estimate how that impact varies across countries' average temperature change between the 1900s and 2000s decades.

Figure 11 shows the marginal effect of the temperature interactions across 1950–1980 average temperature. I see significant negative impacts in the lower three-fourths of the temperature distribution, while estimates for the upper fourth are positive but much noisier. This suggests that climate change especially affects the trade cost of countries adapted to colder climates. For these countries, a given shift in temperature is especially large compared to their baseline. Figure 12 shows the corresponding coefficient estimates across 1900s–2000s temperature change. One might expect that countries which see faster climate change have more of an incentive to adapt to it. If this incentive leads to actual adaptation, these countries would then see smaller impacts. I find, however, that the upper half of the distribution — countries which see relatively rapid climate change — see clear negative impacts. Impacts on the lower half are much noisier, though positive. It seems, therefore, that countries have a harder time dealing with rapid changes in their climatic environment. Together, these results suggest that adaptation to the impact of climate change on trade cost is relatively slow. Warmer countries see less of an impact from decade to decade climate change, which suggests that in the long run, countries were able to adapt to their climatic environment. Countries seem unable, however, to adapt to rapid climate change.

4 Mechanisms

My results of course raise the question: Why would climate change affect trade cost? The most obvious mechanism is that sending and receiving goods is an industrial task much like many others. It involves both manual and cognitive labor. It is well established that weather shocks and climate change affect the productivity of both of these kinds of labor and of industrial firms more generally (Adhvaryu, Kala, & Nyshadham, 2019; Carleton & Hsiang, 2016; Huppertz, 2025; Nath, 2020; Somanathan, Somanathan, Sudarshan, & Tewari, 2021; Zhang, Dêschenes, Meng, & Zhang, 2018). Through the same channels that climate change affects manufacturing firms, it can also affect the efficiency of dock and freight operations. Brancaccio et al. (2020), for example, while not focused

on climate, point out that transportation cost, and port efficiency in particular, is malleable.

As I alluded to in Section 2, the sign of the estimated coefficients on the interactions of interest provides suggestive evidence as to which kinds of trade are most affected. The obvious way to read these interactions is: Changes in temperature affect the impact of distance on trade (thus, they affect trade cost). Another, equally valid way to read these interactions, however, is that the impact of climate change is larger for longer-distance trade. (This again reflects the fact that I can only pick up differential impacts of climate change on different kinds of trade flows, as discussed above as well.) One salient difference between shorter- and longer-distance trade is the mode of transportation. Longer distance trade predominantly uses maritime shipping, rather than land freight. Maritime trade could therefore be at a higher risk from climate change because of its dependence on ports.

The basic argument is that most countries only have relatively few seaports and these are exposed to a range of weather hazards. Because climate change makes those hazards more frequent, climate change would also affect seaports. This would be especially detrimental to the cost of maritime trade. Land freight, by contrast, is relatively decentralized: Producers can load goods onto trucks at their production facility, and trucks can use the entire road network to reach their destination. This makes it relatively resilient against adverse weather events affecting specific roads or production locations — other roads or locations can function as substitutes. Because maritime trade does not have this degree of flexibility, climate change would have a larger impact on it.[11]

To support the argument that maritime trade is driving my results, I first discuss the impact of adverse weather events on ports, as reflected in recent research and discussed by practitioners and policymakers. If ports are affected by weather, this suggests a potential impact channel for climate. I then show patterns in my data suggesting that maritime trade in particular is driving my results. This is the strongest evidence I can bring to bear on the mechanism question. In particular, I show that maritime trade captures a clearer impact than pure distance. I also show that trade cost impacts become more pronounced among country pairs for which maritime trade is exogenously more attractive. Finally, I show that using port temperatures instead of country averages yields larger effects when focusing on maritime trade, but not when focusing merely on distance.

[11] Longer distance trade also uses air freight more commonly, and airports could be exposed to weather hazards similarly to seaports. I focus on maritime trade rather than air freight as the mechanism for two reasons. First, I use long-run historic data, and air freight used to be much less common than it is now (indeed, nonexistent for part of my sample). Second, maritime shipping is vastly more important for longer-distance trade even today, accounting for a much larger share of the weight and value transported across countries and across all distances. (See Appendix Figures 35–38.) I cannot rule out, however, that climate impacts on air freight also play a role. This is another potential mechanism, and should perhaps be explored further in the future.

4.1 Weather impacts on ports

Recent work shows that adverse weather events, such as storms, increase port congestion and decrease port efficiency (Astier, 2025; Izauirre et al., 2021; Massoni, 2025; Sytsma, 2020). Bai et al. (2024) show that this feeds through to global supply chain disruption with considerable impacts on economies across the world. As climate change makes these events more frequent, it can thus affect port efficiency in the longer term. I do not have port efficiency data going back long enough to estimate decade to decade impacts of climate change on ports. I therefore largely rely on this existing work to argue that ports are affected by weather. I do, however, replicate the impact of weather on port efficiency these papers find. Specifically, I use data on container port throughput from UNCTAD to estimate

$$\Delta \log (\text{TEU}_{it}) = \beta_1 \Delta T_{it} + \beta_2 \bar{T}_i^{50\text{-}80} \Delta T_{it} + \gamma_i$$

Note that t denotes years here, not decades. The outcome is country i's container port throughput in year t, expressed in twenty-foot equivalents (TEU), i.e., a measure of freight volume processed. I use year-to-year temperature changes ΔT_{it} as my measure of temperature shocks. The impact is allowed to vary by countries' 1950–1980 average temperature.

Figure 13 shows the marginal effect of temperature shocks across 1950–1980 average temperature, as well as a histogram of 1950–1980 average temperature. The impact is negative for most countries and significant especially for hotter countries. Here, of course, I cannot control for productivity changes using country fixed effects. Instead, Figure 14 shows a version of this model controlling for changes in GDP per capita. This is an endogenous regressor, so results are harder to interpret. Even in this specification, however, I find a negative effect for most countries, though with a significant positive impact on the coldest countries.

The larger impact on hotter countries might seem in contrast with the heterogeneous impacts of climate change I discuss above, which seem to affect colder countries more. This could simply be due to the fact that year-to-year weather variation and its impacts are different from a shift in average climatic conditions, though. A one-time shock may have worse effects in a hotter country (which also tend to be poorer), but the disruption from climatic change may still be worse for countries with a colder baseline climate. In any case, overall, these results corroborate recent work showing that severe weather impacts ports. This weather impact provides an impact channel for climate change on ports.

As an additional piece of evidence, note that policymakers are concerned about climate change impacts on ports. UNCTAD notes that seaports are especially affected by rising sea levels and the associated increased risk of storm surges (Asariotis, 2021). The Environmental Defense Fund finds that Hurricane Katrina caused USD 2.2 billion in damages to US port infrastructure, and that climate change increases the frequency and severity of such storms. Furthermore, they find that inland flooding or droughts disrupt the connections between domestic producers, consumers and international ports, making ports less useful as connections to the rest of the world. Additionally, they discuss that heat waves have led to multi-day port shutdowns, for example, in Melbourne, Australia in 2009 (Van Houtven, Gallaher, Woollacott, & Decker, 2022).

Finally, shipping companies and port operators themselves are wary of climate change, and engaging in costly actions to deal with it. Two examples are, first, Maersk, one of the largest international freight operators. They recently engaged the Zurich Insurance Group (specifically its risk management consulting arm) to help plan how to climate-proof ports Maersk operates (McAllister, 2024). " 'In the past decade, we have seen coastal flooding at our terminal in Port Elizabeth, New Jersey; flooding at our Salalah terminal in Oman; a cyclone hit our Pipavav terminal in India; and regular exposure to tropical windstorms to our terminals in Miami, Florida, and Mobile, Alabama,' says Lars Henneberg, VP, Head of Risk Management at Maersk." Second, the Port of Long Beach enacted a Climate Adaptation and Resiliency Plan as far back as 2016. This plan again highlights the risks posed by storm surges, sea level rise, flooding, and heat waves (Port of Long Beach, 2016).

4.2 Maritime trade drives trade cost impacts

I now discuss a number of results which are consistent with the idea that climate change especially affects the cost of maritime trade, which then drives the overall results. Here, I depart from the distance specification and instead interact temperature changes with bilateral variables that capture the likelihood of a given country pair using maritime trade. Of course, I do not observe transport modes and they are endogenous to climate. I can, however, pick some exogenous factors that would make it more likely that a given country pair uses maritime trade.

First, Table 3 shows results for gravity estimations similar to my main specification (2), but using an indicator which is equal to one if the only connection between the origin and destination is via the ocean. Almost all trade between these country pairs thus has to use maritime shipping. As might be expected, since this sea-only trade indicator is correlated with distance, I find similar

patterns to my main results — climate change at the origin, especially, significantly increases the cost of trade for country pairs which are separated by an ocean. The second column of the table instead uses a long distance indicator. This is equal to one if the distance between the origin and destination country is equal to or larger than the median distance between country pairs that have only an ocean-based connection. The pattern I find is similar, but note that the coefficient on the origin interaction is less precisely estimated. In the third column, I horse race the two, including both sets of interactions. Now, the sea only indicator comes out as clearly significant, while the long distance variable is no longer significant. This provides a first indication that maritime trade, rather than generic long distance trade, is driving the results.

If impacts on seaports are driving my results, one might expect that focusing on climate change at ports would yield even larger effect estimates. At the same time, seaports also rely on inland waterways for incoming and outgoing shipping, which is why droughts can be an issue for international shipping. Focusing on too small an area around the port could miss those impacts. I investigate this using current port locations and calculating average temperatures in a circular buffer around those ports.[12] I then calculate, for each country, the average decadal changes across ports, unweighted and using current (2019) port-level total outflows from the UNCTAD port locations data as weights. Where weights are missing, I fill in using the unweighted average. For countries without ports, I use country-level averages. For countries that no longer exist, I match them to ports within their borders. Since country shapes and port locations come from different data sources, they can be slightly offset from each other. I therefore calculate the 90th percentile of distance between countries and their ports in the data ($\approx$1 km) and match no longer existing countries to all ports within that distance. Some current ports may of course not have existed in the past. For those, I am then just measuring coastal temperatures at locations that are suitable for ports, which still focuses more on maritime trade than country average temperatures.

I estimate my main specification (2) using average temperatures at ports for different buffer sizes, using both log distance and the sea only indicator to measure the impact of trade costs. Figure 16 shows the results. The top row of panels is for distance, while the bottom row uses the sea only indicator. The left column of panels uses outflow weights to average port temperatures within country, while the right column just averages. Each coefficient is for a different buffer size as indicated on the horizontal axis. I also include the results using country average temperatures for comparison. For distance, using port temperature makes no difference. Estimated coefficients

[12] I thank a referee for suggesting to look at port temperatures.

are very similar and there is no discernible pattern to the results. For the sea only indicator, port temperatures find effects that are larger in magnitude and more significant. They become more so as I increase the buffer size, though they stabilize at around 50 km to 100 km buffers. (This could be due to issues like droughts and other inland disasters being missed by a small area around the port. Note also that port buffers include changes in ocean temperatures, which country average temperatures miss. These could be especially important to capture changes in storm patterns, for example.) The fact that port temperatures lead to larger impact estimates when focusing specifically on sea-only connections again strongly suggests that maritime trade, operating through seaports, is driving the effects I find.

Focusing on country pairs separated by an ocean is of course a very conservative way to assess the impact of maritime trade — plenty of country pairs have a land route between them, but due to the length of that route maritime trade is preferable. This specification treats such routes the same as shorter, and hence more attractive, land routes. As a less conservative approach to isolating the impact on maritime trade, I conduct two sets of 'donut hole' estimations. Here, I use the fact that maritime trade becomes more attractive not only with distance, but especially with the number of land borders which would need to be crossed. This is because border crossings incur fixed costs (Anderson & van Wincoop, 2003). I calculate, for each origin-destination pair, the number of land borders which would have to be crossed to get from one to the other. (For country pairs with no land based connection, the number is infinite, Land borders$_{ni} = \infty$.)

I then estimate regressions similar to my main specification (2), but using an indicator which is zero if the origin and destination are neighbors (Land borders$_{ni} = 1$) and one if the number of land borders between the origin and destination is at least a minimum gap size b (Land borders$_{ni} \geq b$), for $b = 2, 3, \ldots, 10$. Country pairs which are not neighbors and have fewer than b land borders in between them (i.e., Land borders$_{ni} \in (1, b)$) are discarded. (These countries are in the donut hole.)

I thus compare country pairs with progressively costlier land trade routes, where maritime trade is therefore more attractive, to neighboring country pairs. If it is true that climate change especially affects maritime trade, I would expect to see negative coefficients on the interaction between the minimum gap size indicator and origin and destination temperature change once country-pairs are separated by a few borders. I would further expect these coefficients to remain roughly stable as I increase the minimum gap size b beyond that threshold. I estimate this using both country average temperatures and port temperatures. For ports, I use the 100 km buffer where effects seemed to stabilize in Figure 16, averaging without outflow weights. If port temperatures again pick up a

stronger impact, this would also suggest that maritime trade is driving the results.

Figure 17 shows the estimated coefficients on the interaction between minimum gap size for each estimation. The left panel uses country averages while the right uses port temperatures. At only two land borders I do not yet see a difference, but coefficients become progressively more negative and are significant once four land borders would need to be crossed. Note that, though confidence intervals become slightly wider (because larger donut holes mean fewer total observations), coefficients remain fairly stable across larger minimum gap sizes. Coefficients are larger in magnitude using port temperatures, especially for destination impacts. I thus see a significant increase in trade cost for country pairs where maritime trade is more attractive. Consistent with the idea that maritime trade is preferable once a threshold number of borders have to be crossed, this difference becomes more pronounced up to a point, but then stays relatively stable. Additionally, port temperatures find a somewhat stronger impact. This suggests I am not picking up some kind of correlate of distance, but rather a differential impact on maritime as opposed to land based trade.

Note also that interacting temperature change with an indicator for international compared to domestic trade instead might look a lot like the first sets of coefficients in the figure, comparing neighboring countries with those separated by only two land borders. I find no significant impact there. This highlights that approaches which work well for estimating the impacts of non-discriminatory trade policy may not work well in the climate context. Here, the impact is driven by physical processes and their impacts on workers and infrastructure, rather than by national policy which begins and ends at a country's borders.

To again highlight that this pattern is not simply because larger gap sizes mean larger distances, Figure 18 shows results for a similar estimation, but using indicators for the distance between origin and destination being equal to or greater than the median distance $\bar{d}_b$ for a given land border bin $b = 2, 3, \ldots, 8$. Here, I first calculate median distance among neighboring countries. I then calculate the median distance between country pairs which are at least $b - 1$ land borders apart. I compare those further apart country pairs to the close country pairs, discarding countries in between the two distances. For example, at $b = 2$, I compare countries which are further apart than the median distance among neighboring countries to those below that distance. I stop at bin eight since sample sizes become smaller and convergence becomes an issue. Here, coefficients again show only a much noisier version of the maritime trade analysis. I find significant and negative effects for some, but not all comparisons, with insignificant effects for larger distances. Port temperatures also show a less clear pattern, actually finding smaller effects for origin countries. This suggests that maritime

trade picks up a pattern in the data that, while correlated with distance, more clearly focuses on the underlying mechanism.

As discussed in Section 2.3, another way to see that maritime connections may pick up additional information compared to raw distance also provides a final conservative robustness check. The main identification threat I face comes from a specific source of heterogeneity — countries with different climatic environments might have different trade cost levels and trends for spurious reasons, e.g., due to their geographic location. Flexibly controlling for this does not change my results much. I could also control for country-pair fixed effects and pair-specific time trends in (2). This controls for unknown patterns of heterogeneity (and absorbs the level effect of distance), and can be especially useful in the trade policy context, where exogeneity is harder to argue for. Note that this approach severely reduces power — it leaves only very little variation to identify a climate impact, since I see only a relatively limited number of decades per country pair. Because I can instead control flexibly for the source of confounding variation and preserve power, I prefer my main approach above.

Nevertheless, as a final robustness check, I estimate the model with pairwise fixed effects, using both log distance and the sea-only trade indicator. This model does not converge for my full sample, so I subset to the 1950s and onwards. This model converges, but relies on an even smaller number of decades per country pair. Figure 15 presents the results, together with the results of estimating (2) for each model for comparison. Notably, the interactions of interest in the distance specification become small and insignificant. For the sea-only indicator, only the origin interaction is significant in the main specification to begin with and it remains significant with pairwise fixed effects (though it becomes somewhat smaller in magnitude).

This suggests that the sea-only indicator focuses more strongly on the country pairs which see an impact from climate change, compared to using raw distance. That is why it remains significant even in the very demanding pairwise fixed effects (and trends) specification, unlike the distance interaction. This highlights the value of moving from a parsimonious trade cost measure like distance, which still fits the climate context, to a measure focused on one specific mechanism, in this case, sea-only connections. I would caution against overinterpreting this result, though, since it relies on relatively small subsamples of the overall data and because I have to drop a number of decades to even get the model to converge.

To summarize, impacts on ports, and through these on maritime trade, are a plausible mechanism behind my main results. This is consistent with (i) the coefficient estimates I find in my main results (maritime trade is also longer-distance trade), (ii) recent research on the impact of weather

on ports, policymakers' concerns regarding the vulnerability of seaports to climate change and port operators' costly adaptation efforts, and (iii) the additional reduced form analyses showing that where maritime shipping is more attractive (or even the only option) I clearly see a differential impact on trade cost and using port temperatures reinforces this pattern.

5 Welfare impacts

I now use a workhorse model of international trade (Eaton & Kortum, 2002) to assess the welfare implications of my reduced form results. I begin by using my gravity resultsto estimate the change in ϕ_{nit} we would observe if we moved to the climate of a different decade $s \neq t$. I can do this by plugging temperature changes between decade s and t into the specification for the bilateral resistance term (1) to obtain a counterfactual ϕ'_{nit}, using primes to denote counterfactual quantities. As a simple, illustrative example, consider the basic distance-based specification (2). Using hats to denote the ratio of counterfactual to observed quantities, the change in bilateral resistance would be

$$\hat{\phi}_{nit} \equiv \frac{\phi'_{nit}}{\phi_{nit}} \overset{(1)}{=} d_{ni}^{\delta_1(T_{is}-T_{it})+\delta_2(T_{ns}-T_{nt})} \tag{5}$$

Importantly, all non-temperature covariates remain constant — I simply estimate the change in bilateral resistance stemming from the changed temperature variables.

To estimate the change in bilateral resistance terms, I use a specification that is consistent with maritime trade being the main driver of the impacts I observe. Specifically, I estimate the impact of climate change on trade for countries with at least four land borders between them, compared to trade between direct neighbors. This is similar to the coefficient in Figure 17 for $b = 4$. I now also allow for the impact of trade cost to vary by origin and destination 1950–1980 average temperature, however, as I did in (4). This allows countries' climatic environment to drive their reactions to climate change. Using a measure of average temperature to do this is common in the literature on climate impacts on productivity (e.g., Burke et al., 2015, 2026; Ramey et al., 2025). I estimate

$$\begin{aligned}\mathbb{E}\left(X_{nit}|\mathbf{D}_{nit}\right) = \exp\Bigg\{&\gamma_{it} + \xi_{nt} + \alpha_t b_{ni}^4 + \left(\sum_{D=1}^{10} \alpha_D b_{ni}^4 + \tau_D t b_{ni}^4\right) + \delta_1 b_{ni}^4 \Delta T_{it} + \delta_2 b_{ni}^4 \Delta T_{nt} \\ &+ \delta_3 b_{ni}^4 \Delta T_{it} \bar{T}_i^{50\text{-}80} + \delta_4 b_{ni}^4 \Delta T_{nt} \bar{T}_n^{50\text{-}80} + \mathbf{C}'_{nit}\boldsymbol{\beta}_t\Bigg\}\end{aligned} \tag{6}$$

Here, b_{ni}^4 is an indicator for the minimum number of land borders between n and i being four or

greater. I estimate this using only direct neighbors and countries with at least four land borders between them, like for the donut hole specification above. This creates a clear comparison between likely maritime and likely non-maritime routes. Figure 19 shows the marginal effect of origin and destination temperature across 1950–1980 average temperature. I find almost exclusively negative impacts, with larger impacts on colder countries but not necessarily significant differences across most of the distribution. I can back out the change in bilateral resistance from these results, similar to (5). For neighboring countries and those with two or three land borders between them, I assume there is no impact of climate change on trade cost, consistent with the idea that maritime trade is the driving mechanism.

To go from the changes in bilateral resistance to an implied welfare impact, I need to specify a model of international trade. This is necessary because I have to discipline how wages and prices adjust in this counterfactual. I use the well-established model of Eaton and Kortum (2002) to estimate the 2010s welfare change when I undo the impact of climate change since previous decades. Under the Eaton and Kortum (2002) model, the bilateral resistance term is equal to $\phi_{nit} = \tau_{nit}^{-\theta}$ where τ_{nit} is a measure of how difficult it is to ship goods from i to n and $\theta > 0$ measures productivity dispersion in the Fréchet distribution of technology underlying the model. To estimate welfare impacts, I rewrite the model in changes (Dekle et al., 2008). The core object I need to estimate welfare impacts are changes in trade shares $\pi_{nit} \equiv X_{nit}/X_{nt}$, where $X_{nt} \equiv \sum_{i=1}^{N} X_{nit}$ is the destination country's total expenditure for decade t. Changes in trade shares $\hat{\pi}_{nit}$ resulting from changes in trade cost $\hat{\tau}_{nit}$ are

$$\hat{\pi}_{nit} = \frac{\hat{A}_{it}(\hat{\tau}_{nit}\hat{w}_{it})^{-\theta}}{\sum_{k=1}^{N} \pi_{nkt}\hat{A}_{kt}(\hat{\tau}_{nkt}\hat{w}_{kt})^{-\theta}} \tag{7}$$

Here, $\hat{A}_{it}$ is the change in country i's productivity for period t (also from the Fréchet distribution underlying technology) and $\hat{w}_{it}$ is the change in country i's wage for period t. The resulting welfare change, letting W_{it} denote country i's welfare in decade t, is

$$\hat{W}_{it} \equiv \frac{W'_{it}}{W_{it}} = \hat{A}_{it}^{\frac{1}{\theta}}\hat{\pi}_{iit}^{-\frac{1}{\theta}} \tag{8}$$

Though I use a specific structural model, as Arkolakis et al. (2012) show, a broad class of international trade models yield similar specifications for the welfare impact. Note that I do not incorporate feedback loops between economic output and emissions, as an integrated assessment model would (for trade-focused examples, see Cruz & Rossi-Hansberg, 2024; Shapiro, 2016). I also set aside

health benefits of reduced warming, e.g., through lower exposure to heat and particulate matter pollution from wildfires. These effects are important to take into account when quantifying the social cost of carbon, for example. Taking them into account would certainly lead to larger welfare effects than I estimate here.[13] Setting these additional welfare effects aside, however, still allows me to assess the welfare impact of climate change purely through the trade cost channel, and to compare this to the pure productivity impacts. If ignoring trade cost leads to an underestimate in that comparison, it almost surely leads to an underestimate in a fuller accounting of the complete welfare effects as well. Because my methodology is simple to plug into any other structural gravity framework, it could be embedded in a fuller accounting of the social cost of carbon in the future.

5.1 Welfare impacts of climate change through the trade cost channel

I first assess the impact of climate change through trade cost only, keeping technology unchanged ($\hat{A}_{it} = 1$). For this case, the welfare change simply becomes the change in own trade share raised to a negative power — if own trade share decreases, welfare increases. It is straightforward to back out $\hat{\tau}_{nit}$ from the estimates of $\hat{\phi}_{nit}$. (Note that, as I mentioned in Section 2.3, I only require estimates of the change in trade costs, not their levels.) I can then solve the system of equations (7) for wage changes $\hat{w}_{it}$ that equate counterfactual trade deficits and surpluses with those observed in the data (ensuring goods market clearing in the counterfactual). The resulting counterfactual changes in trade shares $\hat{\pi}_{nit}$ enable me to calculate welfare changes for each country from (8). I set the only unknown parameter $\theta = 4$, which is in line with estimates in Simonovska and Waugh (2014) and in between the 3.6 found in Bernard, Eaton, Jensen, and Kortum (2003), which also coincides with the lower estimate in Eaton and Kortum (2002), and the 4.87 found in Eaton, Kortum, and Kramarz (2011). I discuss sensitivity to this key parameter choice below. Finally, solving the model requires choosing a normalization. I fix world GDP at its 2010s value.

I use the 2010s as my reference period. Because this estimation requires domestic trade shares, which the TRADHIST database lacks, I use the ITPD data on trade shares for the 2010s to measure π_{nit}. I then calculate welfare changes resulting from a shift to each previous decade's climate. I do this for all previous decades from the 1880s onwards, since I have global weather coverage beginning at that time. Figure 20 shows the population-weighted mean welfare change across decades, as well

[13] Shapiro (2016) finds the direct welfare impacts of trade are considerably larger than its cost in terms of CO_2 emissions. In that case, I would still be capturing the majority of the welfare impact. This crucially depends on the social cost of carbon, however, and that assessment is based on what is now a relatively low estimate, \$29 for the marginal ton compared to recent estimates of around (or above) \$1,000 (e.g., Bilal & Känzig, 2026; Burke et al., 2026).

as the 5th and 95th percentile of welfare changes.[14]

Looking at the results for the 1910s, I estimate that the average country would see an 1.7 percent increase in welfare if we reverted trade cost increases due to climate change over the last 100 years. This is a relatively sizable effect, comparable to estimates of the impact of climate change based on plot-level or firm-level data (Costinot et al., 2016; Nath, 2020) but compatible with direct country-level estimates of the productivity impact (Bilal & Känzig, 2026; Burke et al., 2015; Ramey et al., 2025). I discuss this comparison to productivity impacts in more detail below.

The welfare impact of trade cost changes tends to be larger when switching to earlier climates. This is because temperatures increase over time and reverting to an earlier period's climate thus results in a larger temperature change. For example, the mean welfare increase for the earliest decade, the 1880s, is estimated to be 2.0 percent, whereas for the 1950s I estimate an average welfare increase of 1.2 percent. For the most recent decade, the 2000s, I estimate an 0.1 percent welfare increase, on average. Across decades, almost all countries see an increase in welfare — the 5th percentile of welfare changes is consistently positive. At the 95th percentile, welfare impacts are as high as 4.9 percent in the 1880s counterfactual.

The magnitude of these effects of course depends on the choice of dispersion parameter θ. A lower θ increases dispersion of productivity, leading to stronger comparative advantage and larger gains from trade. This creates larger welfare impacts. As discussed above, I use $\theta = 4$, which is in line with a number of previous findings (e.g., Bernard et al., 2003; Eaton et al., 2011; Simonovska & Waugh, 2014). It is smaller than the central calibration of 8.28 in Eaton and Kortum (2002), also used in Dekle et al. (2008), however. As a robustness check, I therefore also estimate counterfactuals using $\theta = 8.28$. Appendix Figure 40 shows the results. The pattern of welfare gains over time remains similar. Levels change, though: As expected, doubling θ roughly halves welfare gains.

5.2 Drivers and distribution of welfare gains

Figure 21 shows a map of welfare gains across countries for the 1910s counterfactual. There is considerable heterogeneity across the globe, with larger gains in countries as diverse as, for example, South Africa, Angola, Tunisia, Chile, Argentina, Uruguay, Panama, Canada, Ireland, Iceland and Australia. I highlight two key determinants of the welfare gains.

The most obvious factor are climate trends: I would expect that countries which see more climate

[14] Appendix Table 5 shows the same information in table form. Appendix Figure 39 and Appendix Table 6 show versions of these results without population weights. As I discuss below, larger economies benefit less from trade cost reductions, so the unweighted average welfare changes and percentiles are somewhat higher.

change or whose neighbors see more climate change would benefit more from undoing those changes. Figure 22 shows welfare changes in the 1910s counterfactual across countries' own temperature change between period the 1910s and the 2010s. Figure 23 shows welfare changes across the inverse distance weighted change in other countries' change in temperatures, which is calculated as

$$\text{Inverse distance weighted change}_{it} \equiv \frac{1}{\sum_{n \neq i} d_{ni}} \sum_{n \neq i} d_{ni} \Delta T_{nt}$$

where ΔT_{nt} is country n's change in temperature between period t and the 2010s. This measures climate change in the rest of the world, but weighted towards countries' neighbors. Interestingly, both measures of climate trends are somewhat negatively correlated with welfare gains. These temperature measures are, of course, correlated. Figure 24 highlights this, showing inverse distance weighted temperature changes across changes in countries' own temperatures between the 1910s and 2010s. This could mask how own and others' climate trends affect welfare gains.

To disentangle their effects, Table 4 shows results for regressions of welfare impacts $\hat{W}_{it}$ across periods on country characteristics. These regressions include period fixed effects to analyze correlates of welfare change within period, and country fixed effects where possible to isolate effects of temperature changes from those of heterogeneity due to baseline temperature, for example. Standard errors are clustered at the country level, but note that these welfare outcomes are generated regressors, which I ignore here. I treat these results as interesting suggestive evidence for directions of change, not as a true test for significantly different welfare effects. The first column highlights that, with country fixed effects in place, countries' own temperature change is positively, but weakly correlated with welfare gains. The second column shows that inverse distance weighted temperature change is weakly negatively correlated with welfare gains. Column three combines the two and shows that once I take both changes into account, countries' own temperature changes are strongly positively correlated with welfare changes. Surrounding countries' temperature changes, however, are strongly negatively correlated with welfare gains.

The negative impact of neighbors' temperatures may seem counterintuitive, but there is a simple explanation. When country i and its neighbor j both see large temperature changes, they both see rising trade cost and become less attractive trade hubs. Reversing that change benefits both. When only j sees climate change, both countries still see an *absolute* increase in trade cost. Country i, however, sees a reduction in *relative* trade cost — i's cost of exporting and importing falls relative to that of j. This relative cost reduction benefits i. Reversing climate change lowers absolute trade

cost for both countries, but increases i's relative cost. That makes reversing climate change less beneficial for i. The key takeaway is the importance of considering neighbors' climate trends, not just countries' own climate trends, to assess potential welfare impacts of climate change.

A second key factor is the size of the economy. Figure 25 shows the estimated welfare impacts of returning to the climate of the 1910s across countries' 2010s log GDP. Larger economies tend to benefit slightly less from reversing the impact of climate change on trade cost. As Figure 26 shows, however, welfare gains are, if anything, positively correlated with GDP per capita. That larger (i.e., higher GDP) economies benefit less is not surprising: They are less reliant on international trade, since they have larger domestic markets. (See Figure 27, which shows that larger economies tend to have higher own trade shares.)

To show that this is driving the difference in welfare gains by GDP, the fourth column of Table 4 shows a regression of welfare gains on log 2010s GDP. (Since this does not vary within country, I have to drop country fixed effects here.) Across periods, GDP and welfare gains are negatively correlated. The fifth column of Table 4 additionally controls for countries' 2010s own trade share. Once I take that into account, it becomes clear that larger economies actually benefit more from trade cost decreases. Not surprisingly, though, a larger own trade share is correlated with lower welfare gains, because international trade is less important for the economy. The final column of the table controls for countries' 1950–1980 average temperature, which does not change results much. All told, larger economies benefit less from undoing climate change impacts on trade cost because they are less reliant on international trade.

5.3 Comparing trade cost and productivity channels

As mentioned above, the welfare gains I find are sizable — the average 1910s welfare gain of 1.7 percent is comparable in magnitude to estimates of the welfare effects of climate change through agricultural and overall productivity (Costinot et al., 2016; Nath, 2020). It is only about ten percent, however, of the direct estimates of the productivity impact of climate change, depending on the exact specification (e.g., Burke et al., 2015, 2026; Ramey et al., 2025), and about five percent of the global welfare loss from one degree of global warming found in Bilal and Känzig (2026).

To compare welfare impacts through the productivity and trade cost channels, I estimate the impact of climate change on productivity with a setup similar to Ramey et al. (2025). I use decade to decade variation, however, to line up with the trade cost estimation and my focus on climate

change, using GDP data from TRADHIST to be consistent with those results. I estimate

$$\Delta \log (y_{it}) = \beta_1 \Delta T_{it} + \beta_2 \bar{T}_i^{50\text{-}80} \Delta T_{it} + \gamma_i + \varepsilon_{it} \tag{9}$$

where y_{it} is country i's average GDP per capita during decade t, as a measure of productivity. I allow for the effect of temperature to vary with 1950–1980 average temperature. I do not include lagged impacts since existing estimates of the impact of weather shocks on GDP have not gone beyond a ten-year horizon, and I am working on a decadal scale. Figure 28 shows the marginal effect of temperature shocks across 1950–1980 average temperature. I find consistently negative impacts of decadal temperature changes on productivity, with more negative impacts in colder countries (but not generally significant differences across baseline temperature). I then back out implied technology changes $\hat{A}_{it}$, using that in the model, GDP per capita is

$$y_{it} \equiv \frac{w_{it}}{\mathcal{P}_{it}} = \Gamma \left(\frac{\theta - \sigma + 1}{\theta} \right)^{\frac{1}{\sigma - 1}} \left(\frac{A_{it}}{\pi_{iit}} \right)^{\frac{1}{\theta}}$$

where $\mathcal{P}_{it}$ is the optimal CES price index and Γ indicates the Gamma function (Eaton & Kortum, 2002). In general, this depends on all countries' temperature shocks through the trade network. This is reflected in the denominator of (7), showing that changes in trade shares depend on all countries' productivity and trade cost changes. In autarky, however, which I can approximate by taking trade cost to infinity, I have $d_{nit} \to +\infty \, \forall \, i, n \neq i$ and $\pi_{iit} \to 1$. Then, $\hat{y}_{it} = \hat{A}_{it}^{\frac{1}{\theta}}$. The reduced form (9) estimates $\log (\hat{y}_{it})$, and I can back out $\hat{A}_{it} = \exp \{\theta \log (\hat{y}_{it})\}$.

The insight that trade between countries creates a problem with this estimation is due to, and brought out in much more generality in, Astier, Barrows, Calel, and Ollivier (2025). They show that productivity regressions of this form violate the stable unit treatment value assumption (SUTVA), rendering them generally invalid. They also find, however, that these regressions do better for past climate change than for projecting out into the future. I rely on this finding, and the fact that these estimations are widespread in the literature, to use (9) as my productivity benchmark. Very importantly, note that my trade cost regressions control for shocks at all origins and destinations using fixed effects and thus control for changes via the trade network. They therefore do not suffer from this SUTVA violation.

To compare trade cost and productivity impacts, I estimate three different counterfactuals, undoing the impact of climate change (i) *only* on trade cost, as above, (ii) *only* on productivity, and (iii) on *both* trade cost and productivity. This allows me to compare the welfare implications of both

channels and assess whether we meaningfully underestimate the welfare impact of climate change when we focus exclusively on productivity. Figure 29 shows average welfare gains across these three counterfactuals for the 1910s climate. I break the welfare gains up by small (below median 2010s GDP) and large economies, since smaller countries are more affected by the trade cost effect, as discussed above. Gains from increased productivity alone are considerably larger than gains from trade cost alone, as would be expected from most trade models. (Larger countries benefit more in the productivity counterfactual solely because of the estimated larger productivity impact on colder countries seen in 28. Because that differential effect is very noisily estimated, I would not overinterpret the difference here.) Importantly, welfare gains from the combined counterfactual are somewhat larger than those from the productivity-only counterfactual. This is especially true for smaller economies, which see a larger additional welfare gain from the combined counterfactual. Overall, this shows that focusing on productivity alone means underestimating the welfare impacts of climate change to some degree.

To quantify the underestimate, Figure 30 shows a histogram of the welfare gain ratio between the combined and productivity only counterfactuals. That is, this shows the welfare gain from undoing both trade cost and productivity impacts divided by the welfare gain from only undoing productivity impacts. The average country (population-weighted) has about a nine percent larger welfare gain from also undoing trade cost changes, but the ratio can reach 20 percent or more for some economies. Figure 31 shows a histogram of the difference between the two welfare changes, rather than the ratio. The average welfare gain difference is about two percentage points, which is slightly larger than the average welfare gain through trade cost alone. The two channels are thus somewhat complementary.[15]

As discussed above, the welfare impact of climate change varies depending on countries' trade openness as well as their exposure to climate change. Figure 32 shows a world map of the welfare gain ratio, highlighting the global distribution of additional welfare gains. Smaller economies see larger additional gains, but openness to trade and location matter. The UK and Ireland, for example, both of which are relatively open to international trade (and especially reliant on maritime trade), stand out compared to continental Europe. Australia also sees larger additional gains. Of course, the relative welfare impacts of trade cost and productivity are also driven by the dispersion parameter

[15] Of course, reducing trade cost and increasing productivity at the same time would always lead to an impact larger than the sum of its parts. Due to the heterogeneity by 1950–1980 average temperature, however, it would have been possible for the estimates to produce any pattern of additional gains, including a negative correlation between trade cost and productivity changes.

θ. Appendix Figures 41–42 show average welfare gains and the distribution of welfare gain ratios using the more conservative $\theta = 8.28$ calibration. With this alternative parameter value, I still find a five percent underestimate of the impact of climate change, again with larger differences of up to 15 or even 25 percent for a number of countries.

This simple counterfactual exercise suggests that, while productivity is the main driver of climate impacts, ignoring the trade cost channel I study here leads to an appreciable underestimate of the overall effect. This underestimate is larger, and thus more important to take into account, for smaller economies, including rich countries which are especially open to trade or especially reliant on maritime trade. When studying the impacts of climate change on those countries specifically, one should consider taking trade cost impacts into account.

6 Conclusion

I show, using an augmented gravity specification, that decade-to-decade temperature change at the origin or destination country increase bilateral trade cost. Adaptation to these impacts seems slow. The effect appears to be driven by impacts on maritime trade, potentially because seaports are especially vulnerable to extreme weather events and hence to damages from climate change.

The welfare impacts of this are non-negligible: Using the Eaton and Kortum (2002) model, I find that average welfare during the 2010s would have been about 1.7 percent higher if climate change had not increased trade cost over the preceding 100 years. This is purely due to the trade impact of climate change; it sets aside feedback loops between trade-related emissions and output and health impacts of climate change, for example. Welfare gains depend not only on countries' own climate trends, but also on their neighbors' trajectories: When country i's neighbors face more drastic climate change than i itself, country i experiences a relative trade cost reduction. Reverting that change thus benefits i less, since its relative position declines. Poor and rich countries benefit equally. Benefits are especially large for smaller economies, though, which are more reliant on international trade. A simple counterfactual exercise shows that ignoring the trade cost channel I highlight leads to a nine percent underestimate of the welfare impact of climate change. An alternative, more conservative calibration of the model's key parameter still yields a five percent underestimate.

Since I only rely on an augmented gravity specification, the effect of climate change on trade cost I demonstrate in this paper can easily be included in estimations of the overall impact of climate change. This is especially true for estimations based on the broad class of trade models that allow

for gravity estimation to be solved separately from the rest of the model (Arkolakis et al., 2012). For these models, the impact on trade costs can be estimated via the simple reduced-form approach I use here and incorporated into counterfactuals by rewriting the model in changes (Dekle et al., 2008). This simple methodology may aid future analyses of the impact of climate change.

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Tables

Table 1: Specification check

Variable	Origin interaction ($\tilde{d}_{ni}\Delta T_{it}$)	Destination interaction ($\tilde{d}_{ni}\Delta T_{nt}$)
$\tilde{d}_{ni}$	0.001 [0.372]	0.001 [0.224]
Origin-decade FE	Yes	Yes
Destination-decade FE	Yes	Yes
Observations	327,091	327,182
Clusters	28,993	28,993

Note: The outcomes are interactions between log distance $\tilde{d}_{ni}$ and *origin* or *destination* temperature changes ΔT_{ct}, $c \in \{i, n\}$. (The change in decadal mean temperature in country c between decades t and $t-1$ in °C.) Temperature changes are winsorized at the 1st and 99th percentiles. Decades t are the decades from 1820 to 2020. Standard errors are clustered by country pair, p-values provided in brackets.

Table 2: Gravity estimation results

Variable	Basic model	$T_{50\text{–}80}$ deciles	$T_{50\text{–}80}$ deciles & time trend	CC deciles	CC deciles & time trend	Lat. deciles	Lat. deciles & time trend	Weighted distance	Decade $\geq$ '50s	Drop CHN	Full interaction	$\mathbf{C}^{+}_{nit}$	Domestic trade	Residual T	Benchmark
$\tilde{d}_{ni} \times 2010s$	−0.550 [0.000]	−0.594 [0.000]	−0.489 [0.000]	−0.707 [0.000]	−0.961 [0.000]	−0.704 [0.000]	−1.001 [0.000]	−0.587 [0.000]	−0.545 [0.000]	−0.618 [0.000]	−0.549 [0.000]	−0.535 [0.000]	0.136 [0.000]	−0.577 [0.000]	−0.565 [0.000]
$\tilde{d}_{ni} \times \Delta T_{it}$	−0.101 [0.000]	−0.112 [0.000]	−0.108 [0.000]	−0.063 [0.010]	−0.071 [0.003]	−0.172 [0.000]	−0.192 [0.000]	−0.145 [0.000]	−0.132 [0.000]	−0.098 [0.001]	−0.105 [0.000]	−0.134 [0.000]	−0.097 [0.000]	−0.221 [0.000]	
$\tilde{d}_{ni} \times \Delta T_{nt}$	−0.077 [0.003]	−0.065 [0.007]	−0.054 [0.024]	−0.041 [0.071]	−0.048 [0.027]	−0.129 [0.000]	−0.128 [0.000]	−0.110 [0.000]	−0.098 [0.002]	−0.077 [0.006]	−0.082 [0.007]	−0.106 [0.000]	−0.070 [0.000]	−0.218 [0.000]	
$\tilde{d}_{ni} \otimes$ decade	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$\tilde{d}_{ni} \otimes T_{50\text{–}80}$ decile$_i$	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes T_{50\text{–}80}$ decile$_n$	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes T_{50\text{–}80}$ decile$_i \times$ decade	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes T_{50\text{–}80}$ decile$_n \times$ decade	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ CC decile$_i$	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ CC decile$_n$	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ CC decile$_i \times$ decade	No	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ CC decile$_n \times$ decade	No	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ Lat. decile$_i$	No	No	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ Lat. decile$_n$	No	No	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ Lat. decile$_i \times$ decade	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No
$\tilde{d}_{ni} \otimes$ Lat. decile$_n \times$ decade	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No
$\mathbf{C}_{nit} \otimes$ decade	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$\mathbf{C}_{nit} \otimes [\Delta T_{it}, \Delta T_{nt}]$	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No
$\mathbf{C}^{+}_{nit} \otimes$ decade	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No
$\tilde{d}_{ni}\mathbb{I}[i = n] \otimes$ decade	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
Origin-decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Destination-decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	326,747	326,747	326,747	326,747	326,747	326,747	326,747	293,461	237,391	323,031	326,747	326,747	294,944	324,433	327,550
Clusters	28,993	28,993	28,993	28,993	28,993	28,993	28,993	22,118	23,078	23,078	28,993	28,993	28,993	28,943	28,993

Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. $\tilde{d}_{ni} \equiv \log(d_{ni})$ is the log of the great circle distance d_{ni} between the origin and destination countries in km. Since the coefficient on that variable is allowed to vary across decades, I only report the coefficient for the latest period, $\tilde{d}_{ni} \times 2010s$. ΔT_{ct} is the change in decadal mean temperature in country c between decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. $\mathbf{C}_{nit}$ contains a common language indicator, sharing a border indicator and two indicators for current and past colonial relationships, taking decade means for all variables within each origin-destination pair. Decades t are the decades from 1820 to 2020. $T_{50\text{–}80}$ *decile*$_c$ is country c's decile of average temperature between the 1950s and 1980s. *CC decile*$_c$ is country c's decile of average temperature change (i.e., climate change) between the 1900s and 2000s decades. *Lat. decile*$_c$ is country c's decile of latitude. Models with these decile controls include separate trade cost levels and, for the *time trend* version, trade cost time trends for each decile. $\mathbf{C}^{+}_{nit}$ includes indicators for both countries being in the EU, OECD and GATT. *Weighted distance* uses population-weighted great circle distance instead of the unweighted measure (this is missing for countries which no longer exist, so the observation count is lower). *Decade* $\geq$ *'50s* only uses decades from the 1950s onwards. *Drop CHN* drops China from the estimation sample. *Residual T* uses temperature residuals obtained from an AR(3) estimation with heterogeneity by 1950–1980 average temperature, akin to Ramey, Nath, and Klenow (2025). I use $\otimes$ to indicate a variable being interacted with a set of fixed effects (e.g., letting the effect of distance vary by decade). Standard errors are clustered by country pair, p-values provided in brackets.

Table 3: Mechanisms behind main results

Variable	Sea-only trade	Long distance trade	Comparison
Sea only$_{ni} \times$ 2010s	−0.563		−0.418
	[0.000]		[0.000]
Sea only$_{ni} \times \Delta T_{it}$	−0.178		−0.227
	[0.003]		[0.000]
Sea only$_{ni} \times \Delta T_{nt}$	−0.032		−0.085
	[0.548]		[0.138]
Long distance$_{ni} \times$ 2010s		−0.429	−0.294
		[0.000]	[0.000]
Long distance$_{ni} \times \Delta T_{it}$		−0.164	−0.017
		[0.012]	[0.806]
Long distance$_{ni} \times \Delta T_{nt}$		−0.082	0.014
		[0.166]	[0.826]
Sea only$_{ni} \otimes$ decade	Yes	No	Yes
Long distance$_{ni} \otimes$ decade	No	Yes	Yes
$\mathbf{C}_{nit} \otimes$ decade	Yes	Yes	Yes
Origin-decade FE	Yes	Yes	Yes
Destination-decade FE	Yes	Yes	Yes
Observations	326,747	326,747	326,747
Clusters	28,993	28,993	28,993

Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. Sea only$_{ni}$ is an indicator for the country pair being separated by an ocean. Since the coefficient on that variable is allowed to vary across decades, I only report the coefficient for the latest period, Sea only$_{ni} \times$ 2010s. Long distance$_{ni}$ is an indicator for the distance between the origin and destination being greater than or equal to the average distance between country pairs that only have an ocean-based connection (that is, pairs which have Sea only$_{ni} = 1$). Again, I only report the coefficient for the latest period. ΔT_{ct} is the change in decadal mean temperature in country c between decades t and $t-1$ in $^{\circ}$C. Temperature changes are winsorized at the 1st and 99th percentiles. $\mathbf{C}_{nit}$ contains a common language indicator, sharing a border indicator and two indicators for current and past colonial relationships, taking decadal means for all variables within each origin-destination pair. Decades t are the decades from 1820 to 2020. I use $\otimes$ to indicate a variable being interacted with a set of fixed effects (e.g., letting the effect of the sea only indicator vary by decade). Standard errors are clustered by country pair, p-values provided in brackets.

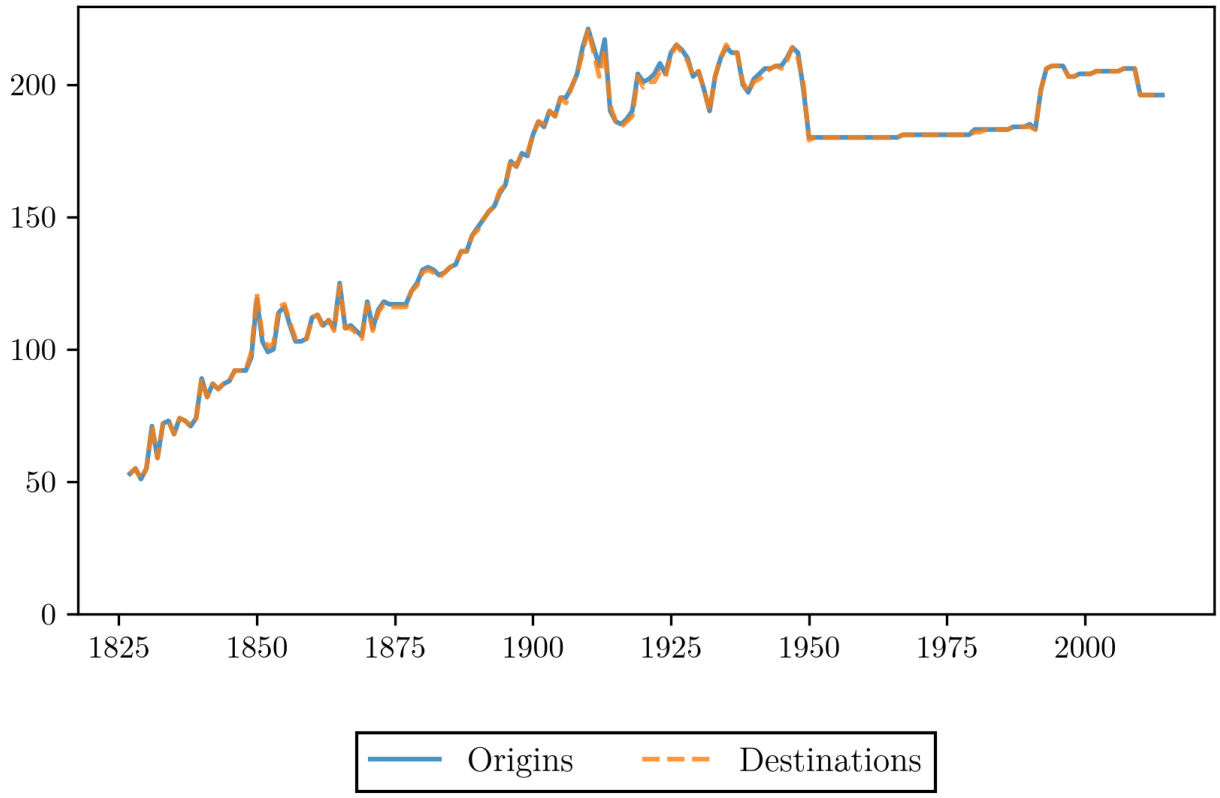
Table 4: Correlates of welfare changes

Variable	$\hat{W}_{it}$	$\hat{W}_{it}$	$\hat{W}_{it}$	$\hat{W}_{it}$	$\hat{W}_{it}$	$\hat{W}_{it}$
Own ΔT	0.228		0.585			
	[0.231]		[0.024]			
Inverse distance weighted ΔT		−0.315	−1.674			
		[0.568]	[0.027]			
Log 2010s GDP				−0.071	0.169	0.112
				[0.140]	[0.002]	[0.082]
2010s own trade share (%)					−0.042	−0.043
					[0.000]	[0.000]
1950s–'80s $\bar{T}$						−0.035
						[0.100]
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	No	No	No

Note: The outcome $\hat{W}_{it}$ is the welfare change for country i under decade t's climate counterfactual. *Own* ΔT is a country's change in temperature between each decade and the 2010s, whereas the *inverse distance weighted* ΔT for country i is the average change in all other countries' temperatures, weighted by the inverse of their distance to i. *1950s–'80s* $\bar{T}$ is countries' average temperature from the 1950s to the 1980s. Standard errors are clustered by country, p-values provided in brackets.

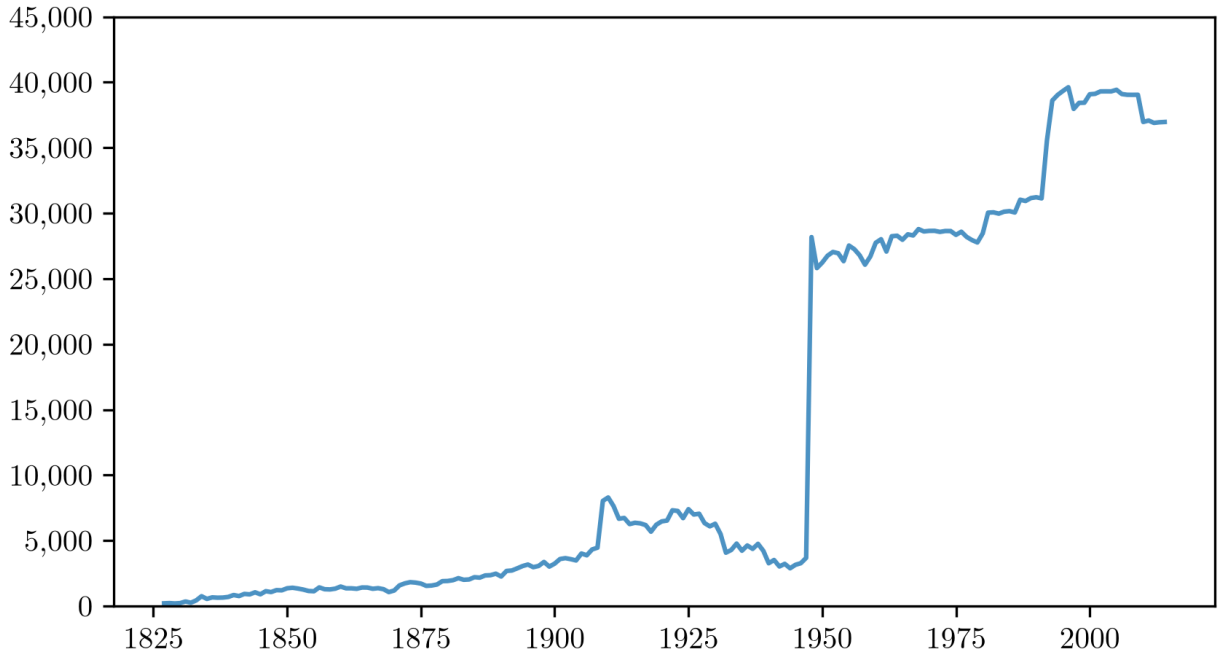
Figures

Figure 1: Trade data country counts by year



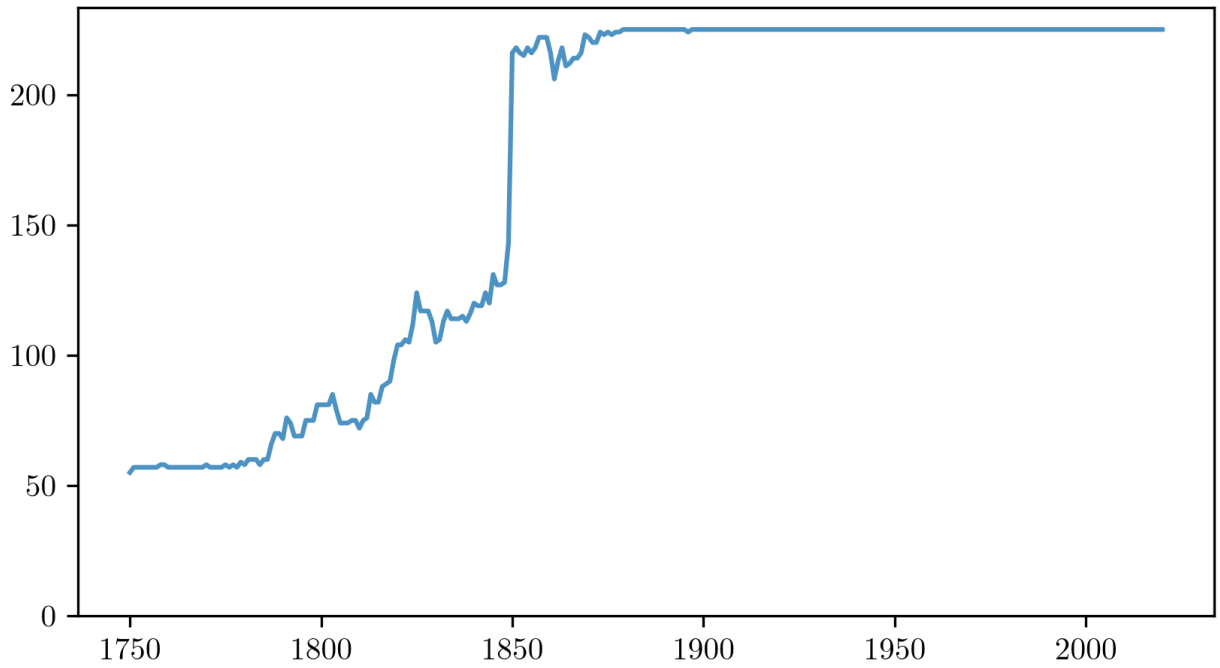
Note: The figure shows the number of countries observed in the TRADHIST trade data by year. I subset to observations with non-missing trade flows and distance information.

Figure 2: Trade flow counts by year



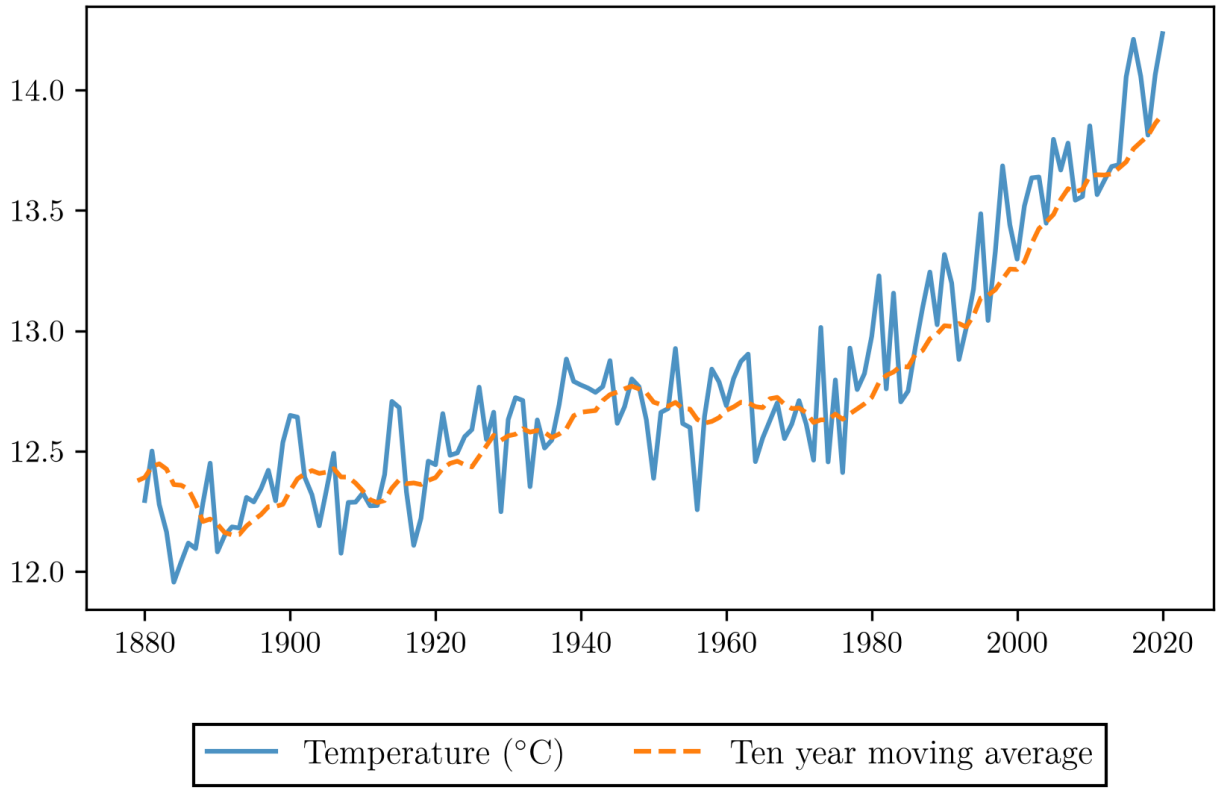
Note: The figure shows the number of trade flows observed in the TRADHIST trade data by year. I subset to observations with non-missing trade flows and distance information.

Figure 3: Weather observation counts for current countries by year



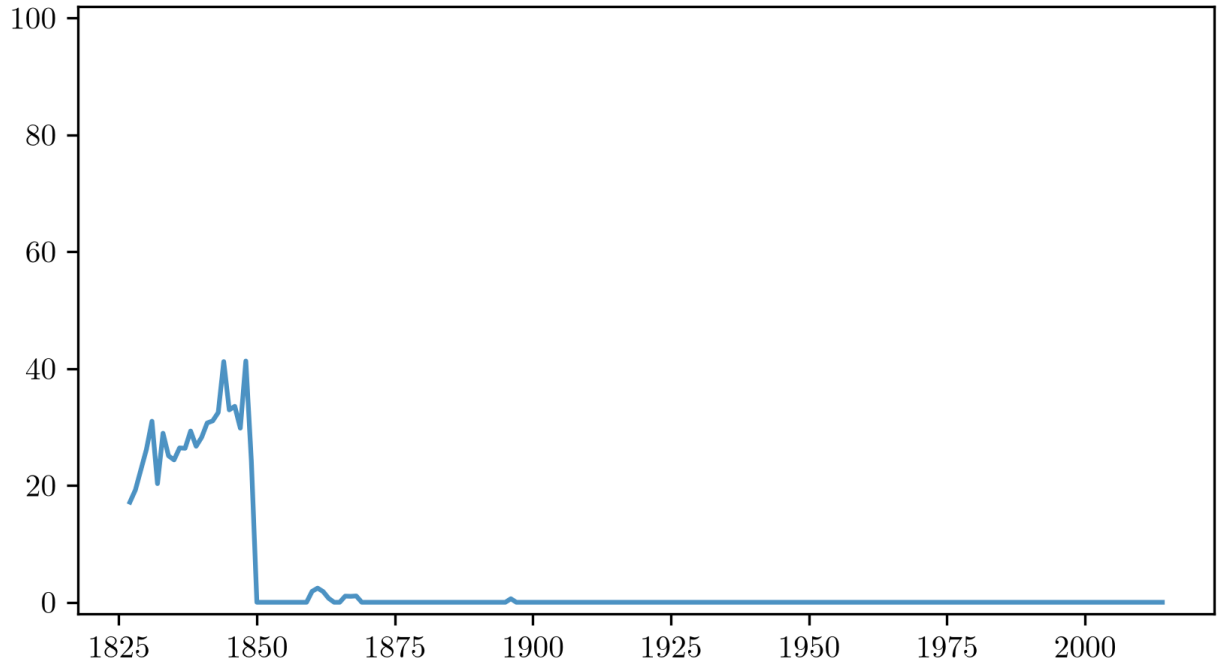
Note: The figure shows the number of countries with non-missing weather observations by year. I subset to countries which currently exist and ever appear in the TRADHIST trade data. (For example, in this plot, I include Germany, which currently exists and appears in the trade data, but not the former West and East Germany, which do appear in the trade data but no longer exist.) The number of countries in the sample therefore does not change over time.

Figure 4: Average temperature (°C) by year



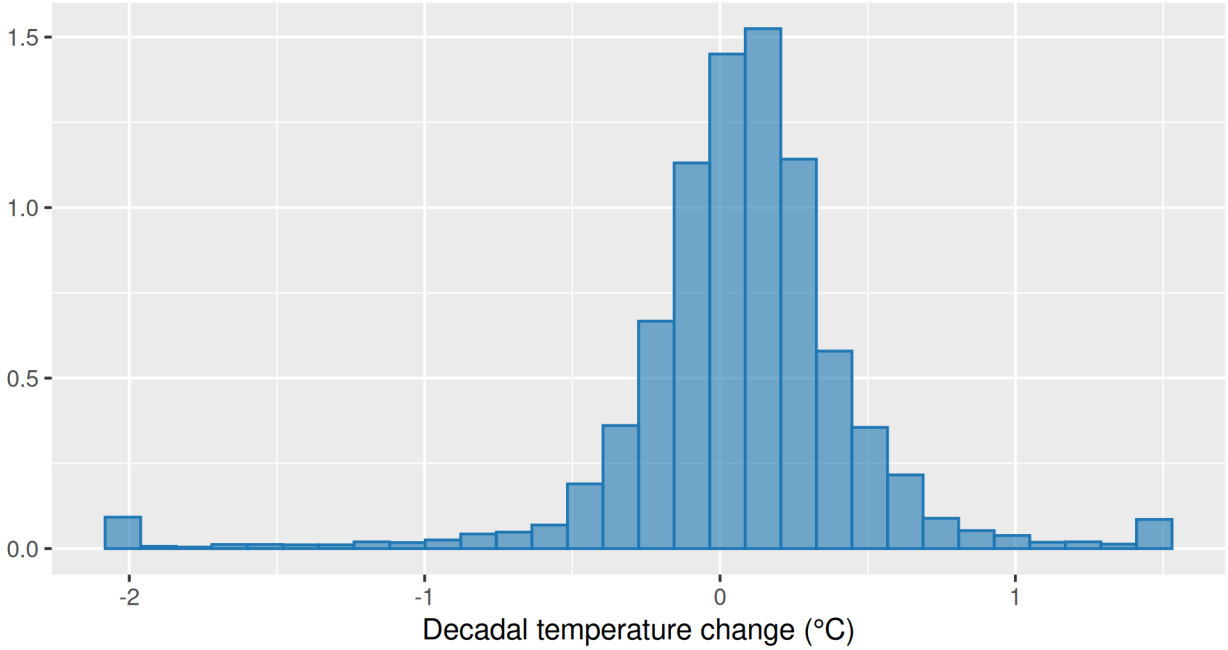
Note: The figure shows the average temperature across years, weighted by country area. The figure starts in 1880, where I have global weather coverage. I subset to countries which currently exist and ever appear in the TRADHIST trade data. (For example, in this plot, I include Germany, which currently exists and appears in the trade data, but not the former West and East Germany, which do appear in the trade data but no longer exist.) The number of countries in the sample therefore does not change over time.

Figure 5: Unmatched trade flows (percent) by year



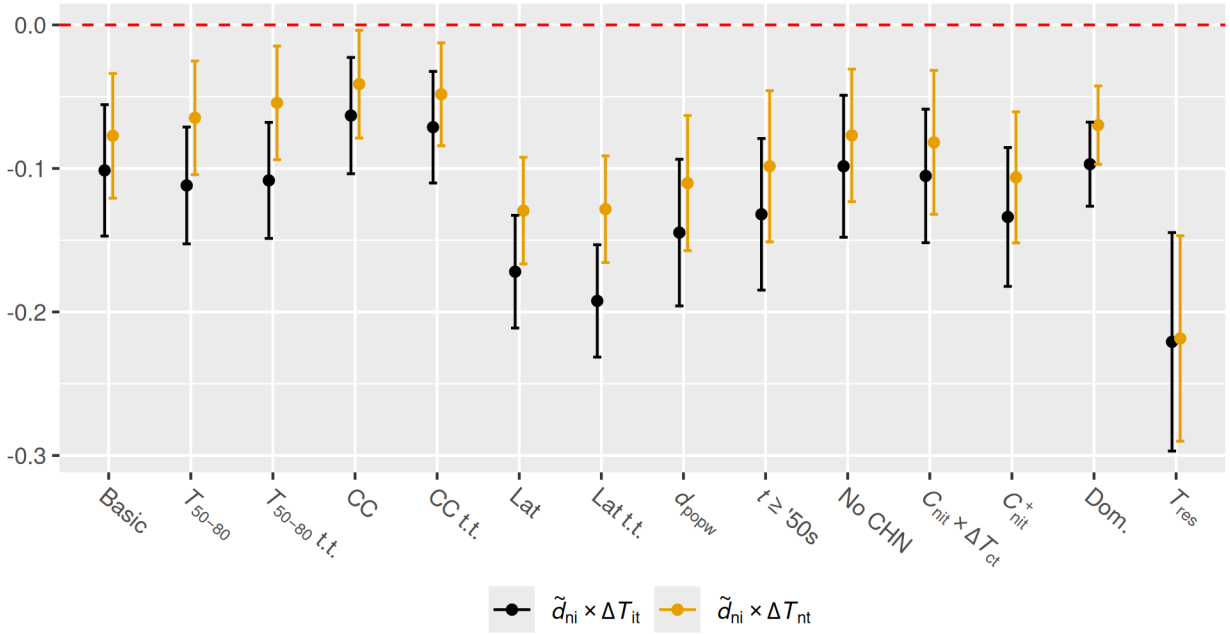
Note: The figure shows the percent of TRADHIST trade observations which cannot be matched to weather information by year. I subset to observations with non-missing trade flows and distance information.

Figure 6: Decadal temperature changes



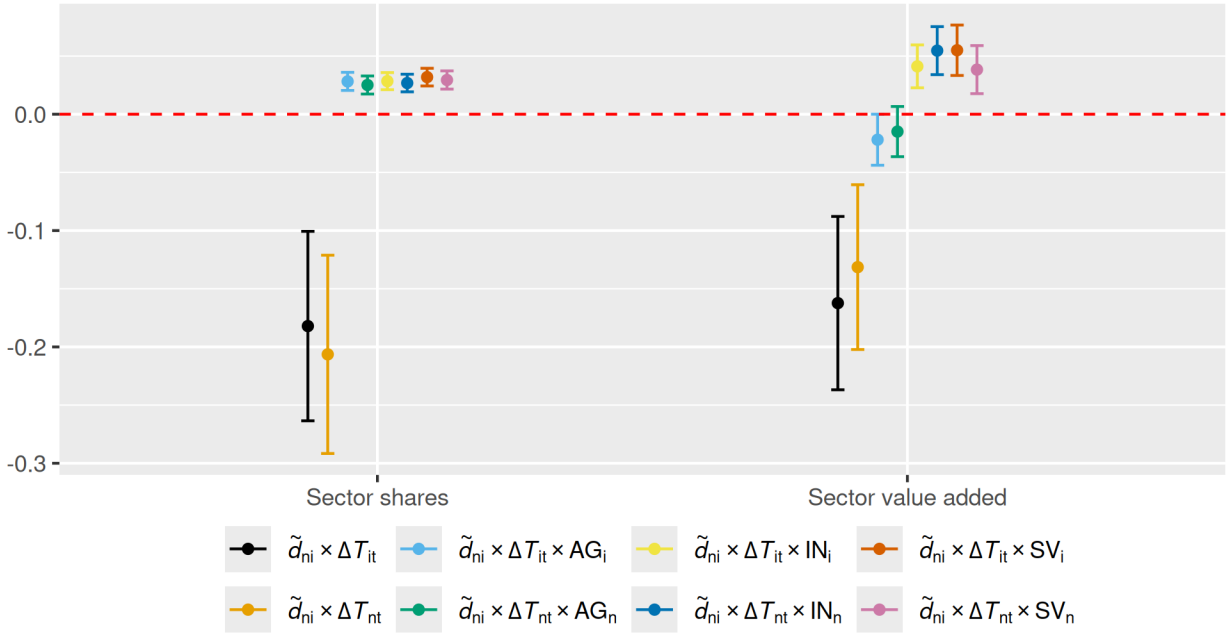
Note: The figure shows the distribution of decadal temperature changes across all countries and decades. As in my gravity estimations, decadal temperature changes are winsorized at the 1st and 99th percentiles.

Figure 7: Coefficients on temperature interactions across specifications



Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. $\tilde{d}_{ni} \equiv \log(d_{ni})$ is the log of the great circle distance d_{ni} between the origin and destination countries in km. ΔT_{ct} is the change in decadal mean temperature in country c between decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. C_{nit} contains a common language indicator, sharing a border indicator and two indicators for current and past colonial relationships, taking decade means for all variables within each origin-destination pair. Decades t are the decades from 1820 to 2020. $T_{50\text{-}80}$ is country c's average temperature between the 1950s and 1980s. CC is country c's average temperature change (i.e., climate change) between the 1900s and 2000s decades. *Lat.* is country c's latitude. These models include separate trade cost levels and, for the *t.t.* (time trend) version, trade cost time trends for each decile of those variables. d_{popw} uses population-weighted great circle distance instead of the unweighted measure (this is missing for countries which no longer exist, so the observation count is lower). $t \geq$ *'50s* only uses decades from the 1950s onwards. *No CHN* drops China from the estimation sample. $C_{nit} \times \Delta T_{ct}$ interacts temperatures not only with distance, but also with the variables included in C_{nit}. C^{+}_{nit} additionally includes indicators for both countries being in the EU, OECD and GATT. *Dom.* proxies for domestic trade by subtracting total exports from GDP and allows for the effect of distance by decade to vary between international and domestic trade, capturing globalization (Yotov, 2012). T_{res} uses temperature residuals obtained from an AR(3) estimation with heterogeneity by 1950–1980 average temperature, similar in spirit to Ramey, Nath, & Klenow (2025). Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 8: Impact by sectoral composition



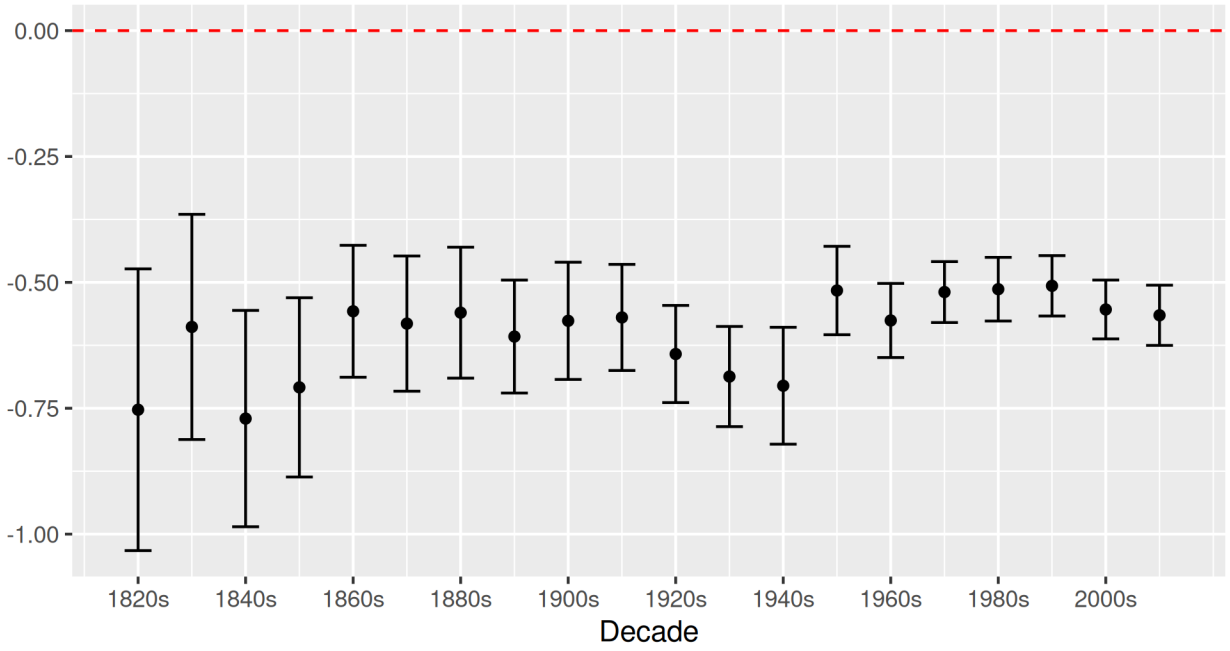
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. $\tilde{d}_{ni} \equiv \log(d_{ni})$ is the log of the great circle distance d_{ni} between the origin and destination countries in km. ΔT_{ct} is the change in decadal mean temperature in country c between decades t and $t-1$ in °C. I interact decade-to-decade temperature changes times distance with the sectoral shares or sectoral value added of the agricultural, industrial and services sector, with miscellaneous/other as the left-out category. Interactions are centered at sample means (in the case of distance, I de-mean by subtracting log mean distance). Temperature changes are winsorized at the 1st and 99th percentiles. Standard errors are clustered by country pair. The shaded areas indicate 90 percent confidence intervals.

Figure 9: Coefficients on log distance across decades



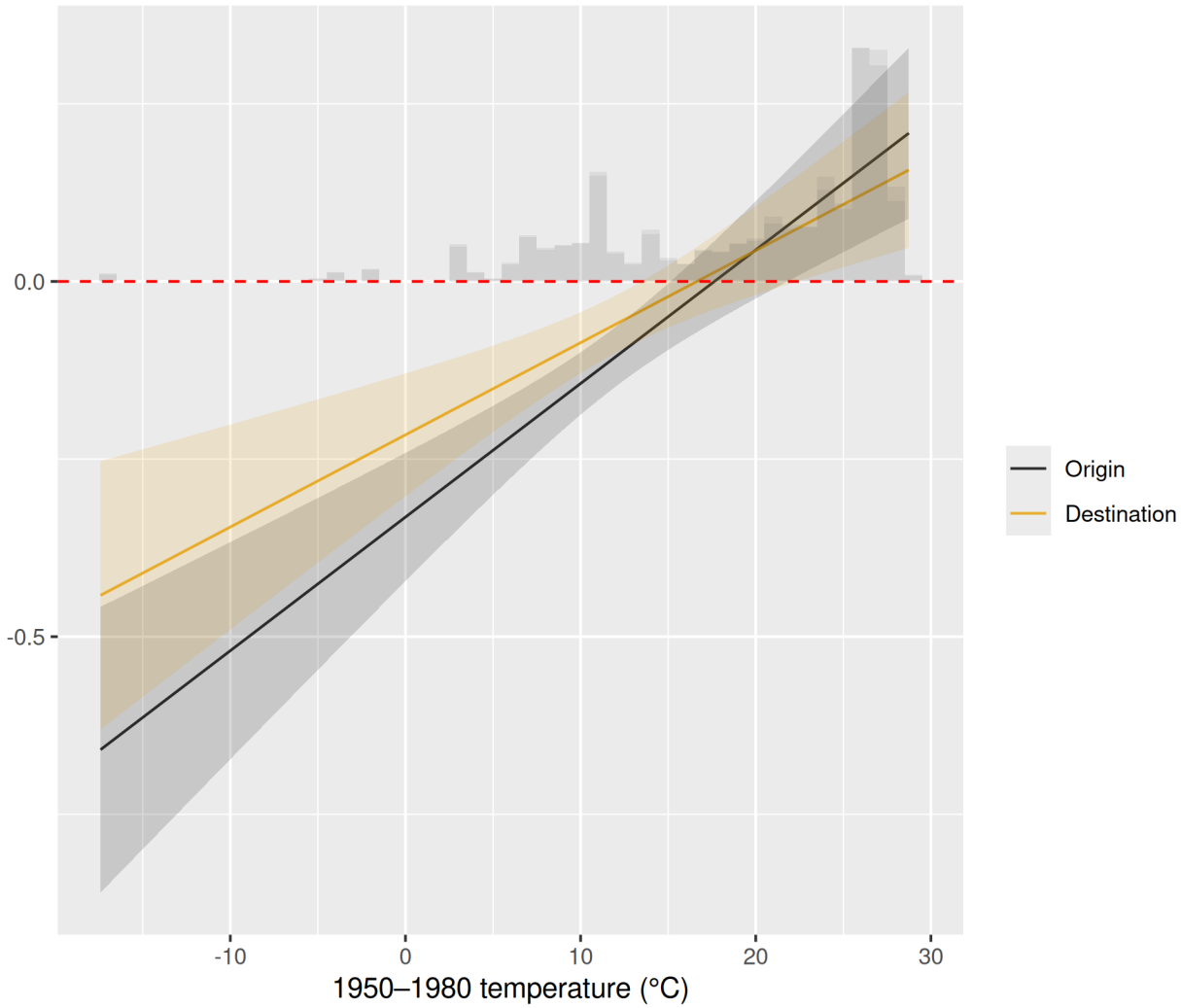
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. Coefficients are for log distance between origin-destination pairs interacted with decade indicators. Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 10: Coefficients on log distance across decades (benchmark excluding temperature variables)



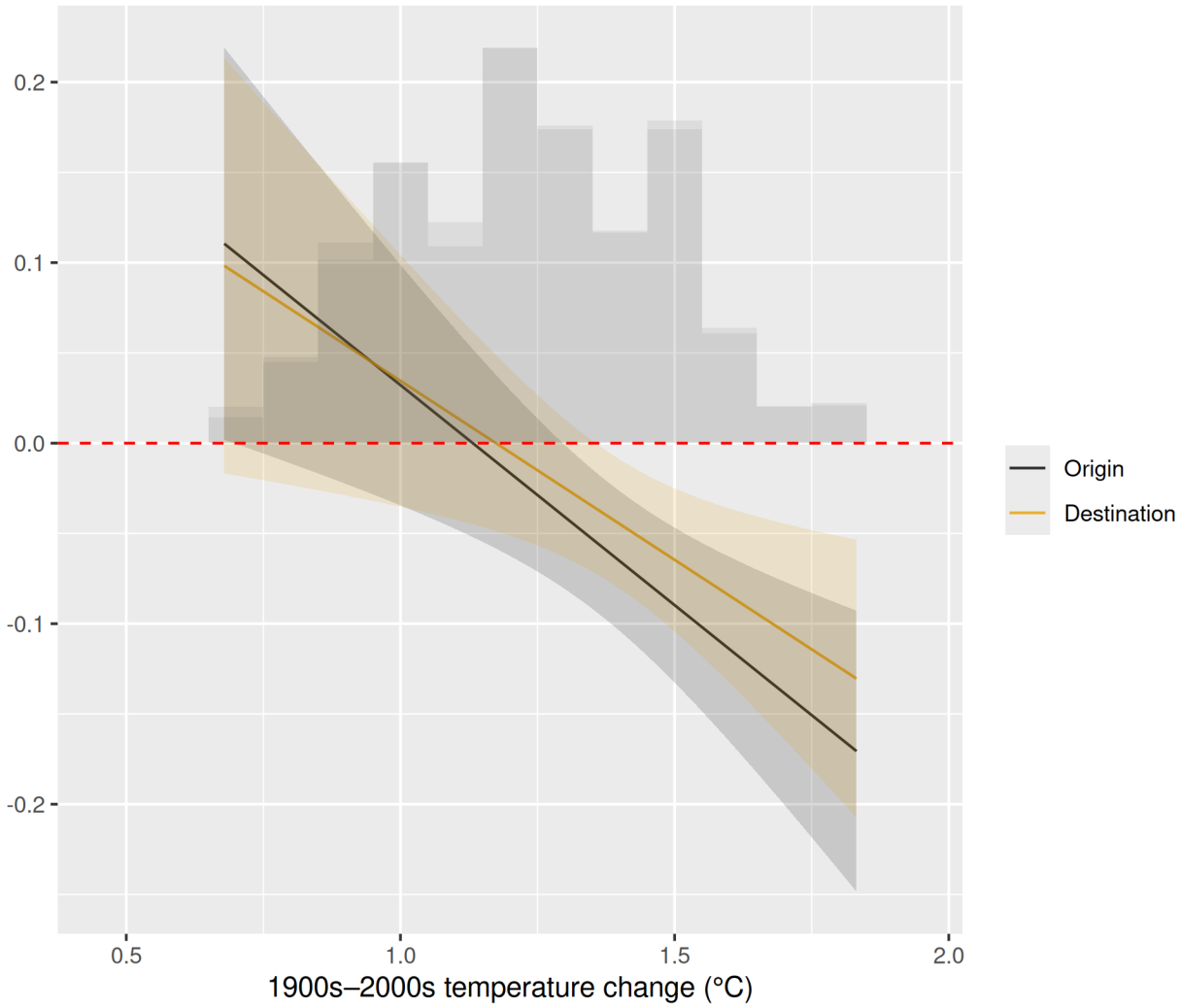
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. Coefficients are for log distance between origin-destination pairs interacted with decade indicators. This benchmark specification does not include origin and destination temperatures. Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 11: Coefficients on temperature interactions across 1950–1980 average temperature



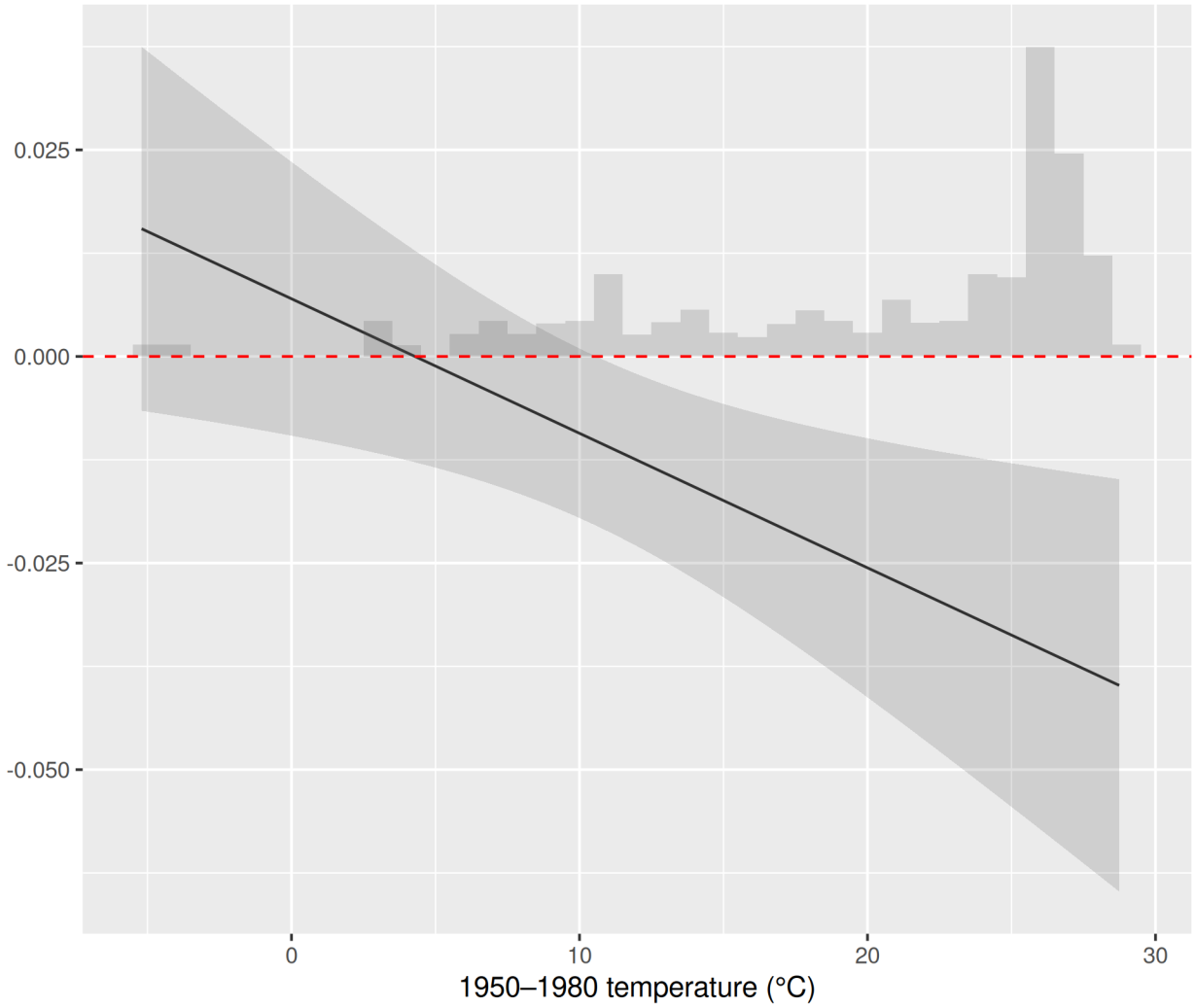
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. I interact decade-to-decade temperature changes with log distance, and allow the effect to vary by 1950–1980 average temperature to assess heterogeneity. The graph shows the change in the marginal effect of distance due to a one degree change in temperature (the change in trade cost due to climate change) across that average temperature. It also shows a histogram of average temperature across countries. Temperature changes are winsorized at the 1st and 99th percentiles. Standard errors are clustered by country pair. The shaded areas indicate 90 percent confidence intervals.

Figure 12: Coefficients on temperature interactions across 1900s–2000s climate change



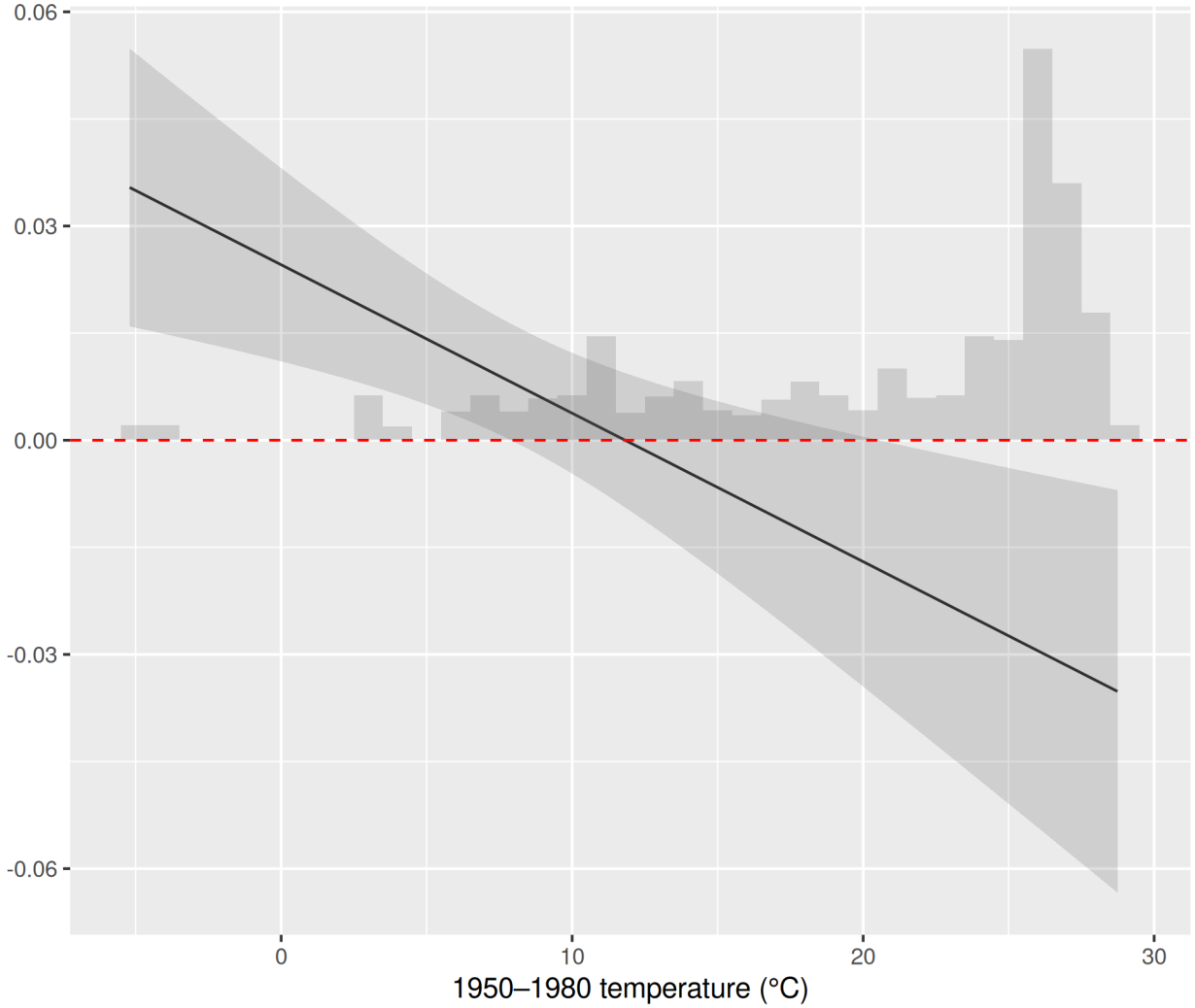
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. I interact decade-to-decade temperature changes with log distance, and allow the effect to vary by countries' temperature change between the first decade of the 1900s and the first decade of the 2000s. The graph shows the change in the marginal effect of distance due to a one degree change in temperature (the change in trade cost due to climate change) across that temperature change. It also shows a histogram of temperature change across countries. Temperature changes are winsorized at the 1st and 99th percentiles. Standard errors are clustered by country pair. The shaded areas indicate 90 percent confidence intervals.

Figure 13: Marginal effect of year-to-year weather shocks on container port throughput



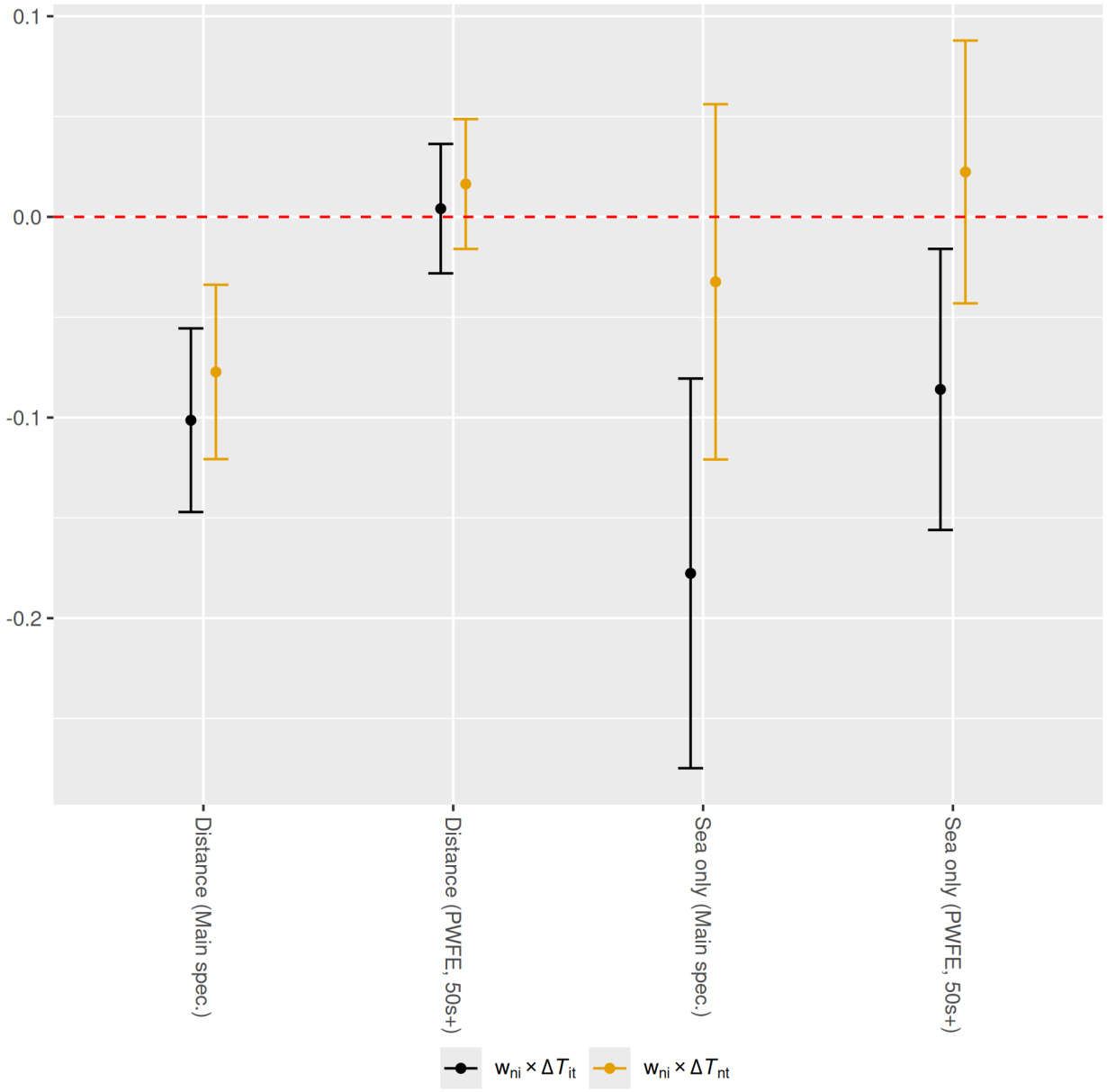
Note: Results are from a regression of log change in container port throughput on temperature changes and country fixed effects. The effect of temperature varies with countries' 1950–1980 average temperature. This graph shows the marginal effect of temperature shocks across 1950–1980 average temperature. The graph also shows a histogram of 1950–1980 average temperature. Standard errors are clustered by country. The shaded area indicates a 90 percent confidence interval.

Figure 14: Marginal effect of year-to-year weather shocks on container port throughput, controlling for change in GDP p.c.



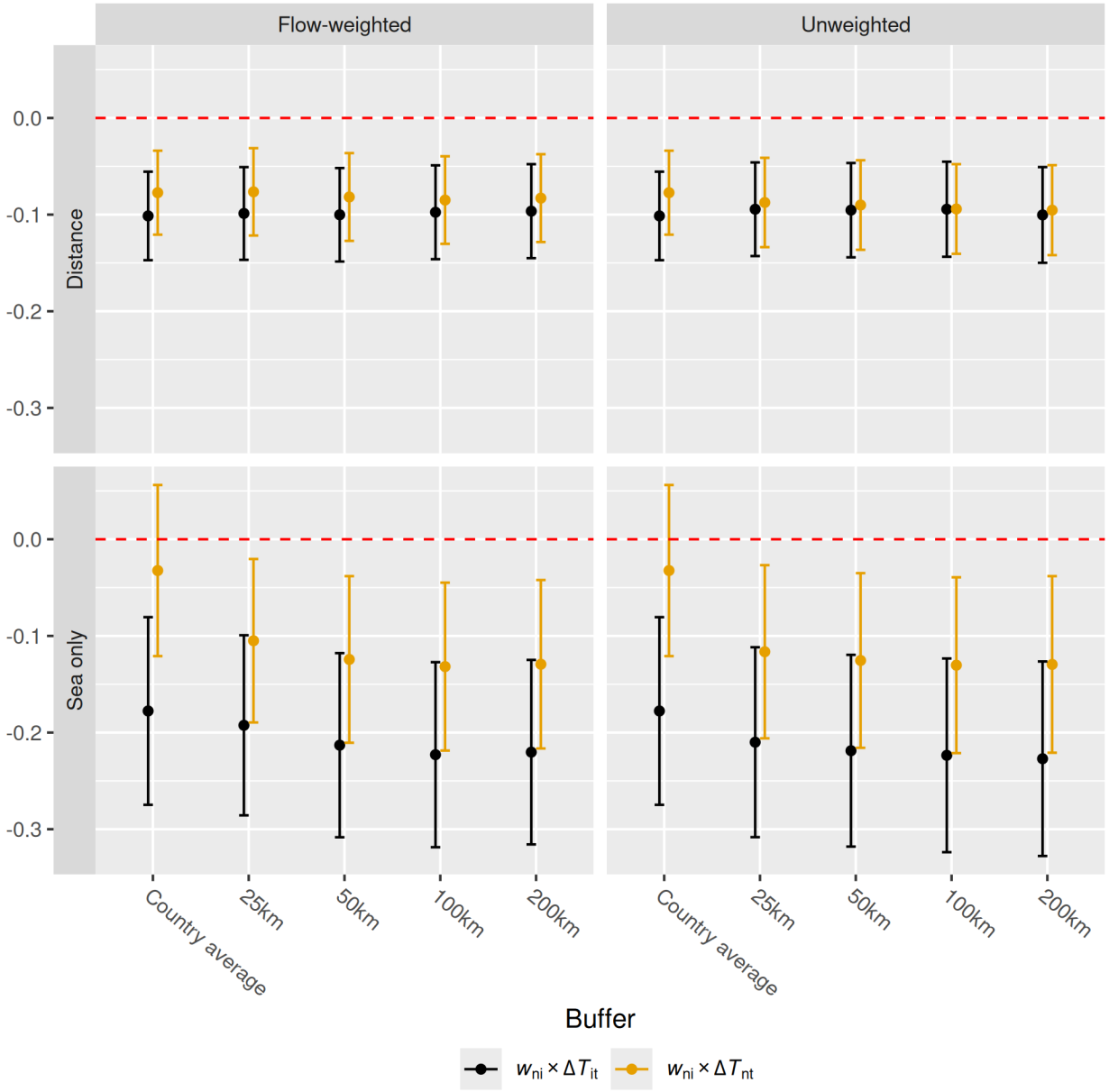
Note: Results are from a regression of log change in container port throughput on temperature changes and country fixed effects, controlling for changes in log GDP per capita. The effect of temperature varies with countries' 1950–1980 average temperature. This graph shows the marginal effect of temperature shocks across 1950–1980 average temperature. The graph also shows a histogram of 1950–1980 average temperature. Standard errors are clustered by country. The shaded area indicates a 90 percent confidence interval.

Figure 15: Estimations including pairwise fixed effects



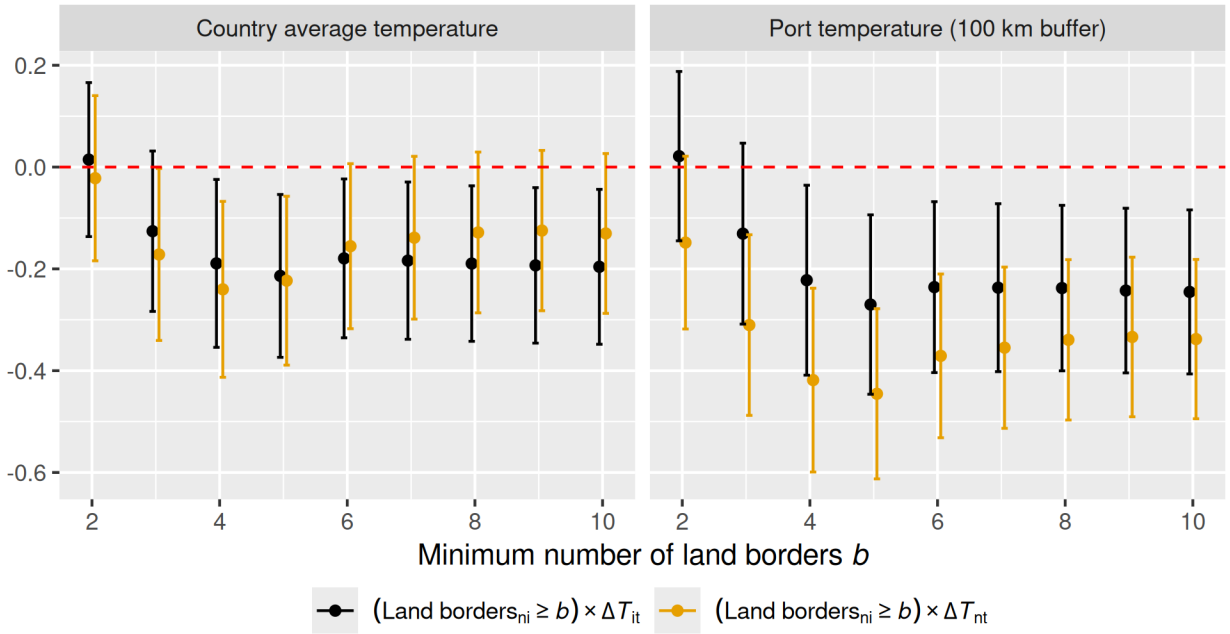
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. w_{ni} is a stand-in for one of two parameterizations of trade cost I consider: *Distance* uses log great circle distance between countries as the main measure of trade cost. *Sea only* uses an indicator for countries where the only connection is via the ocean. ΔT_{ct} is the change in decadal mean temperature in country c between decades t and $t-1$ in °C. *Main spec.* estimations use my standard estimation setup (2) with origin- and destination-period fixed effects. *PWFE, 50s+* estimations use country pair fixed effects and pair-specific time trends. Because that estimation does not converge for the full sample, I have to subset to the 1950s and later periods. Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 16: Coefficient estimates using port temperatures



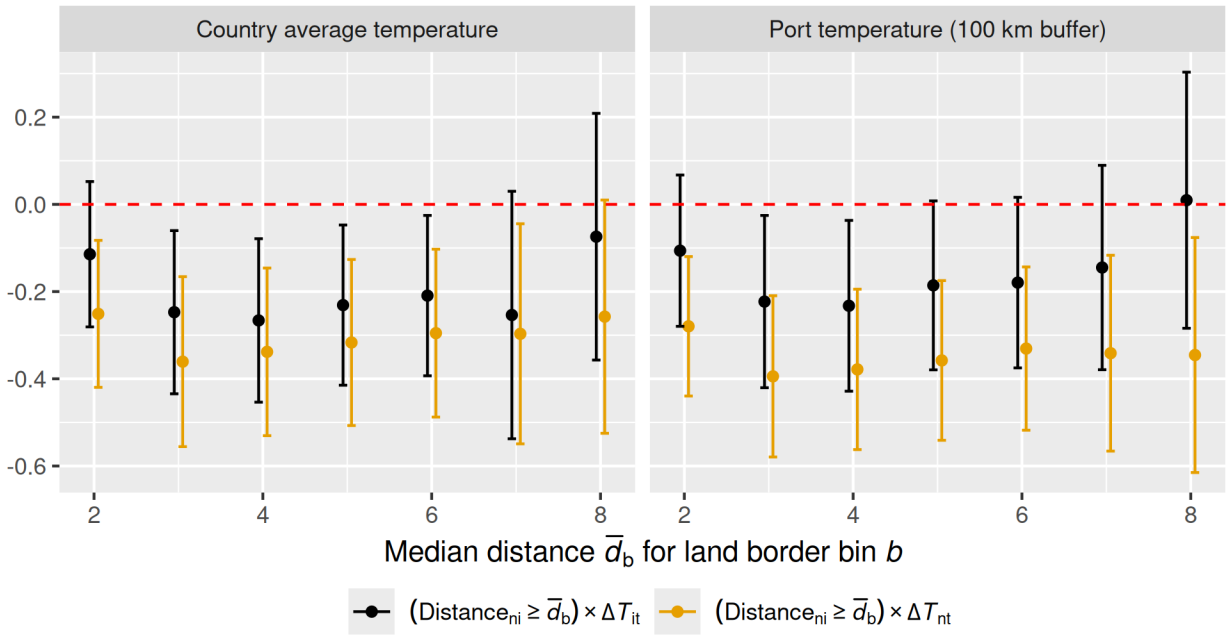
Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. Each row of panels uses a different variable to measure trade cost changes. w_{ni} is a stand-in for those variables. *Distance* uses the log of the great circle distance between the origin and destination countries in km. *Sea only* uses an indicator for countries where the only connection is via the ocean. ΔT_{ct} is the change in decadal mean temperature in country c, or in country c's ports, between decades t and $t-1$ in $^{\circ}$C. Temperature changes are winsorized at the 1st and 99th percentiles. Port temperatures are measured in different buffers around each port, as indicated on the horizontal axis. (*Country average* uses the country's average temperature.) Each column of panels uses a different way to average port temperatures for each country — *unweighted* averages across ports, whereas *flow-weighted* uses 2019 total outflows to weight each port. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 17: Land border crossing donut hole estimation



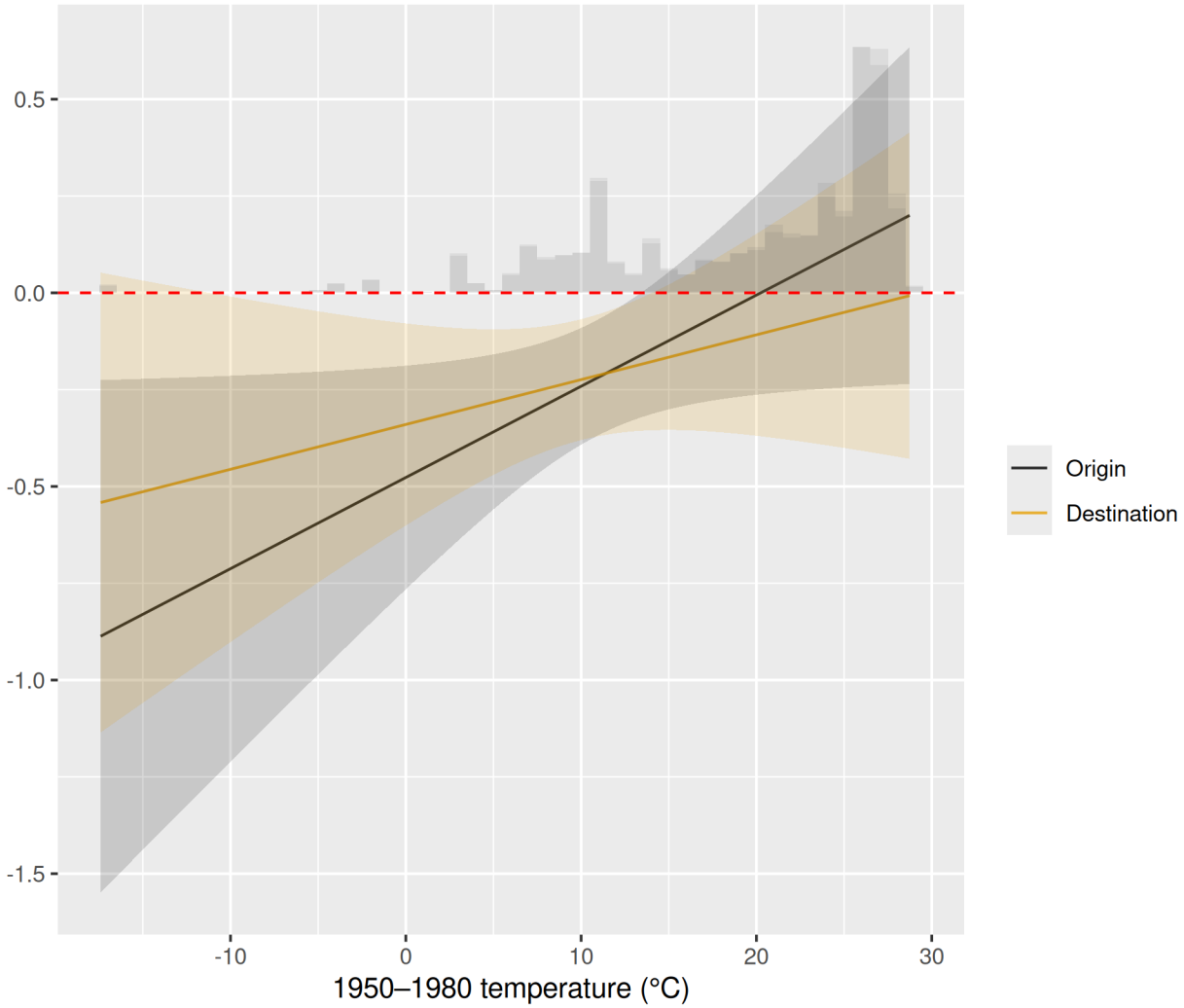
Note: Results are from gravity estimations for decade-level average trade flows between countries, estimated via Poisson pseudo-maximum likelihood to deal with zero flows. Land borders$_{ni} \geq b$ is an indicator; it is equal to one if the number of countries that need to be crossed to get, via a land route, from the origin to the destination country is greater than or equal to b. It is equal to zero if the two countries are neighbors. For all other country pairs (those in the donut hole), the indicator is missing — they are discarded. Each coefficient is from a separate estimation for a different donut hole size. ΔT_{ct} is the change in decadal mean temperature in country c, or in country c's ports, between decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. Port temperatures are measured in a 100 km buffer around ports and averaged for each country (unweighted). Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 18: Distance decile donut hole estimation



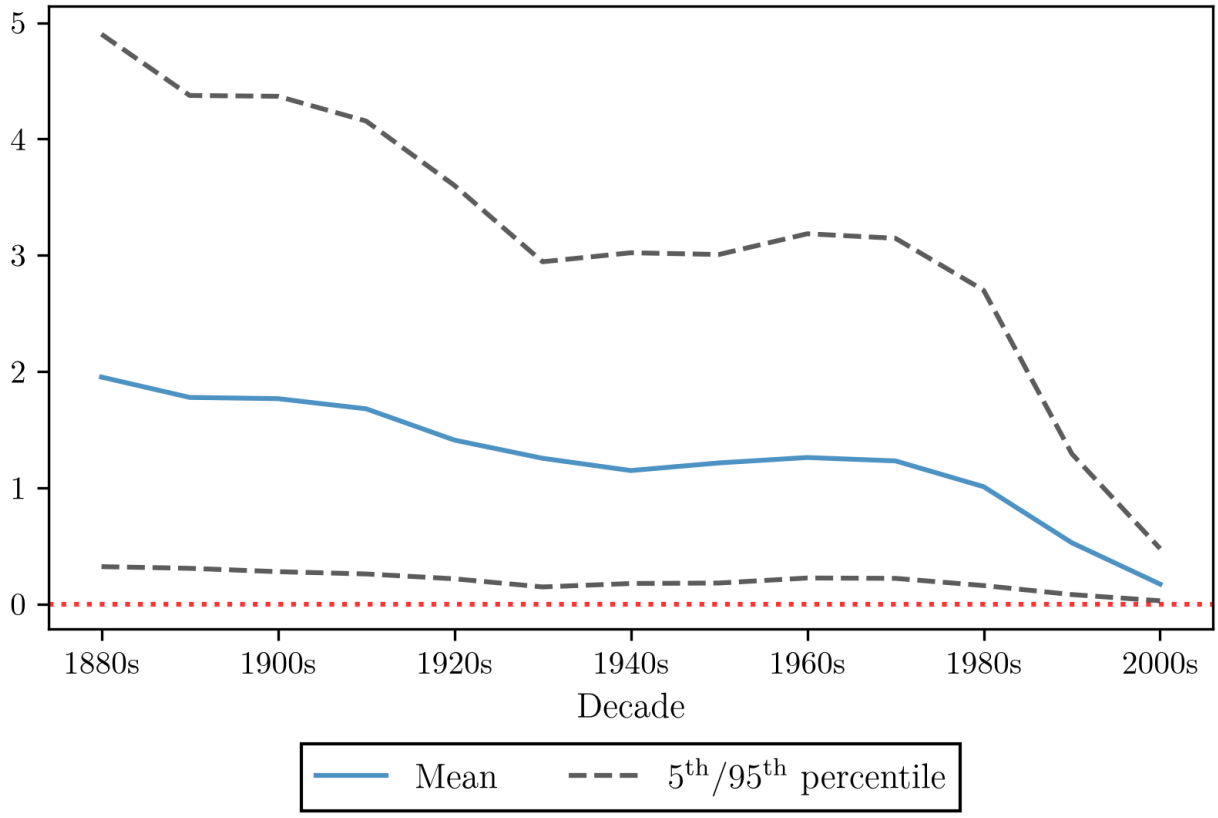
Note: Results are from gravity estimations for decade-level average trade flows between countries, estimated via Poisson pseudo-maximum likelihood to deal with zero flows. $\text{Distance}_{ni} \geq \bar{d}_b$ is an indicator; it is equal to one if the distance between the origin to the destination country is greater than or equal to the median distance between countries separated by $b-1$ land borders. It is equal to zero if the two countries are closer to each other than the median distance between neighboring countries. For all other country pairs (those in the donut hole), the indicator is missing — they are discarded. Each coefficient is from a separate estimation for a different donut hole size. ΔT_{ct} is the change in decadal mean temperature in country c, or in country c's ports, between decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. Port temperatures are measured in a 100 km buffer around ports and averaged for each country (unweighted). Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 19: Coefficients on temperature interactions across 1950–1980 average temperature, donut hole specification using four or more land borders



Note: The outcome are decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. I interact decade-to-decade temperature changes with an indicator for four or more land borders between countries, and allow the effect to vary by 1950–1980 average temperature to assess heterogeneity. The graph shows the change in the effect of having to cross those four borders due to a one degree change in temperature (the change in trade cost due to climate change) across that average temperature. The graph also shows a histogram of that average temperature. Temperature changes are winsorized at the 1st and 99th percentiles. The shaded areas indicate a 90 percent confidence interval.

Figure 20: Population-weighted summary statistics for welfare change (percent) across decades



Note: The mean and percentiles use 2010s population as weights. The 5th and 95th percentile are percentiles of estimated welfare gains, not confidence intervals for the mean.

Figure 21: Welfare change (percent) in 1910s climate counterfactual across countries

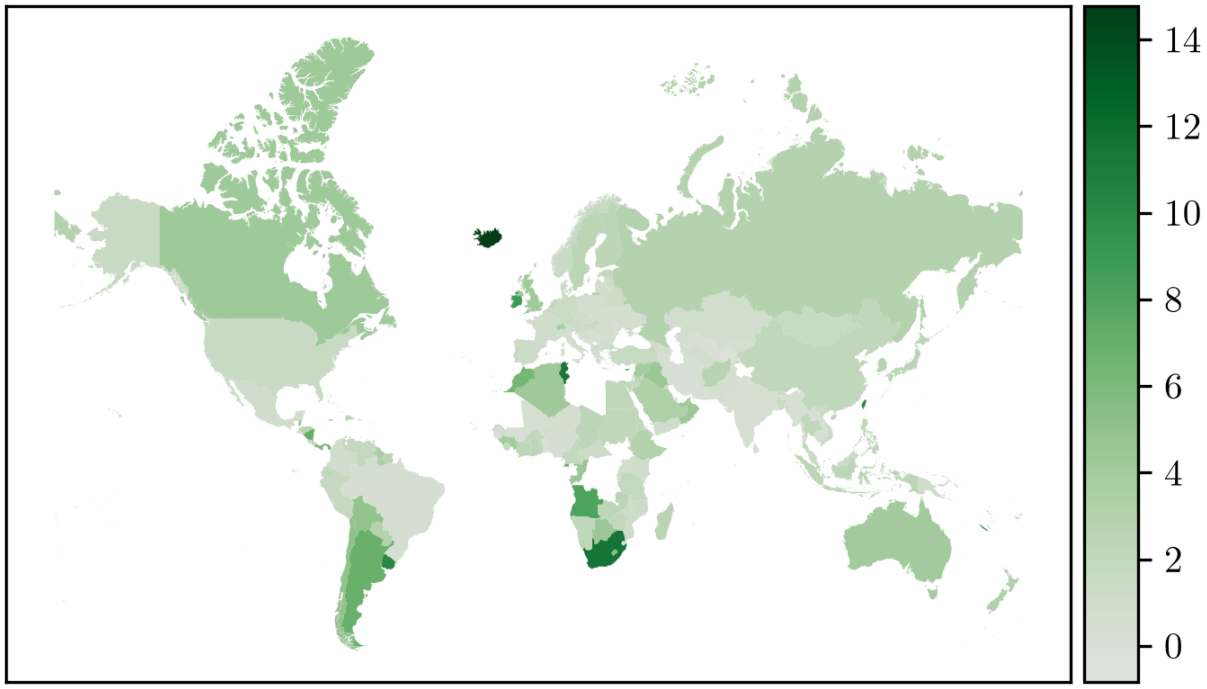
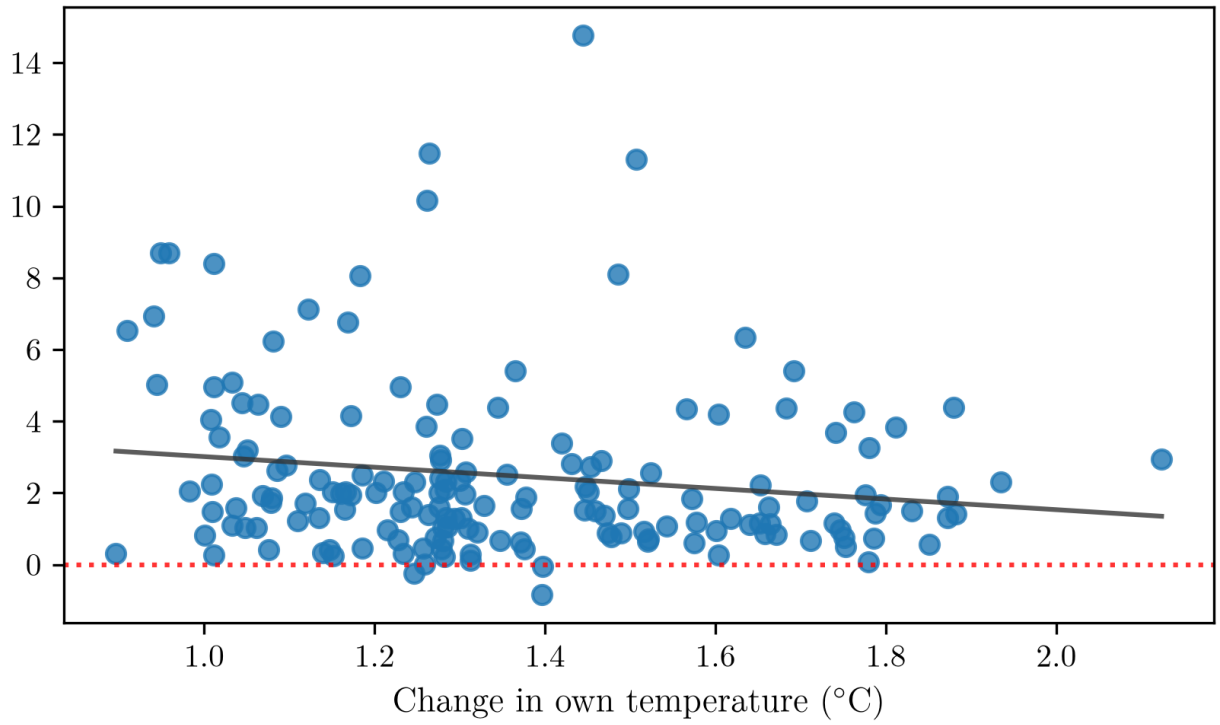
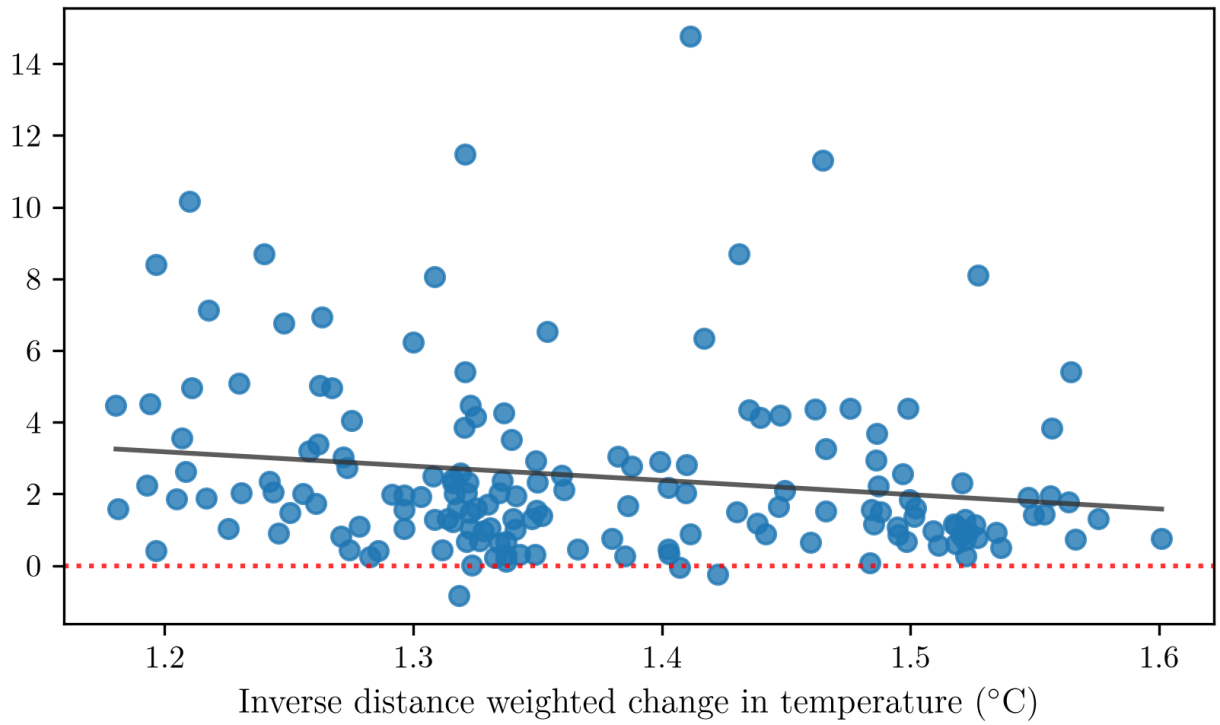


Figure 22: Welfare change (percent) in 1910s climate counterfactual across change in own temperature between the 1910s and 2010s



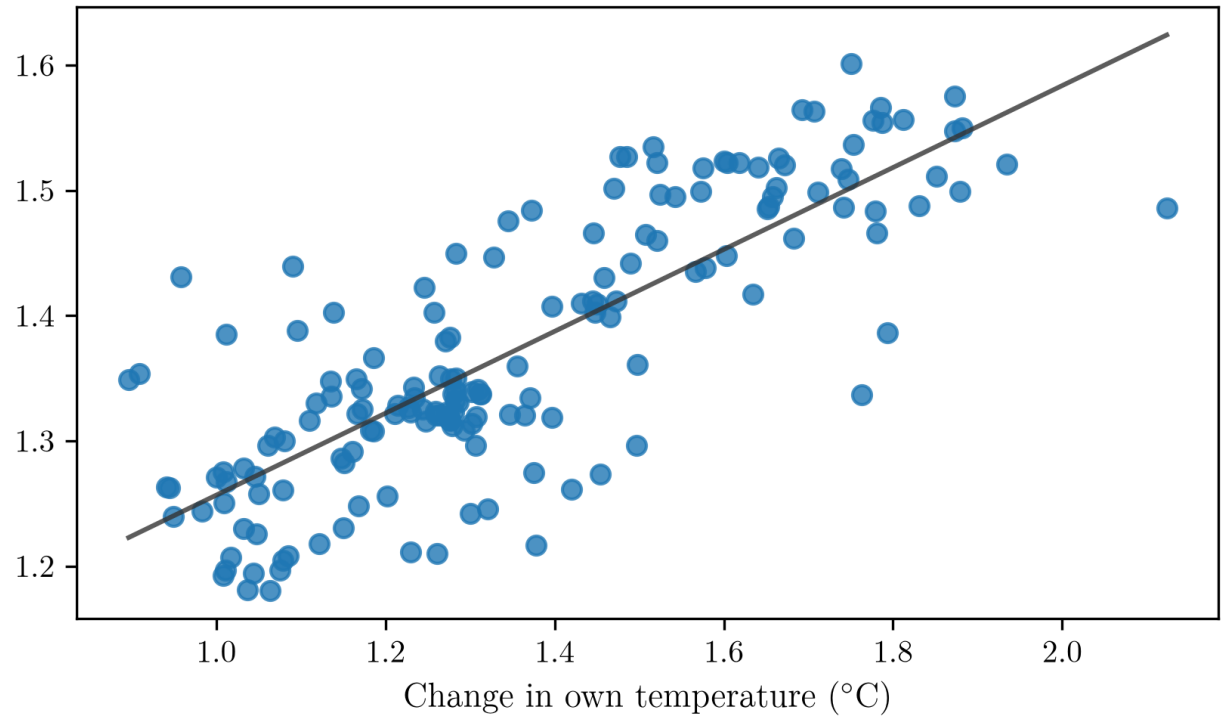
Note: Change in own temperature is the change in country i's own temperature between the 1910s and 2010s. The solid line shows a linear fit.

Figure 23: Welfare change (percent) in 1910s climate counterfactual across inverse distance weighted change in other countries' temperature between the 1910s and 2010s



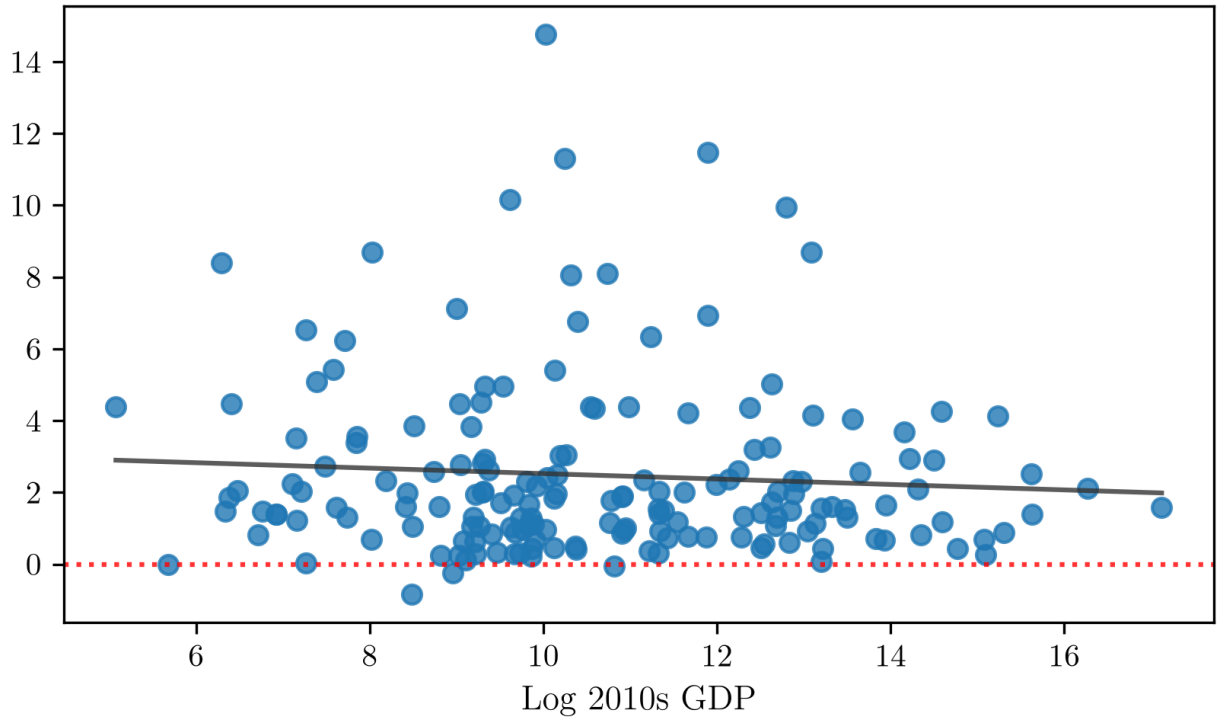
Note: The inverse distance weighted change for country i is the average change in all other countries' temperatures, weighted by the inverse of their distance to i. The solid line shows a linear fit.

Figure 24: Inverse distance weighted change in other countries' temperature between the 1910s and 2010s across change in own temperature between the 1910s and 2010s



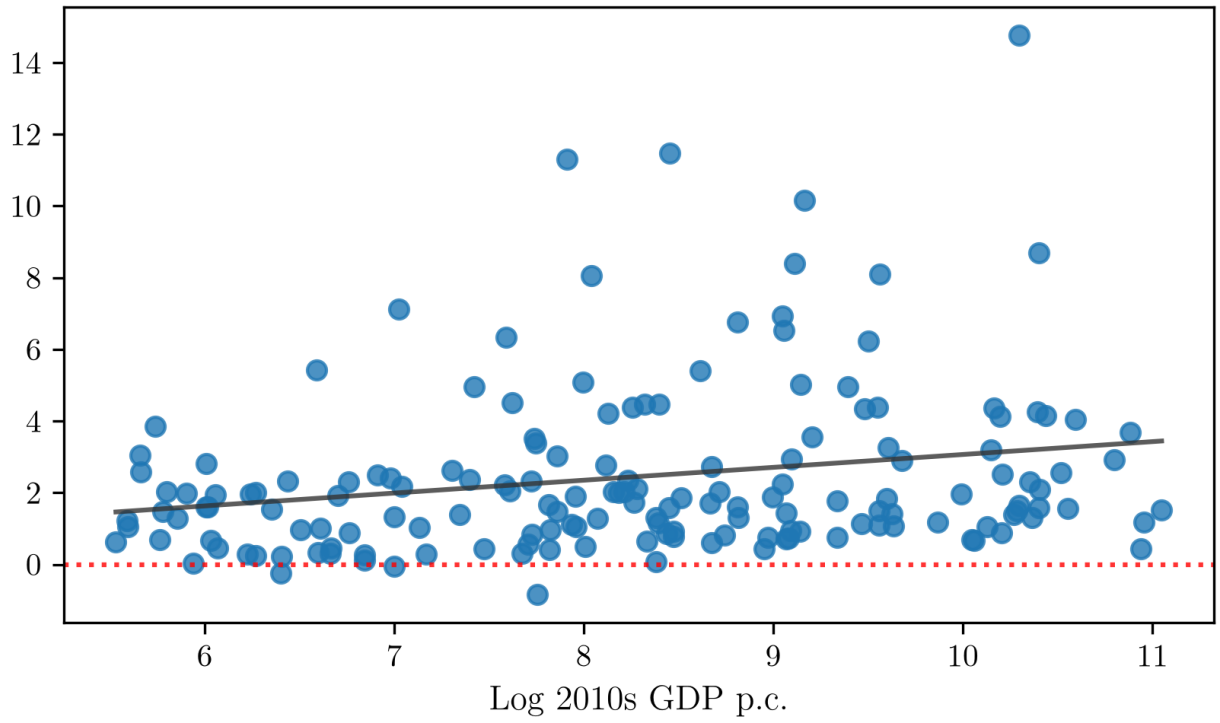
Note: The inverse distance weighted change for country i is the average change in all other countries' temperatures, weighted by the inverse of their distance to i. The solid line shows a linear fit.

Figure 25: Welfare change (percent) in 1910s climate counterfactual across 2010s GDP



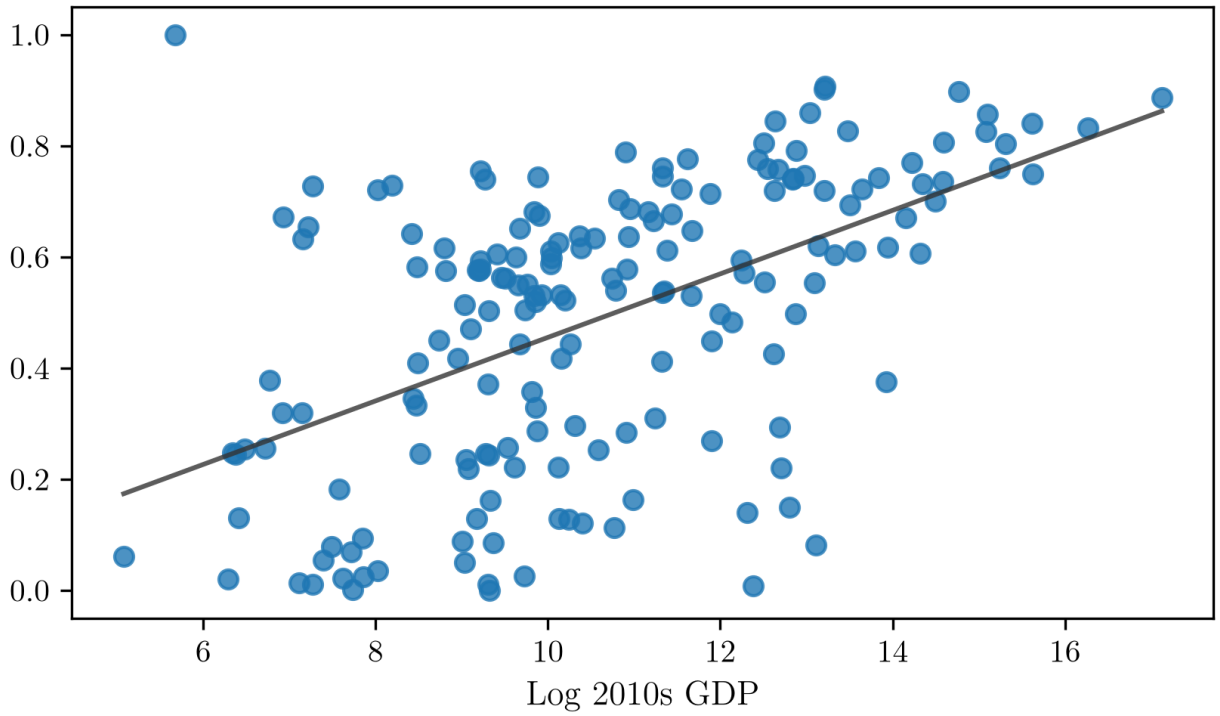
Note: The solid line shows a linear fit.

Figure 26: Welfare change (percent) in 1910s climate counterfactual across 2010s GDP per capita



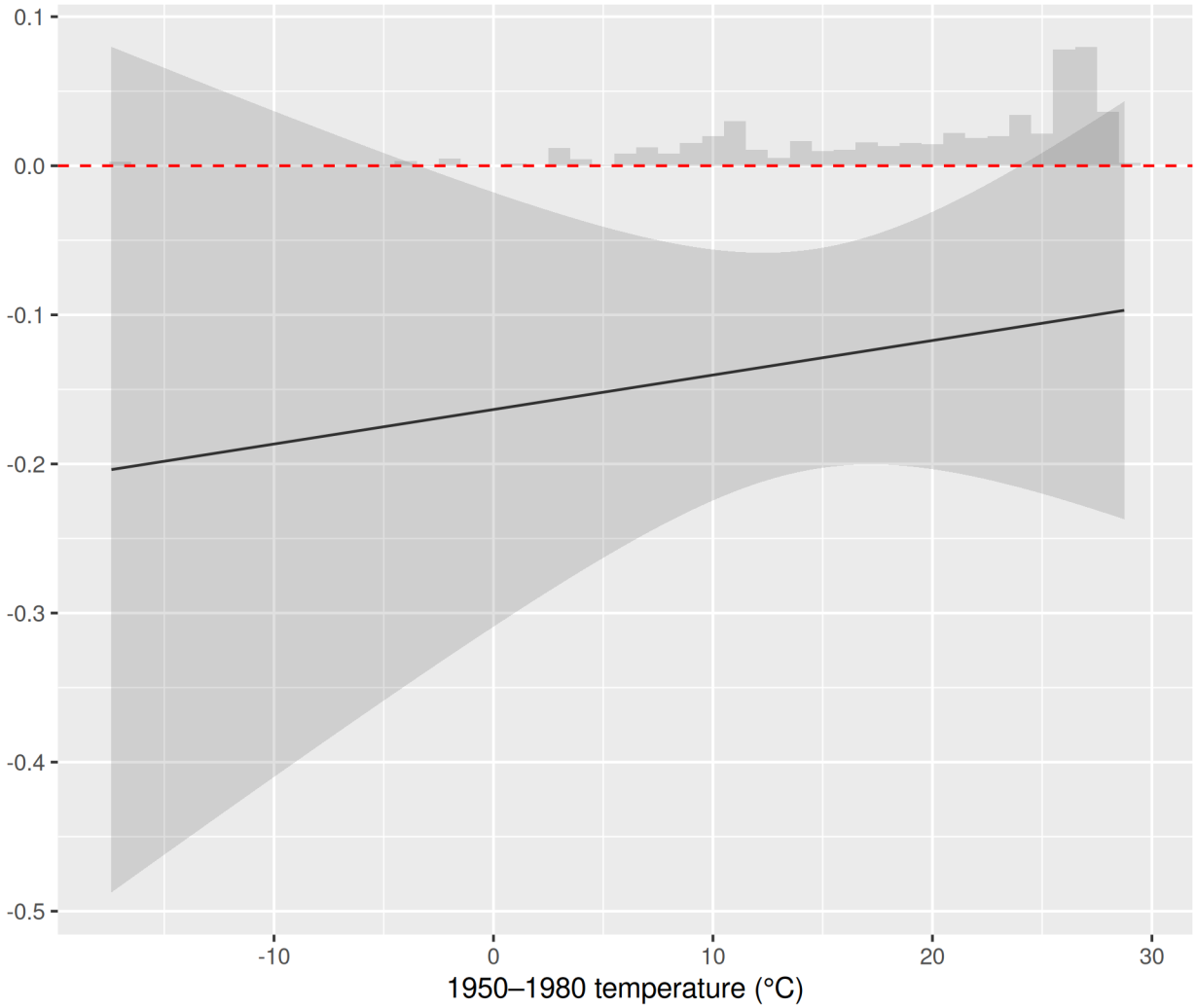
Note: The solid line shows a linear fit.

Figure 27: 2010s own trade share across 2010s GDP



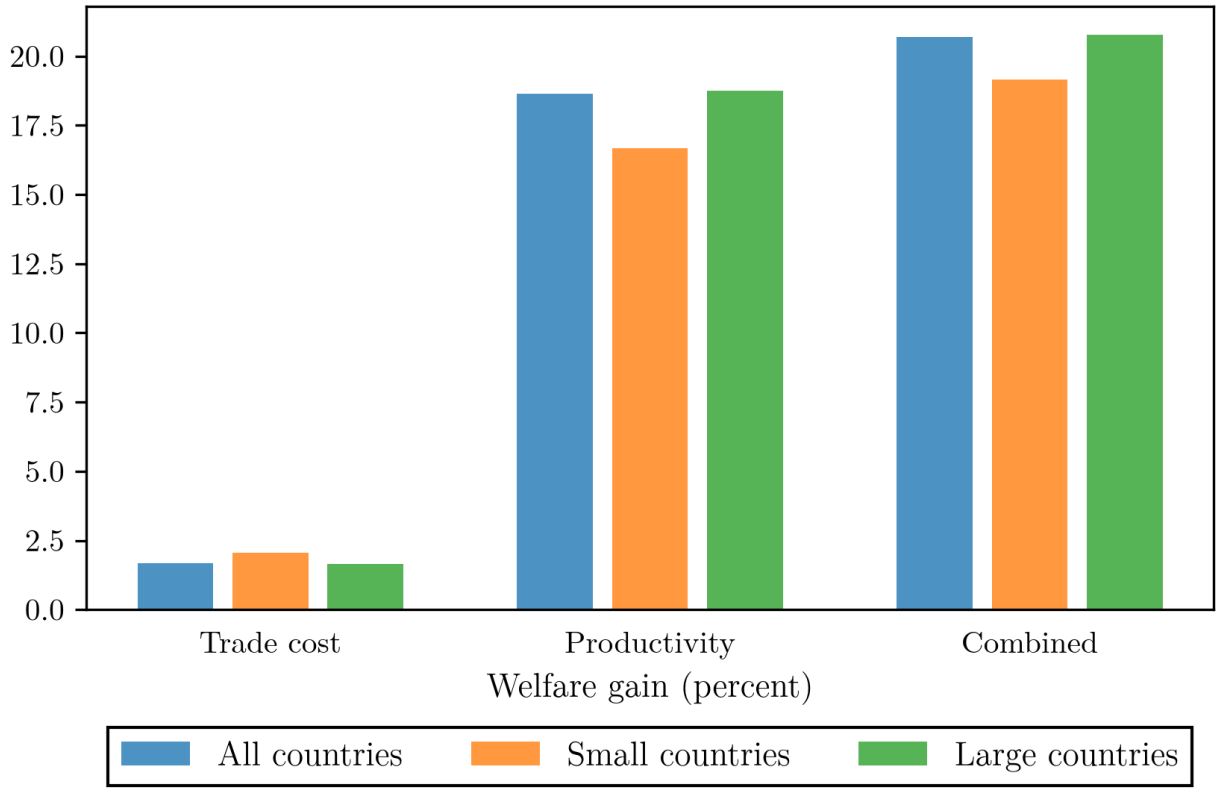
Note: The solid line shows a linear fit.

Figure 28: Marginal effect of decadal temperature changes on log GDP per capita



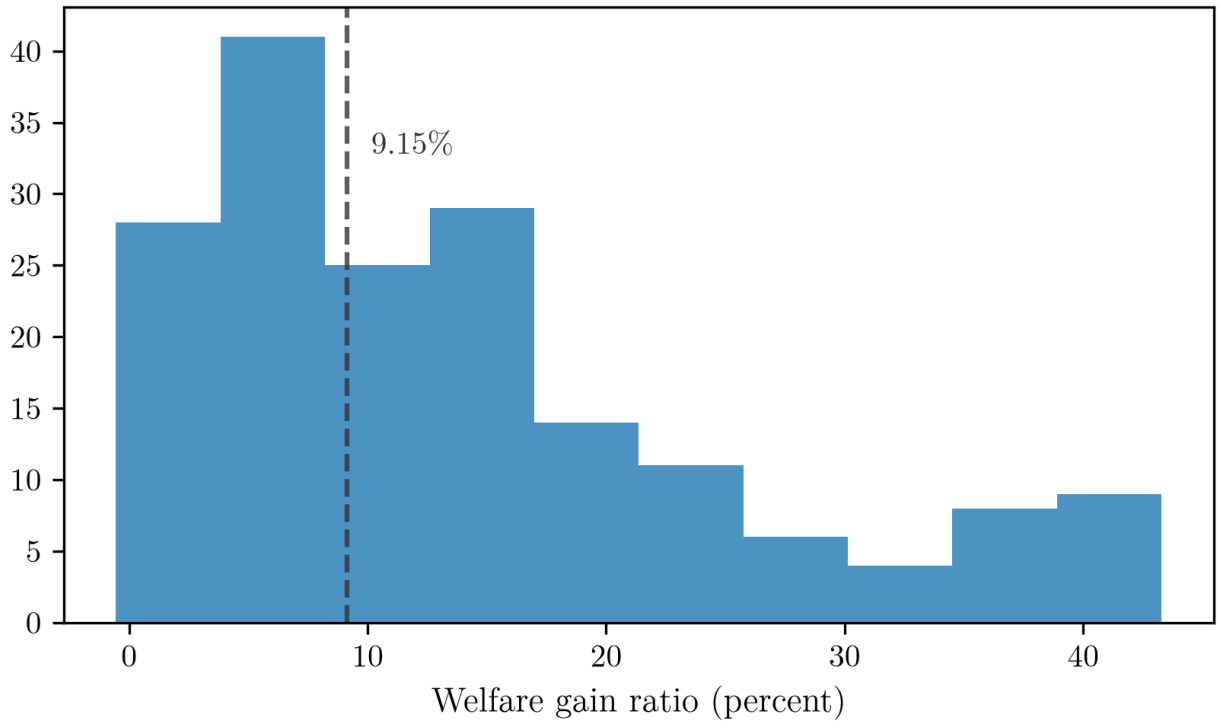
Note: Results are from a regression of log change in decadal average GDP per capita on decadal temperature changes and country fixed effects. The effect of temperature varies with countries' 1950–1980 average temperature. This graph shows the marginal effect of temperature shocks across 1950–1980 average temperature. The graph also shows a histogram of 1950–1980 average temperature. Standard errors are clustered at the country level. The shaded area indicates a 90 percent confidence interval.

Figure 29: Population-weighted average welfare gains (percent) across different scenarios for 1910s climate counterfactual



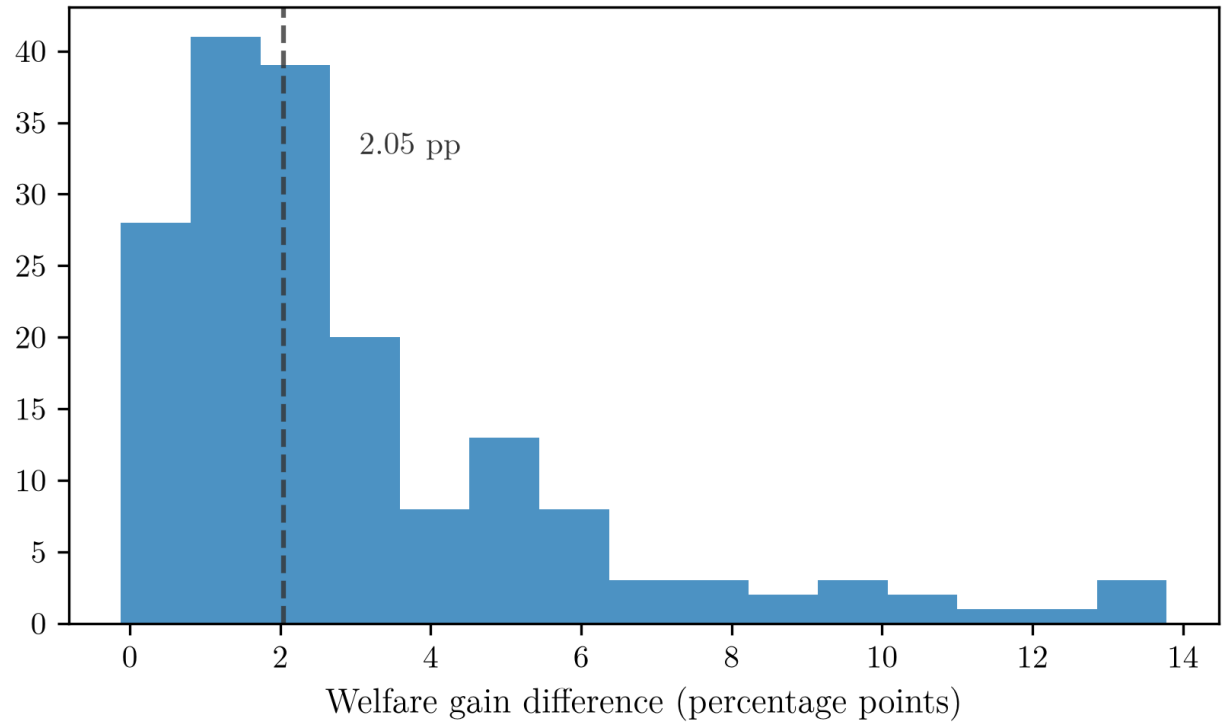
Note: The figure shows population-weighted average welfare gains under each scenario. *Trade cost* undoes the impact of climate change on trade cost. *Productivity* undoes the impact of climate change on productivity. *Combined* undoes both. *All countries* shows the average for all countries in the data. *Small economies* shows the average for countries with below median 2010s GDP. *Large economies* shows the average for countries with above median 2010s GDP.

Figure 30: Additional welfare gains from combined trade cost and productivity change vs. productivity change alone for 1910s climate counterfactual — welfare gain ratio



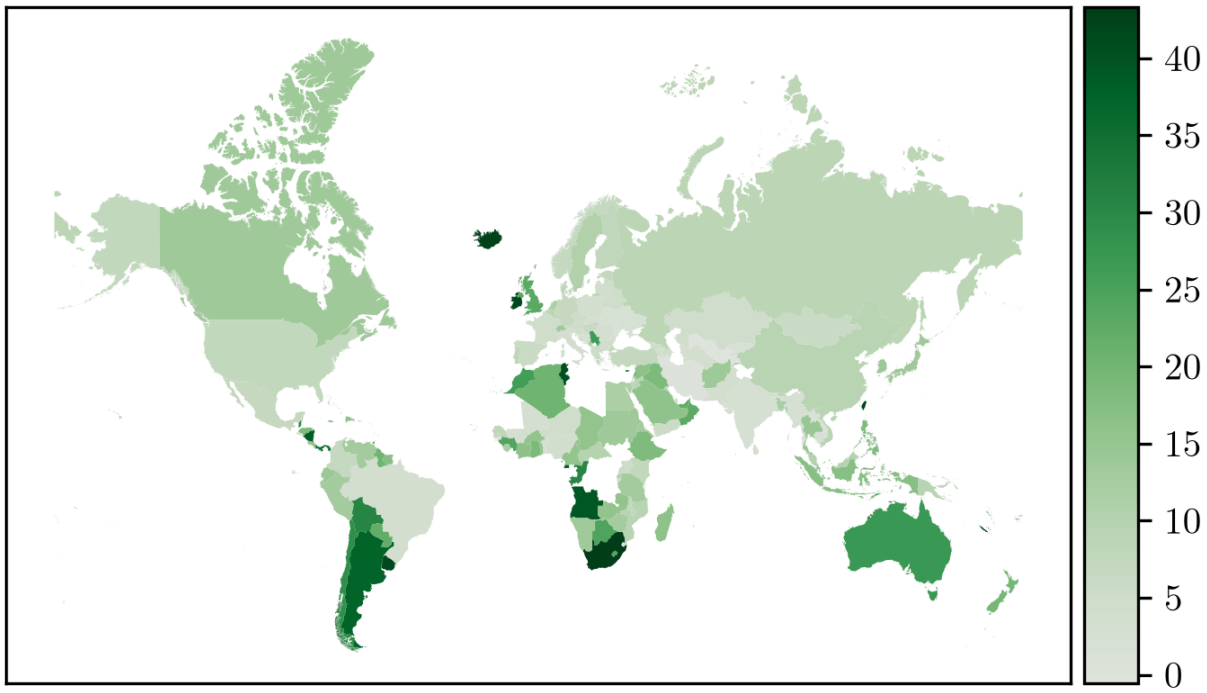
Note: This graph shows a histogram of the welfare gain ratio between the trade cost and productivity and the productivity only counterfactual. The welfare gain ratio is the welfare change from undoing climate change impacts on both productivity and trade networks divided by the welfare change from only undoing the impact on productivity. A welfare gain ratio of 10 percent, for example, means that welfare gains from undoing both effects are 10 percent larger than those from only undoing productivity effects. The dashed line indicates the population-weighted average of the welfare gain ratio.

Figure 31: Additional welfare gains from combined trade cost and productivity change vs. productivity change alone for 1910s climate counterfactual — welfare gain difference



Note: This graph shows a histogram of the welfare gain difference between the trade cost and productivity and the productivity only counterfactual. The welfare gain difference is the welfare change from undoing climate change impacts on both productivity and trade networks minus the welfare change from only undoing its impact on productivity. A welfare gain difference of 10 percentage points, for example, means that welfare gains from undoing both effects are 10 percentage points larger than those from only undoing productivity effects. The dashed line indicates the population-weighted average of the welfare gain difference.

Figure 32: Additional welfare gains from combined trade cost and productivity change vs. productivity change alone for 1910s climate counterfactual across countries



Note: This map shows the global distribution of the welfare gain ratio between the trade cost and productivity and the productivity only counterfactual. The welfare gain ratio is the welfare change from undoing climate change impacts on both productivity and trade networks divided by the welfare change from only undoing the impact on productivity. A welfare gain ratio of 10 percent, for example, means that welfare gains from undoing both effects are 10 percent larger than those from only undoing productivity effects.

Appendix A Additional tables

Table 5: Population-weighted summary statistics for welfare change (percent) across decades

Statistic	1880s	1890s	1900s	1910s	1920s	1930s	1940s	1950s	1960s	1970s	1980s	1990s	2000s
Mean	1.953	1.778	1.767	1.680	1.411	1.254	1.149	1.215	1.261	1.232	1.010	0.528	0.172
p_5	0.323	0.308	0.280	0.260	0.218	0.148	0.178	0.182	0.225	0.222	0.160	0.083	0.029
p_{10}	0.323	0.308	0.280	0.260	0.218	0.148	0.178	0.182	0.225	0.222	0.160	0.083	0.034
p_{25}	0.511	0.446	0.471	0.438	0.371	0.347	0.325	0.343	0.353	0.333	0.303	0.147	0.035
p_{50}	1.949	1.752	1.652	1.572	1.406	1.039	1.207	1.147	1.278	1.212	1.026	0.492	0.161
p_{75}	2.272	2.163	2.172	2.108	1.690	1.698	1.362	1.503	1.562	1.486	1.249	0.594	0.184
p_{90}	3.247	2.867	2.987	2.945	2.471	2.102	1.918	2.144	2.141	2.234	1.693	0.979	0.354
p_{95}	4.902	4.375	4.368	4.155	3.601	2.945	3.022	3.007	3.185	3.147	2.699	1.293	0.478

Note: The table summarizes the estimated percent change in welfare under climate change counterfactuals for each decade. *Mean* reports the average welfare change for each decade, while p_x reports the xth percentile of welfare changes for each decade. The mean and percentiles use 2010s population as weights.

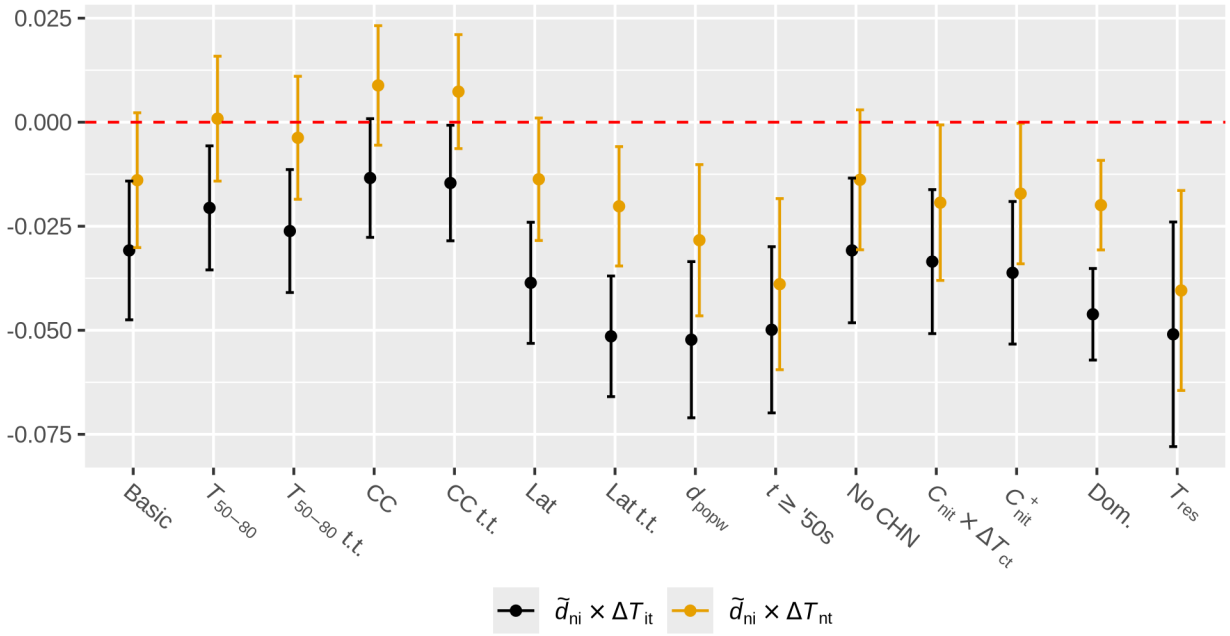
Table 6: Summary statistics for welfare change (percent) across decades

Statistic	1880s	1890s	1900s	1910s	1920s	1930s	1940s	1950s	1960s	1970s	1980s	1990s	2000s
Mean	2.960	2.630	2.622	2.477	2.120	1.805	1.782	1.819	1.938	1.877	1.579	0.847	0.266
p_5	0.332	0.247	0.248	0.248	0.183	0.137	0.175	0.128	0.175	0.184	0.180	0.082	-0.028
p_{10}	0.515	0.436	0.441	0.419	0.361	0.322	0.326	0.285	0.338	0.318	0.282	0.145	0.008
p_{25}	1.141	0.976	1.019	0.919	0.874	0.672	0.725	0.723	0.715	0.711	0.591	0.321	0.062
p_{50}	2.171	1.868	1.829	1.717	1.554	1.253	1.272	1.263	1.437	1.336	1.149	0.564	0.165
p_{75}	3.737	3.188	3.230	3.032	2.679	2.363	2.153	2.353	2.501	2.318	1.944	1.049	0.353
p_{90}	6.160	5.666	5.453	5.277	4.505	4.041	3.906	4.083	3.913	4.143	3.292	1.915	0.694
p_{95}	9.112	8.340	8.223	8.079	6.609	5.469	5.413	5.933	6.218	6.151	4.905	2.518	0.950

Note: The table summarizes the estimated percent change in welfare under climate change counterfactuals for each decade. *Mean* reports the average welfare change for each decade, while p_x reports the xth percentile of welfare changes for each decade.

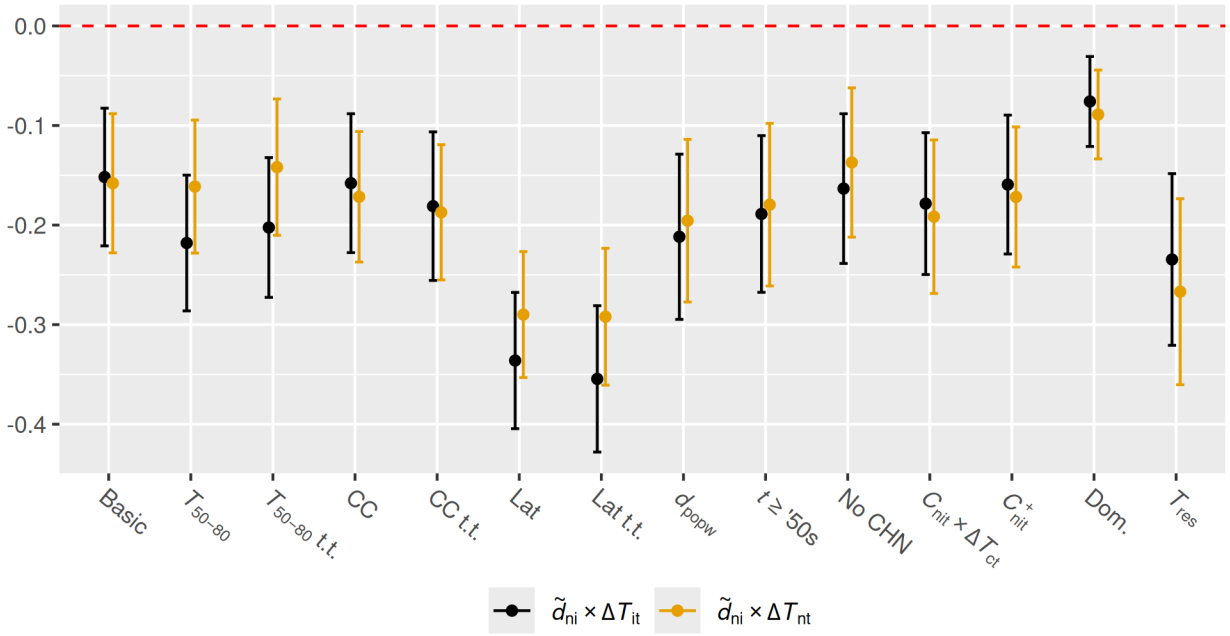
Appendix B Additional figures

Figure 33: Coefficients on temperature interactions across specifications, using half-decades (five-year periods) instead of decades



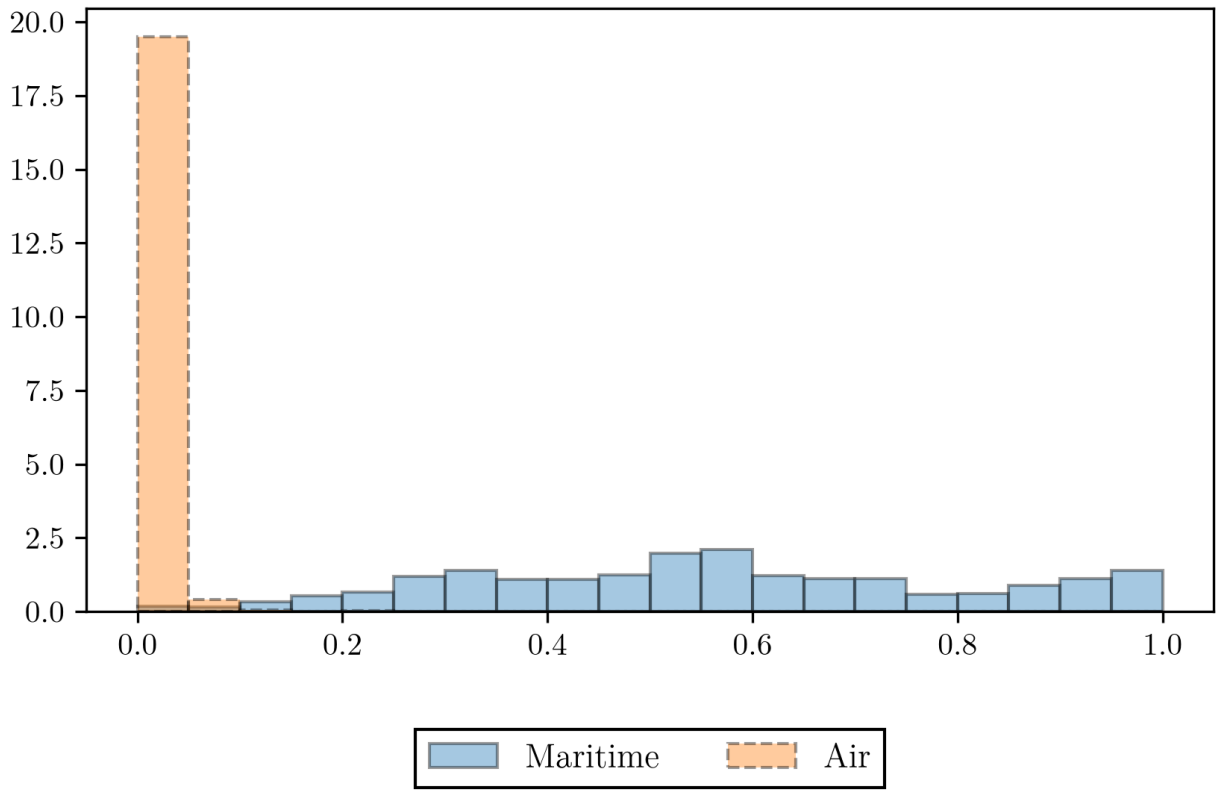
Note: The outcome are half-decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. $\tilde{d}_{ni} \equiv \log(d_{ni})$ is the log of the great circle distance d_{ni} between the origin and destination countries in km. ΔT_{ct} is the change in half-decadal mean temperature in country c between half-decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. C_{nit} contains a common language indicator, sharing a border indicator and two indicators for current and past colonial relationships, taking half-decade means for all variables within each origin-destination pair. Half-decades t are the half-decades from 1820 to 2020. $T_{50\text{-}80}$ is country c's average temperature between the 1950s and 1980s. CC is country c's average temperature change (i.e., climate change) between the 1900s and 2000s decades. *Lat.* is country c's latitude. These models include separate trade cost levels and, for the *t.t.* (time trend) version, trade cost time trends for each decile of those variables. d_{popw} uses population-weighted great circle distance instead of the unweighted measure (this is missing for countries which no longer exist, so the observation count is lower). $t \geq$ *'50s* only uses half-decades from the 1950s onwards. *No CHN* drops China from the estimation sample. $C_{nit} \times \Delta T_{ct}$ interacts temperatures not only with distance, but also with the variables included in C_{nit}. C^{+}_{nit} additionally includes indicators for both countries being in the EU, OECD and GATT. *Dom.* proxies for domestic trade by subtracting total exports from GDP and allows for the effect of distance by half-decade to vary between international and domestic trade, capturing globalization (Yotov, 2012). T_{res} uses temperature residuals obtained from an AR(3) estimation with heterogeneity by 1950–1980 average temperature, similar in spirit to Ramey, Nath, & Klenow (2025). Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 34: Coefficients on temperature interactions across specifications, using three-decades (30-year periods) instead of decades



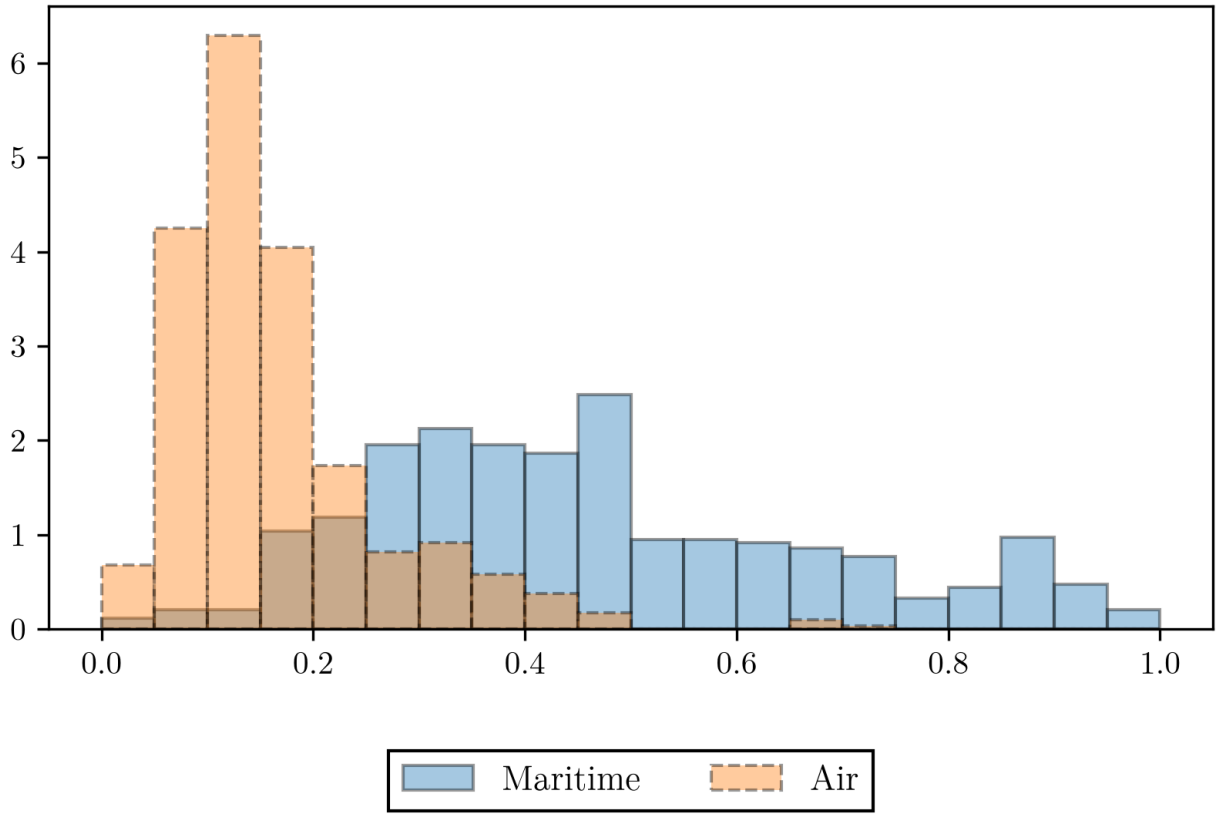
Note: The outcome are three-decade-level average trade flows from country i to country n, winsorized at the 99th percentile. The estimation uses pseudo-Poisson maximum likelihood (PPML) to accommodate zero trade flows. $\tilde{d}_{ni} \equiv \log(d_{ni})$ is the log of the great circle distance d_{ni} between the origin and destination countries in km. ΔT_{ct} is the change in three-decadal mean temperature in country c between three-decades t and $t-1$ in °C. Temperature changes are winsorized at the 1st and 99th percentiles. C_{nit} contains a common language indicator, sharing a border indicator and two indicators for current and past colonial relationships, taking three-decade means for all variables within each origin-destination pair. Three-decades t are the three-decades from 1820 to 2020. $T_{50\text{-}80}$ is country c's average temperature between the 1950s and 1980s. *CC* is country c's average temperature change (i.e., climate change) between the 1900s and 2000s decades. *Lat.* is country c's latitude. These models include separate trade cost levels and, for the *t.t.* (time trend) version, trade cost time trends for each decile of those variables. d_{popw} uses population-weighted great circle distance instead of the unweighted measure (this is missing for countries which no longer exist, so the observation count is lower). $t \geq$ *'50s* only uses three-decades from the 1950s onwards. *No CHN* drops China from the estimation sample. $C_{nit} \times \Delta T_{ct}$ interacts temperatures not only with distance, but also with the variables included in C_{nit}. C^{+}_{nit} additionally includes indicators for both countries being in the EU, OECD and GATT. *Dom.* proxies for domestic trade by subtracting total exports from GDP and allows for the effect of distance by three-decade to vary between international and domestic trade, capturing globalization (Yotov, 2012). T_{res} uses temperature residuals obtained from an AR(3) estimation with heterogeneity by 1950–1980 average temperature, similar in spirit to Ramey, Nath, & Klenow (2025). Standard errors are clustered by country pair. Vertical lines and whiskers indicate 90 percent confidence intervals.

Figure 35: Maritime and air trade shares across countries, 2016–2019, weight-weighted



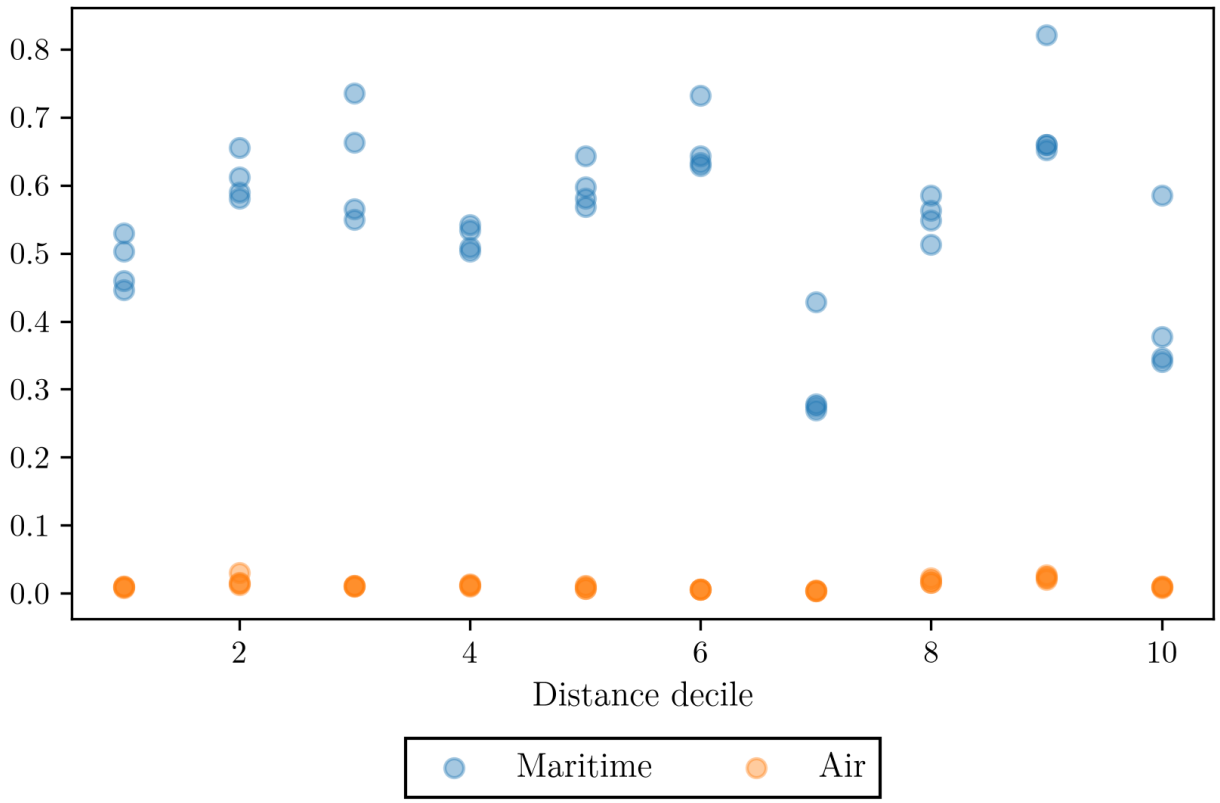
Note: This histogram shows the distribution of the weight-weighted share of maritime and air trade, i.e., what share of the total weight of goods traded each mode transports, across countries and years.

Figure 36: Maritime and air trade shares across countries, 2016–2019, value-weighted



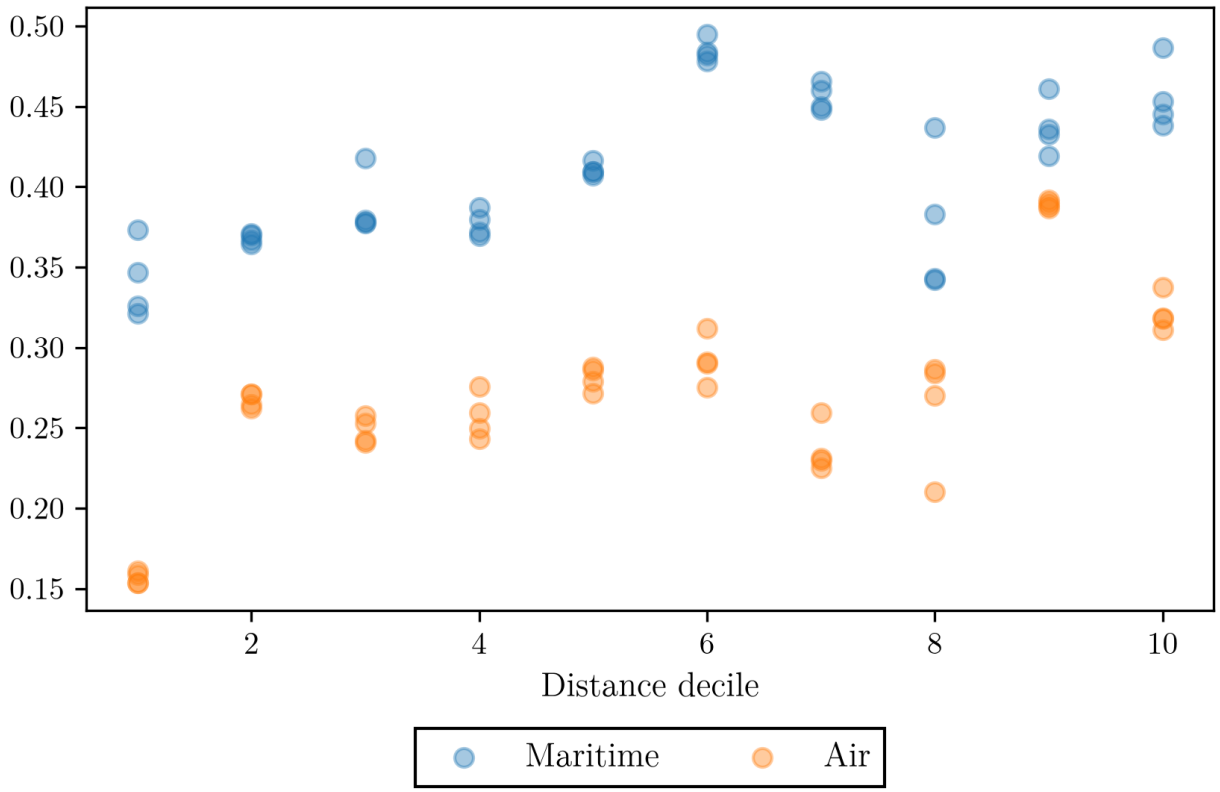
Note: This figure shows the distribution of the value-weighted share of maritime and air trade, i.e., what share of the total value of goods traded each mode transports, across countries and years.

Figure 37: Maritime and air trade shares across distance deciles, 2016–2019, weight-weighted



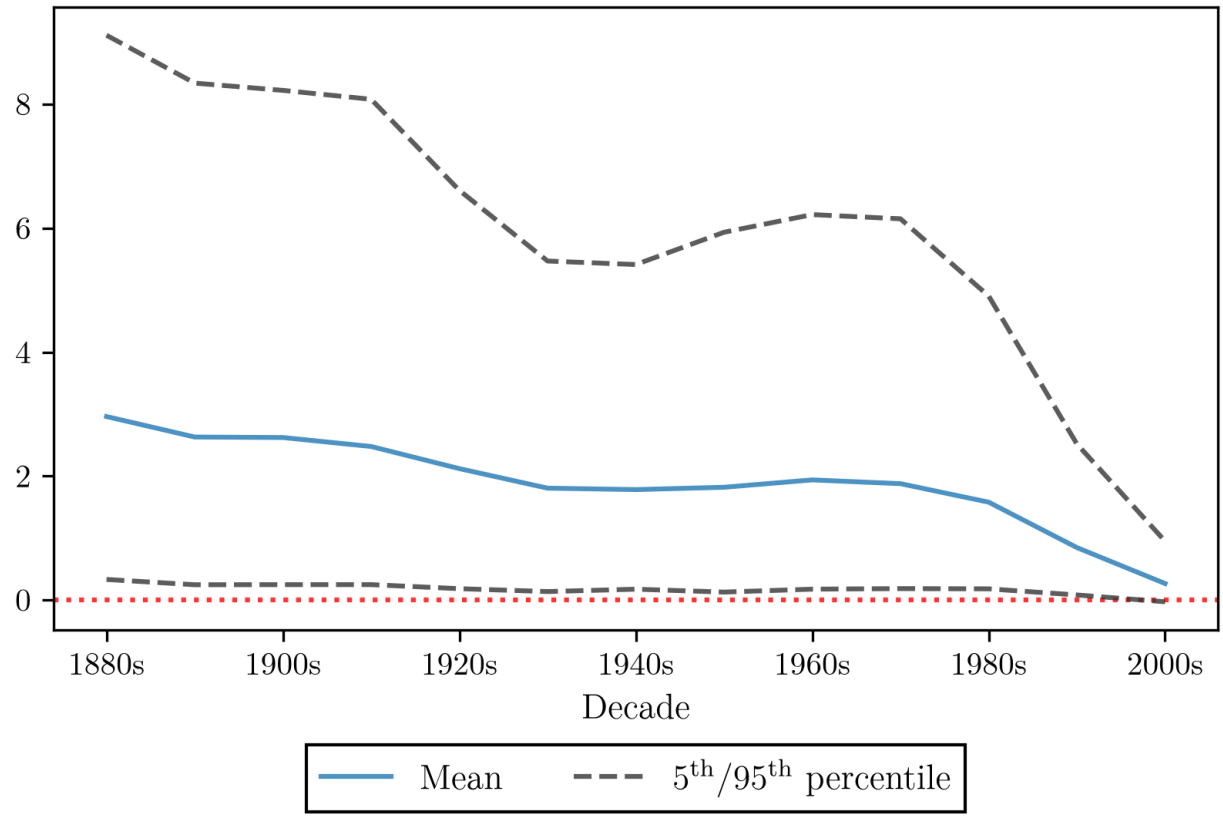
Note: This figure shows the weight-weighted share of maritime and air trade, i.e., what fraction of the total weight of goods traded each mode transports, across distance deciles. Each dot represents a different year. (Deciles are across distances for all trade connections, unweighted.)

Figure 38: Maritime and air trade shares across distance deciles, 2016–2019, value-weighted



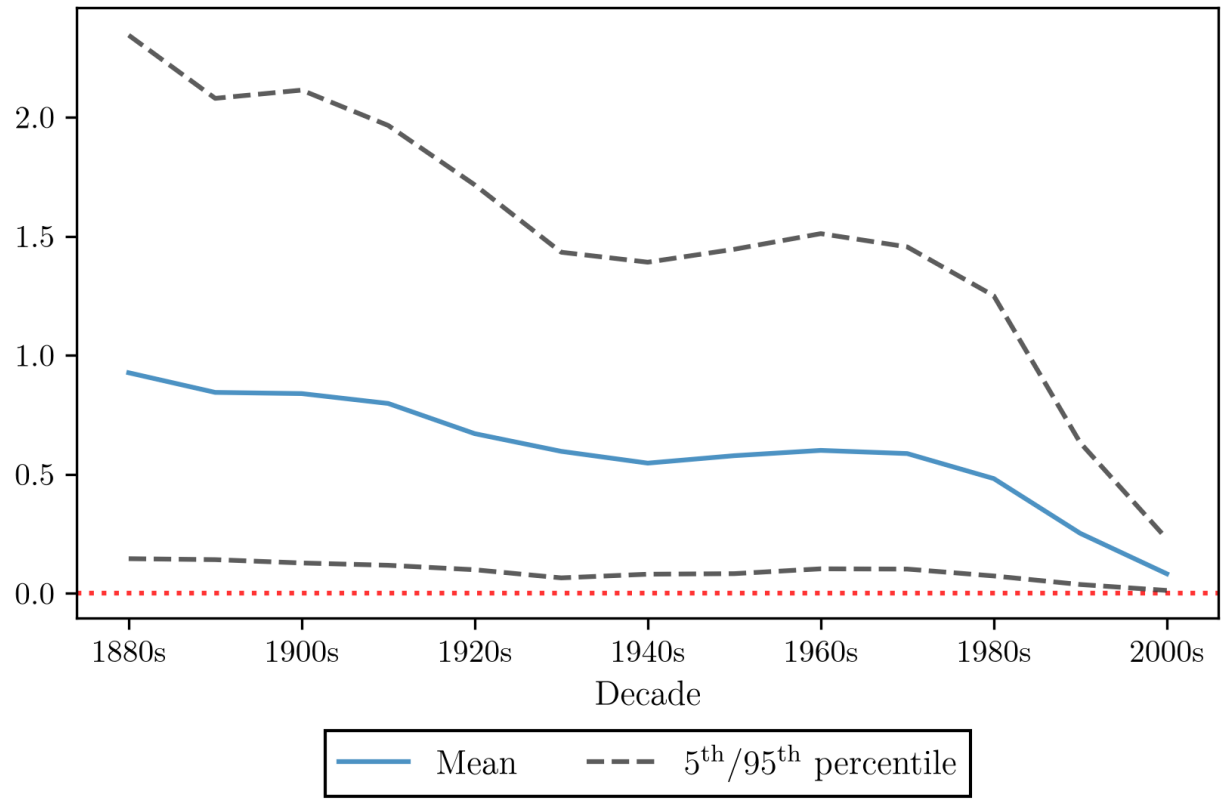
Note: This histogram shows the value-weighted share of maritime and air trade, i.e., what fraction of the total weight of goods traded each mode transports, across distance deciles. Each dot represents a different year. (Deciles are across distances for all trade connections, unweighted.)

Figure 39: Summary statistics for welfare change (percent) across decades



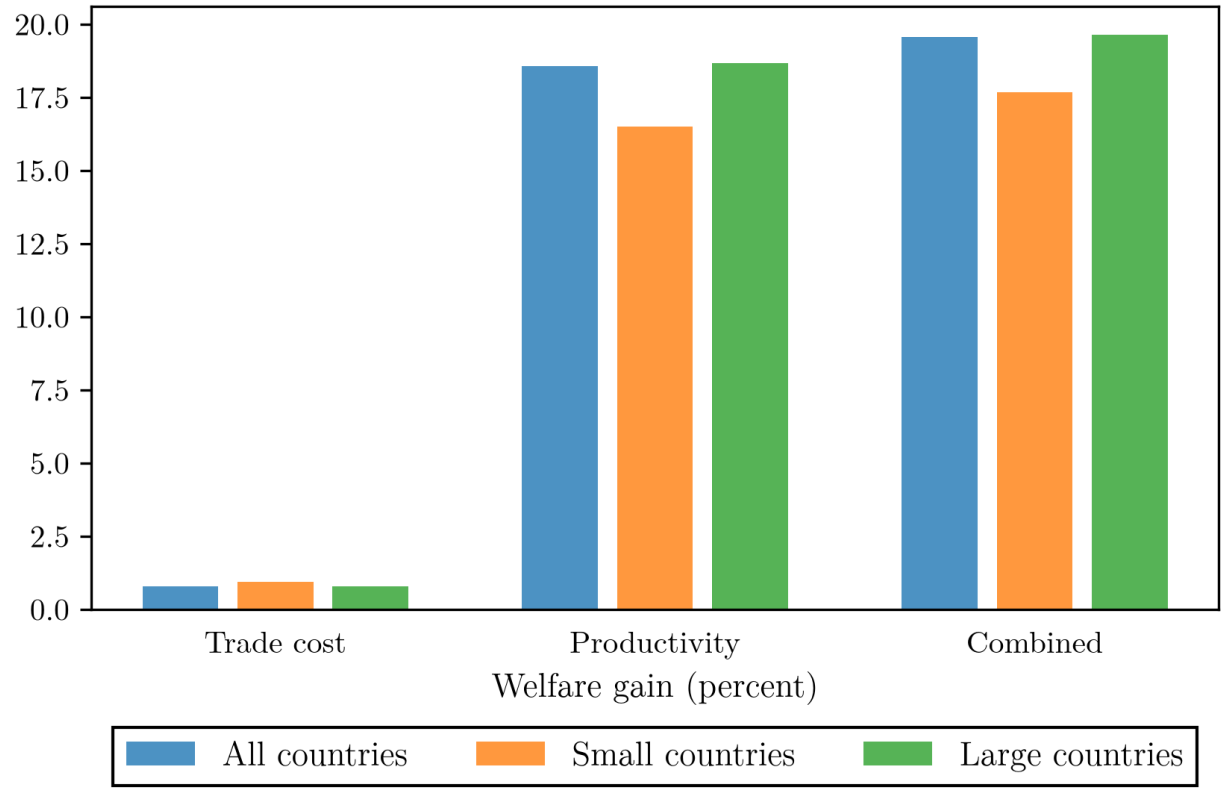
Note: The 5th and 95th percentile are percentiles of estimated welfare gains, not confidence intervals for the mean.

Figure 40: Population-weighted summary statistics for welfare change (percent) across decades, robustness check using doubled dispersion parameter $\theta = 8.28$



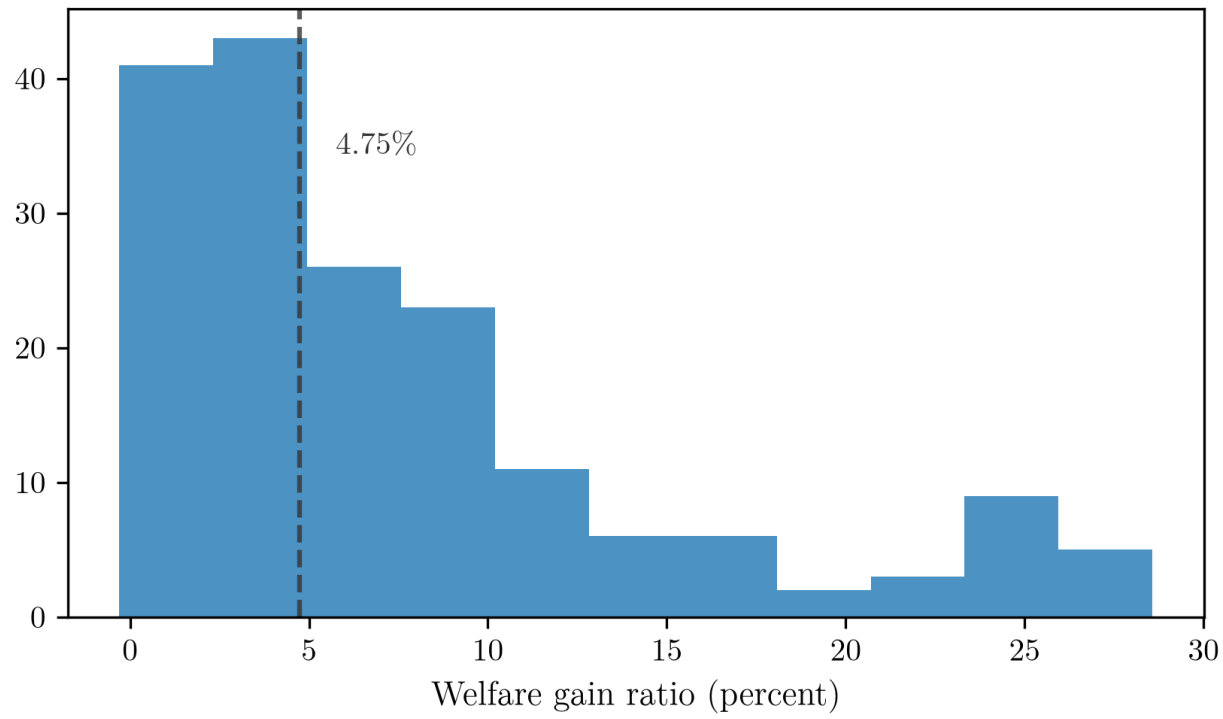
Note: This estimation uses an increased dispersion parameter $\theta = 8.28$ instead of the main model calibration of $\theta = 4$. The mean and percentiles use 2010s population as weights. The 5th and 95th percentile are percentiles of estimated welfare gains, not confidence intervals for the mean.

Figure 41: Population-weighted average welfare gains (percent) across different scenarios for 1910s climate counterfactual, robustness check using doubled dispersion parameter $\theta = 8.28$



Note: This estimation uses an increased dispersion parameter $\theta = 8.28$ instead of the main model calibration of $\theta = 4$. The figure shows population-weighted average welfare gains under each scenario. *Trade cost* undoes the impact of climate change on trade cost. *Productivity* undoes the impact of climate change on productivity. *Combined* undoes both. *All countries* shows the average for all countries in the data. *Small economies* shows the average for countries with below median 2010s GDP. *Large economies* shows the average for countries with above median 2010s GDP.

Figure 42: Additional welfare gains from combined trade cost and productivity change vs. productivity change alone for 1910s climate counterfactual — welfare gain ratio, robustness check using doubled dispersion parameter $\theta = 8.28$



Note: This estimation uses an increased dispersion parameter $\theta = 8.28$ instead of the main model calibration of $\theta = 4$. This graph shows a histogram of the welfare gain ratio between the trade cost and productivity and the productivity only counterfactual. The welfare gain ratio is the welfare change from undoing climate change impacts on both productivity and trade networks divided by the welfare change from only undoing the impact on productivity. A welfare gain ratio of 10 percent, for example, means that welfare gains from undoing both effects are 10 percent larger than those from only undoing productivity effects. The dashed line indicates the population-weighted average of the welfare gain ratio.

Appendix C Simulating the impact of misspecification

This appendix contains simulations to highlight that misspecification of level trade costs does not lead to a biased estimate of the impact of trade cost shocks, as long as the underlying temperature shocks are exogenous. This misspecification could be due to heterogeneous effects of distance or due to generic endogeneity of distance. I simulate data from the data generating process

$$\log(X)_{nit} = \gamma_{it} + \xi_{nt} - \alpha_{ni}\tilde{d}_{ni} - \alpha_{ni}\tilde{d}_{ni}\Delta T_{it} - \alpha_{ni}\tilde{d}_{ni}\Delta T_{nt} + \varepsilon_{nit}$$

Note that the coefficient α_{ni} on distance allows for heterogeneous effects of distance across country pairs. I draw fixed effects γ_{it} and ξ_{nt}, log distance $\tilde{d}_{ni}$ and origin and destination temperature shocks ΔT_{it} and ΔT_{nt} i.i.d. from $\mathcal{N}(0,1)$, where $\mathcal{N}$ is the normal distribution. I then estimate

$$\mathbb{E}(\log(X)_{nit}) = \gamma_{it} + \xi_{nt} + \alpha\tilde{d}_{ni} + \delta_1\tilde{d}_{ni}\Delta T_{it} + \delta_2\tilde{d}_{ni}\Delta T_{nt}$$

This estimation is similar to what I do in the paper (though I can now use OLS to estimate the model since the simulated trade flows never contain zeros). Note that this estimator ignores the potentially heterogeneous effects of distance. I simulate three versions of the model:

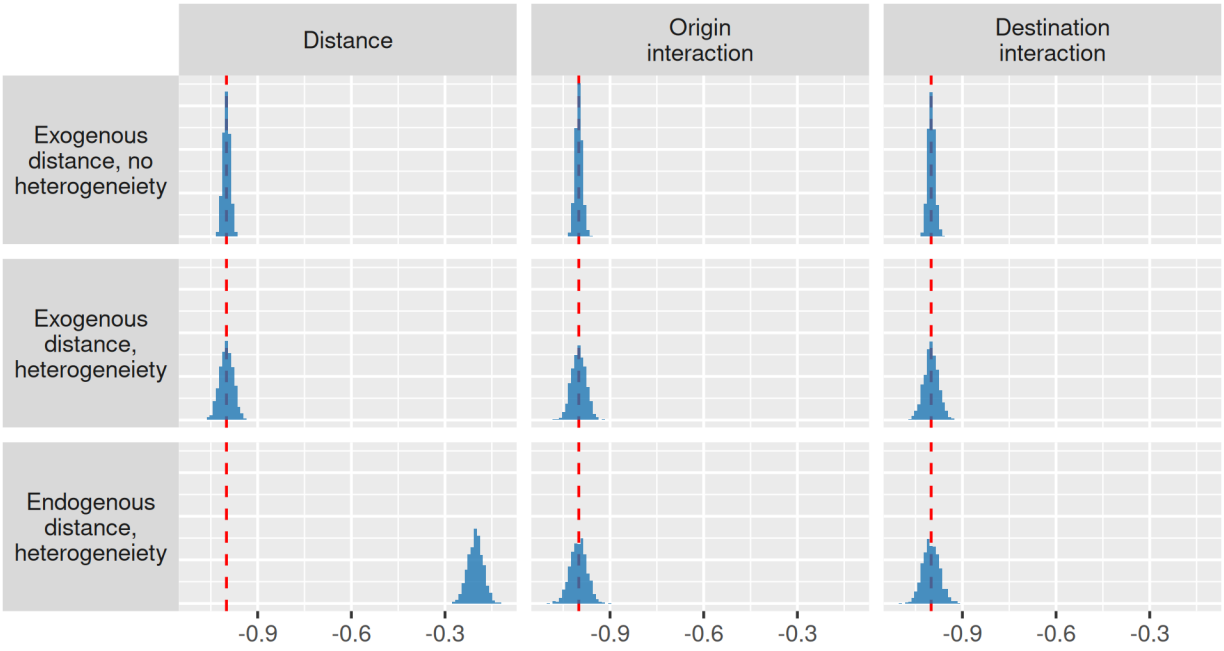
1. The first sets $\alpha_{ni} = 1$ for all country pairs and draws errors $\varepsilon_{nit} \sim \mathcal{N}(0,1)$. With this setup, the estimations thus use the correct functional form.

2. The second version draws errors as before, but draws $\alpha_{ni} \sim \mathcal{U}(0.5, 1.5)$, where $\mathcal{U}$ is the uniform distribution. This adds heterogeneous effects of distance, which the estimator ignores.

3. The final version keeps these heterogeneous effects of distance and additionally generates the error as $\varepsilon_{nit} = u_{nit} + |v_{nit}|\,\tilde{d}_{nit}$, where u_{nit} and v_{nit} are drawn from $\mathcal{N}(0,1)$. This makes distance endogenous, which the estimator again ignores.

Figure 43 shows the results. I present distributions of estimates across 1,000 simulations with 30 countries and ten time periods each. Each panel shows a histogram of coefficient estimates across simulations. The dashed lines indicate the true coefficient of interest. This is -1 in all models. For the first setup, without heterogeneous effects, this is simply the true impact of all three variables. For the two setups with heterogeneity, it is the true average effect across all country pairs (since $\mathbb{E}(\alpha_{ni}) = 1$). Each column shows estimates for a different coefficient: *distance* ($\hat{\alpha}$), *origin interaction* ($\hat{\delta}_1$) and *destination interaction* ($\hat{\delta}_2$). Each row shows a different setup: *Exogenous distance, no*

Figure 43: Histograms of coefficient estimates across simulations



Note: This graph shows the distribution of estimated coefficients on three variables — log distance, the interaction between log distance and origin temperature shocks, and the analogous interaction for destination shocks — across 1,000 simulations with 30 countries and ten time periods each. Each column presents one variable. Each row presents results for a different data generating process. *Exogenous distance* means distance is not correlated with the error term. *No heterogeneity* indicates that all country pairs have the same effect of distance, whereas *heterogeneity* means all country pairs have different coefficients for distance. *Endogenous distance* means distance is correlated with the error term. Shocks are always independent of distance and the error term. The dashed lines indicate the true coefficients (or true average coefficients, for estimations with heterogeneity) used in the data generating process.

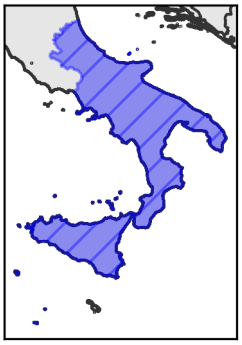
heterogeneity is the first setup described above where distance is exogenous and does not have heterogeneous impacts. The estimator is therefore using the correct functional form. *Exogenous distance, heterogeneity* has distance as exogenous but with heterogeneous effects across country pairs. Finally, *Endogenous distance, heterogeneity* has distance correlated with the error term and features heterogeneous impacts of distance.

As expected, the coefficient on distance becomes biased once I make distance endogenous. Note, however, that the coefficients on the interactions of interest remain centered on the correct true effect, regardless of misspecification. With heterogeneous distance effects, the interactions still capture the correct average effect (though estimates become noisier). Even with endogenous distance, where the distance coefficient itself becomes inconsistent, the interactions remain consistent estimators.

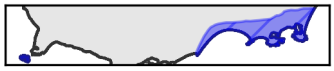
Appendix D Maps of added countries

This appendix shows maps of all TRADHIST reporting units that are not currently in the GADM database. These maps were created by matching existing GADM administrative units at different levels to historical maps of each reporting unit. Map titles are CEPII's additional ISO codes for each reporting unit as found in Table 4 of the TRADHIST documentation, available at https://www.cepii.fr/PDF_PUB/wp/2016/wp2016-14.pdf.

2SICIL



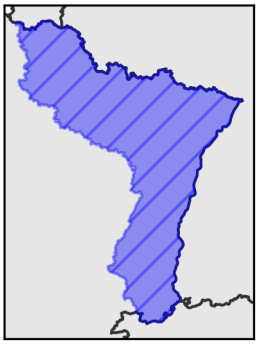
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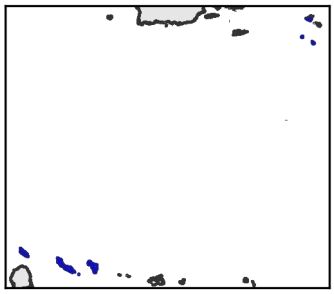
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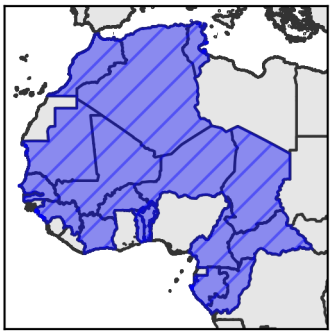
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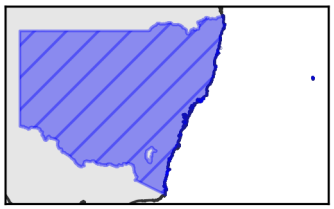
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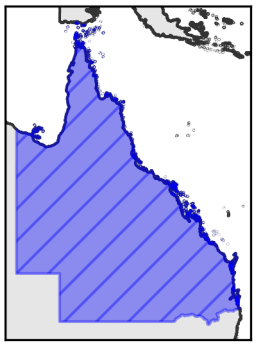
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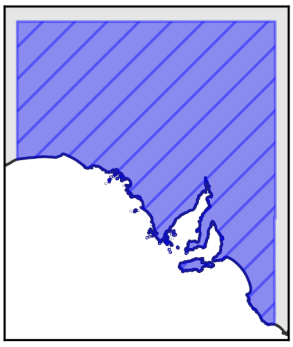
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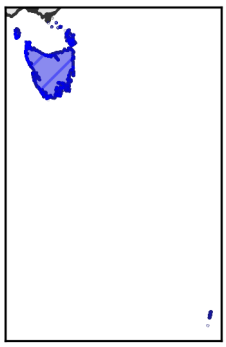
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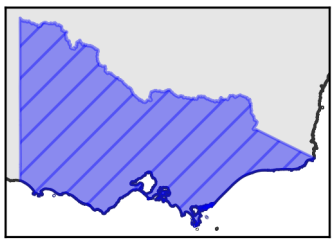
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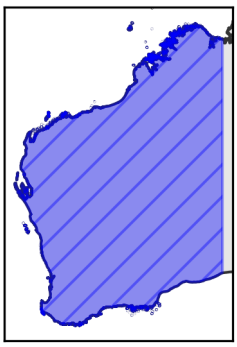
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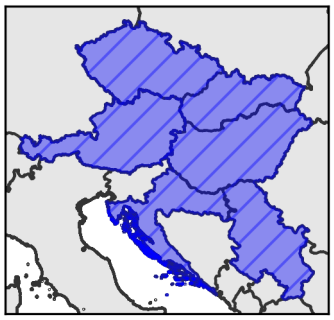
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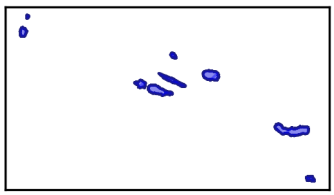
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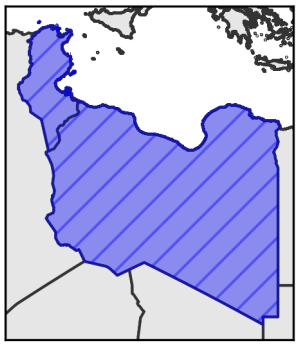
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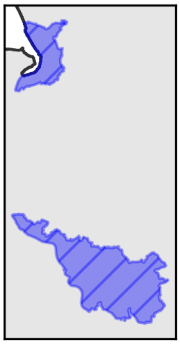
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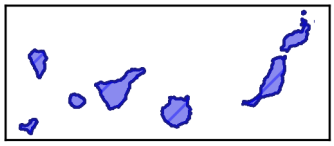
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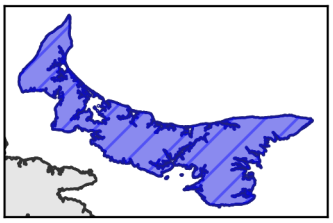
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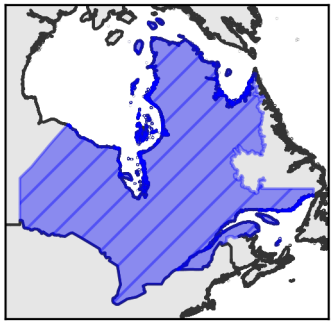
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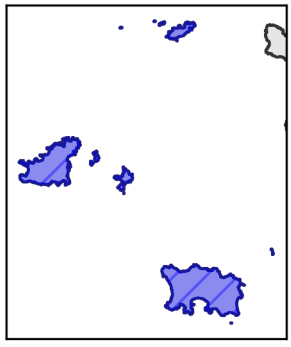
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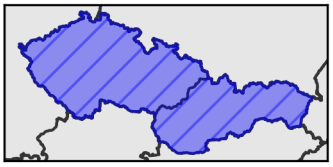
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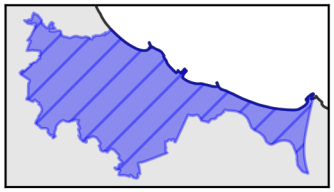
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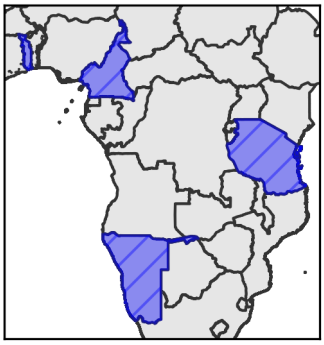
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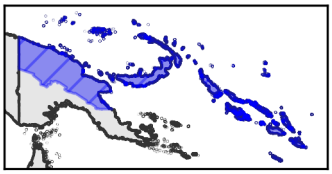
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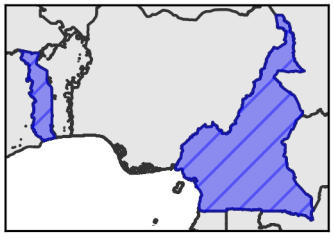
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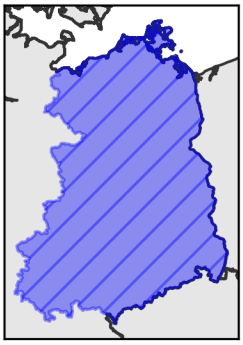
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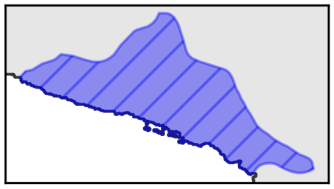
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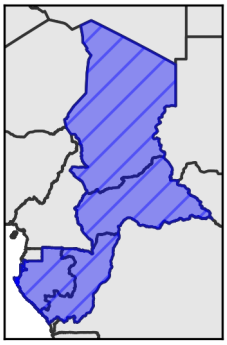
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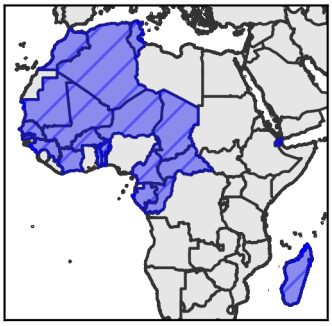
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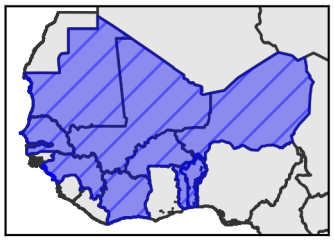
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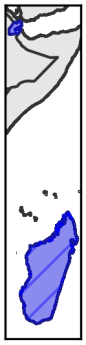
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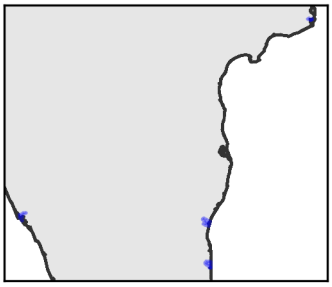
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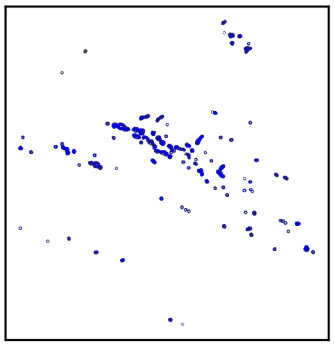
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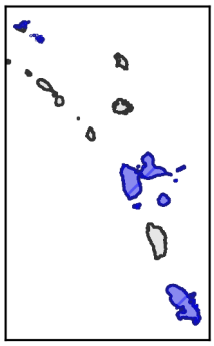
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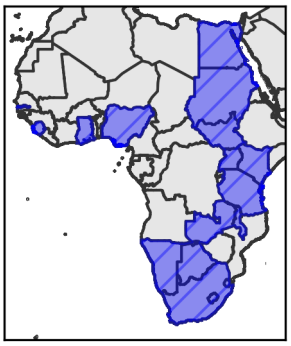
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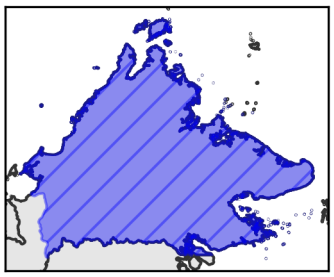
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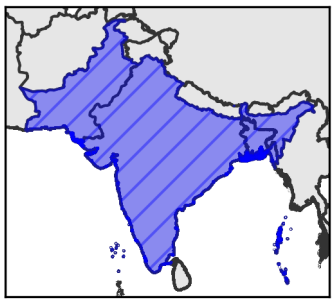
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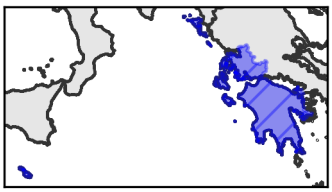
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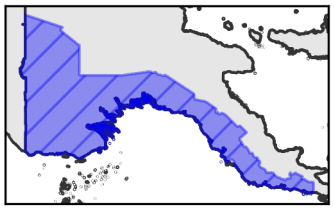
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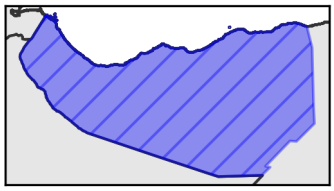
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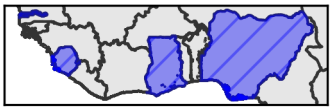
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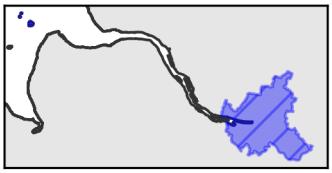
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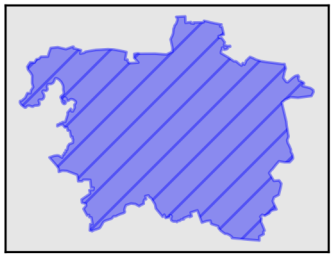
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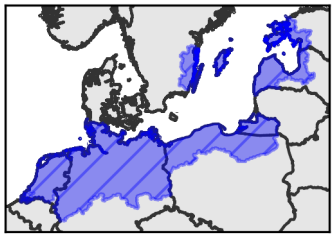
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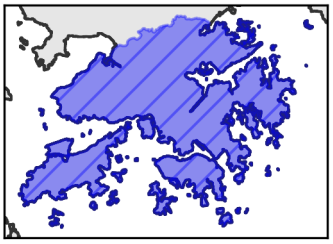
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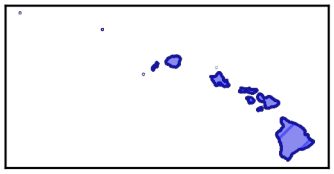
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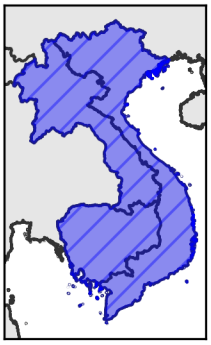
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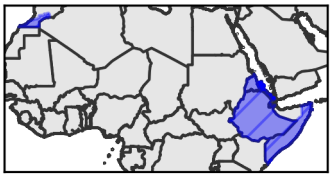
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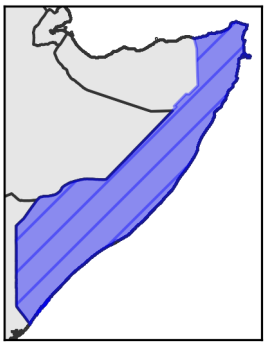
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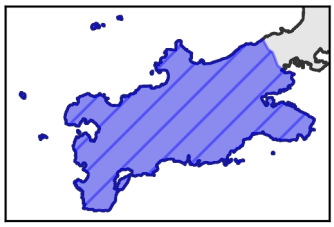
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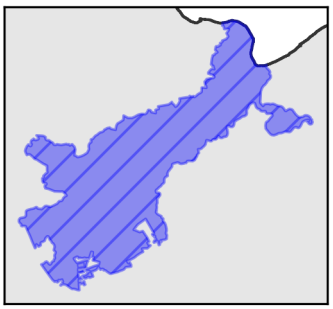
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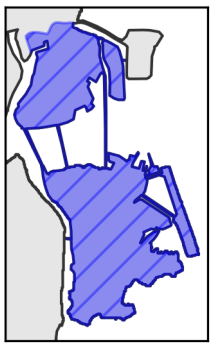
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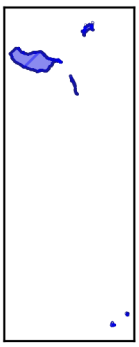
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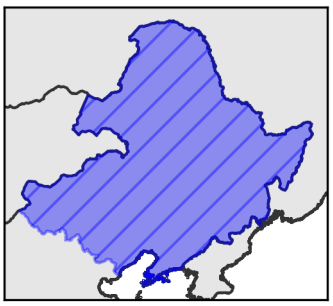
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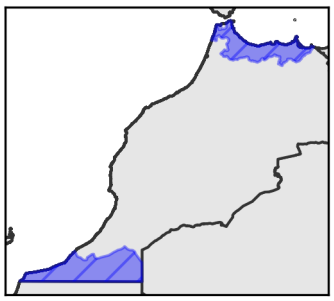
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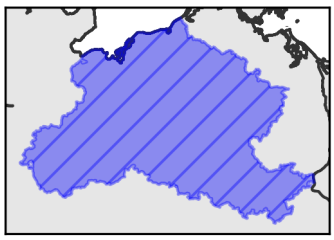
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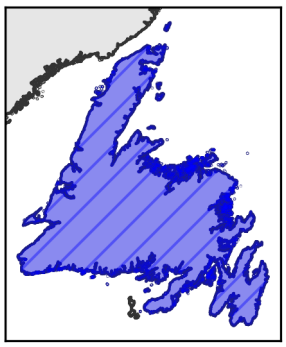
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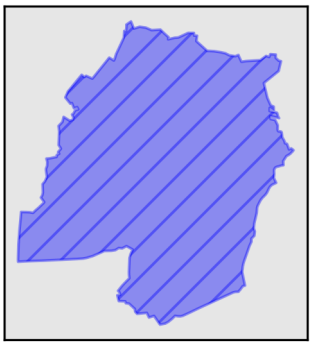
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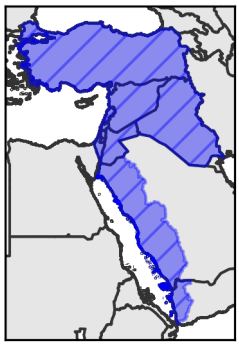
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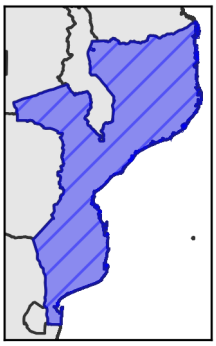
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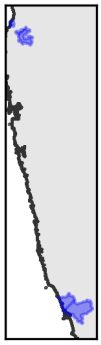
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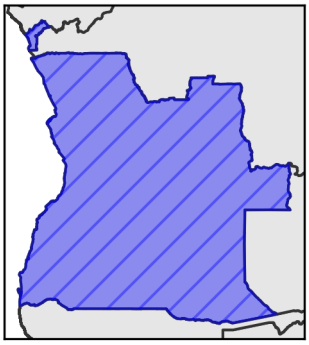
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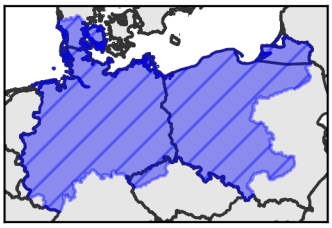
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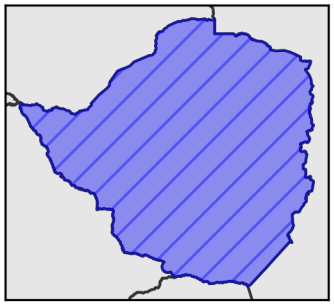
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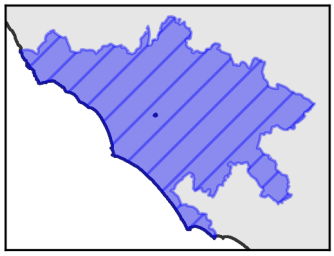
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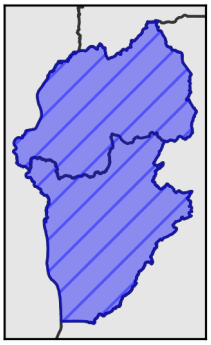
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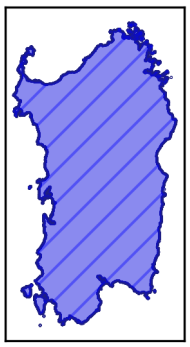
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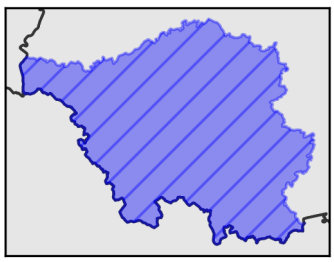
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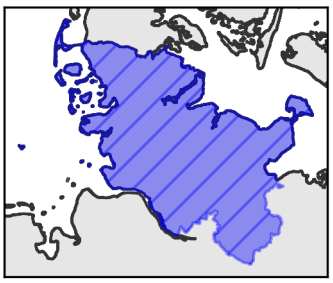
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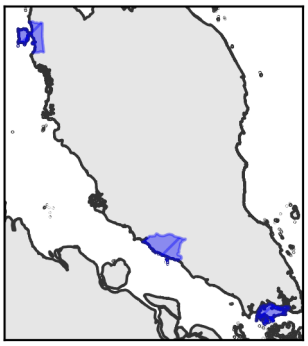
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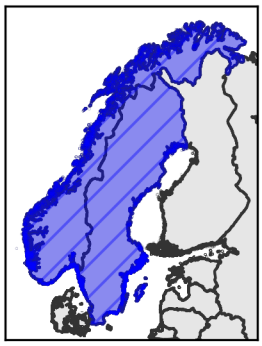
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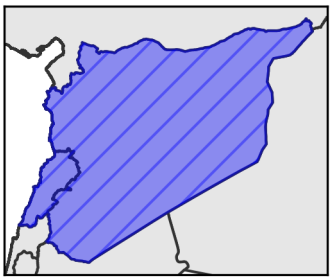
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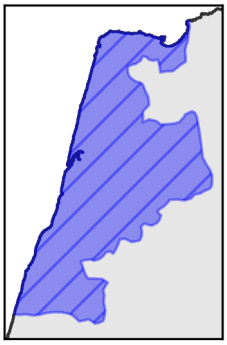
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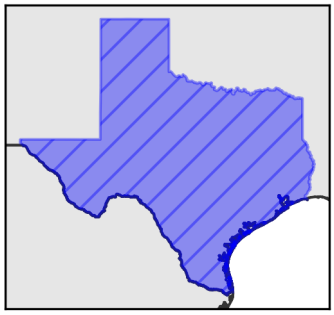
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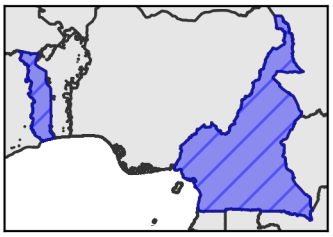
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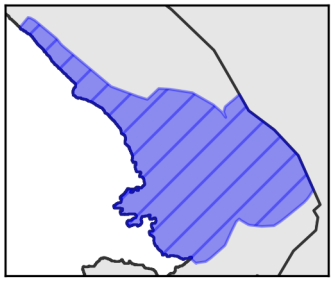
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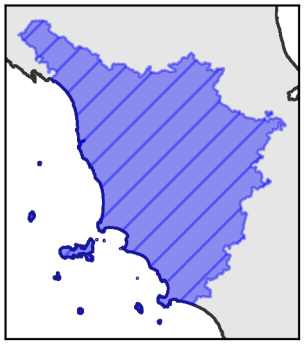
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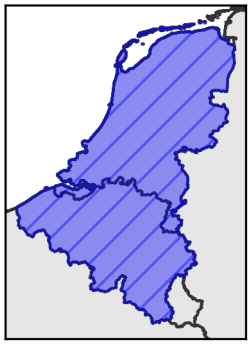
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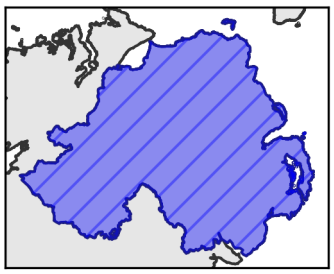
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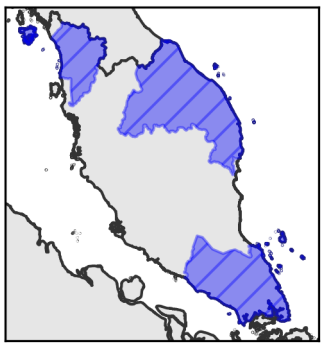
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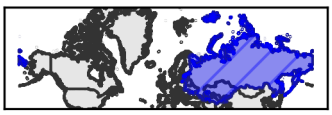
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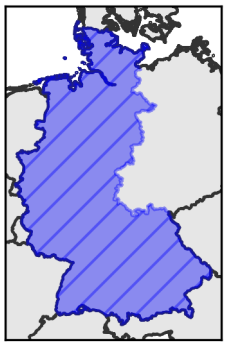
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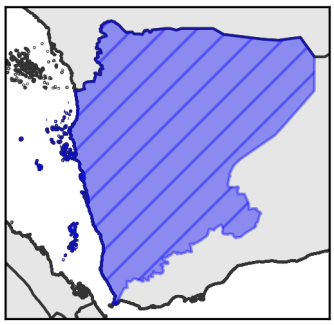
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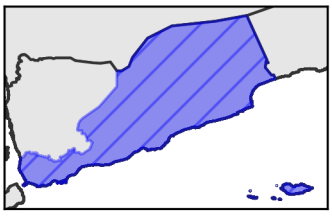
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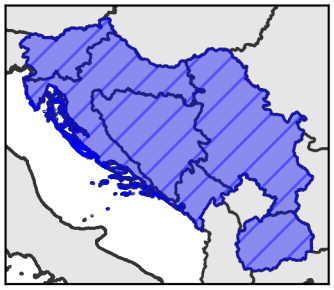
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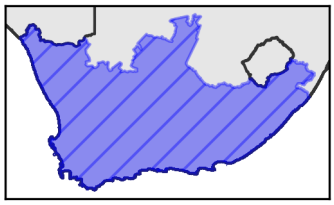
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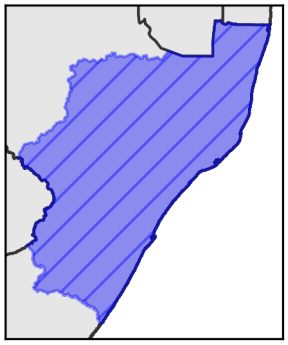
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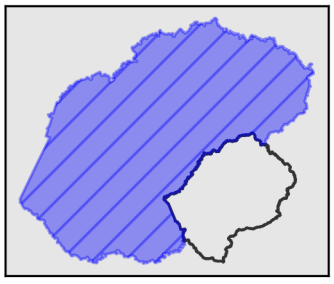
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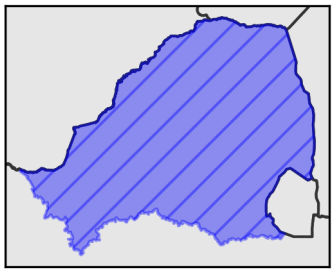
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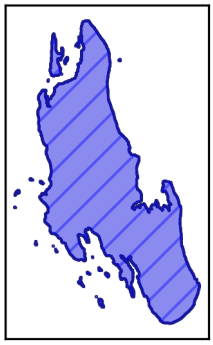
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